

§4 控制系统的硬件设计 /Hardware Design of Control System

本章要求/Target of Chapter：

- 学习汽车电子硬件设计的基本过程/Learn basic design process of automotive electronic hardware；
- 学会使用PROTEL DXP的基本操作/Learn basic operation of Protel DXP;
- 了解电磁兼容性设计的基本背景/Understand basic background of EMC;
- 学会查阅器件的相关文档/Learn working with chip specifications;
- 完成大作业的硬件设计/Finish hardware design of lecture project



§4 控制系统的硬件设计 /Hardware Design of Control System

●ECU照片/Hardware Pictures for ECU

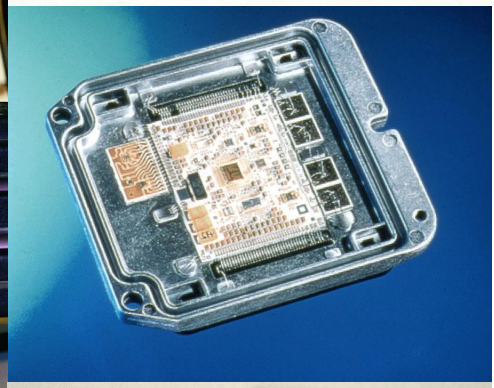
















§4 控制系统的硬件设计 /Hardware Design of Control System

●问题：汽车电子控制系统的大脑ECU是怎么设计出来的/How to design the brain of AECs– ECU?

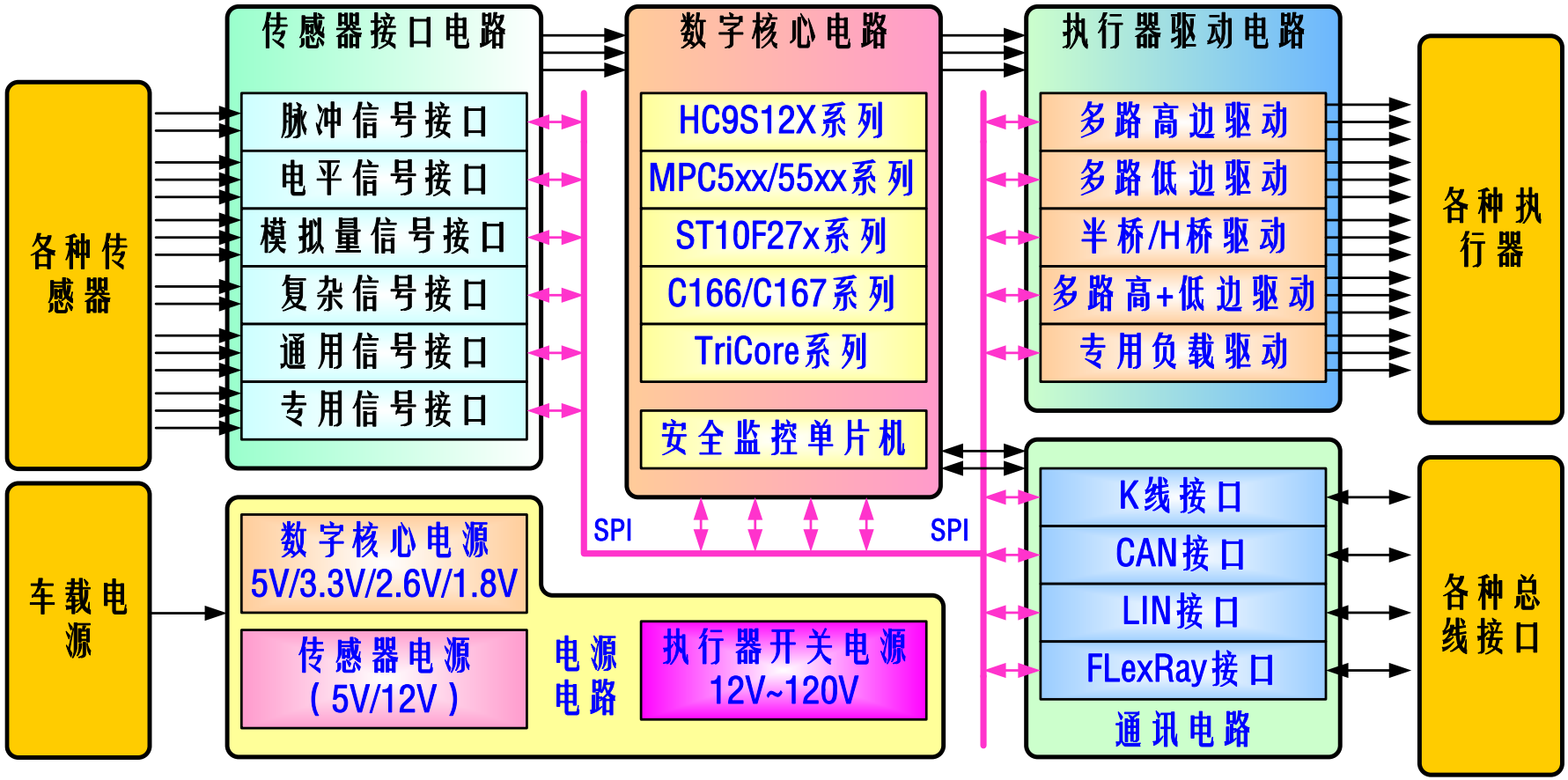
◆图片：汽油机电控系统ECU；/ Pictures : ECU for Gasoline Engine

◆更多图片：自动变速箱，柴油机电控系统/ More Pictures: ECU for gearbox, diesel engine

●结论：硬件设计要抓住硬件的基本框架/ Conclusion: key point for hardware design: essential framework of ACEs



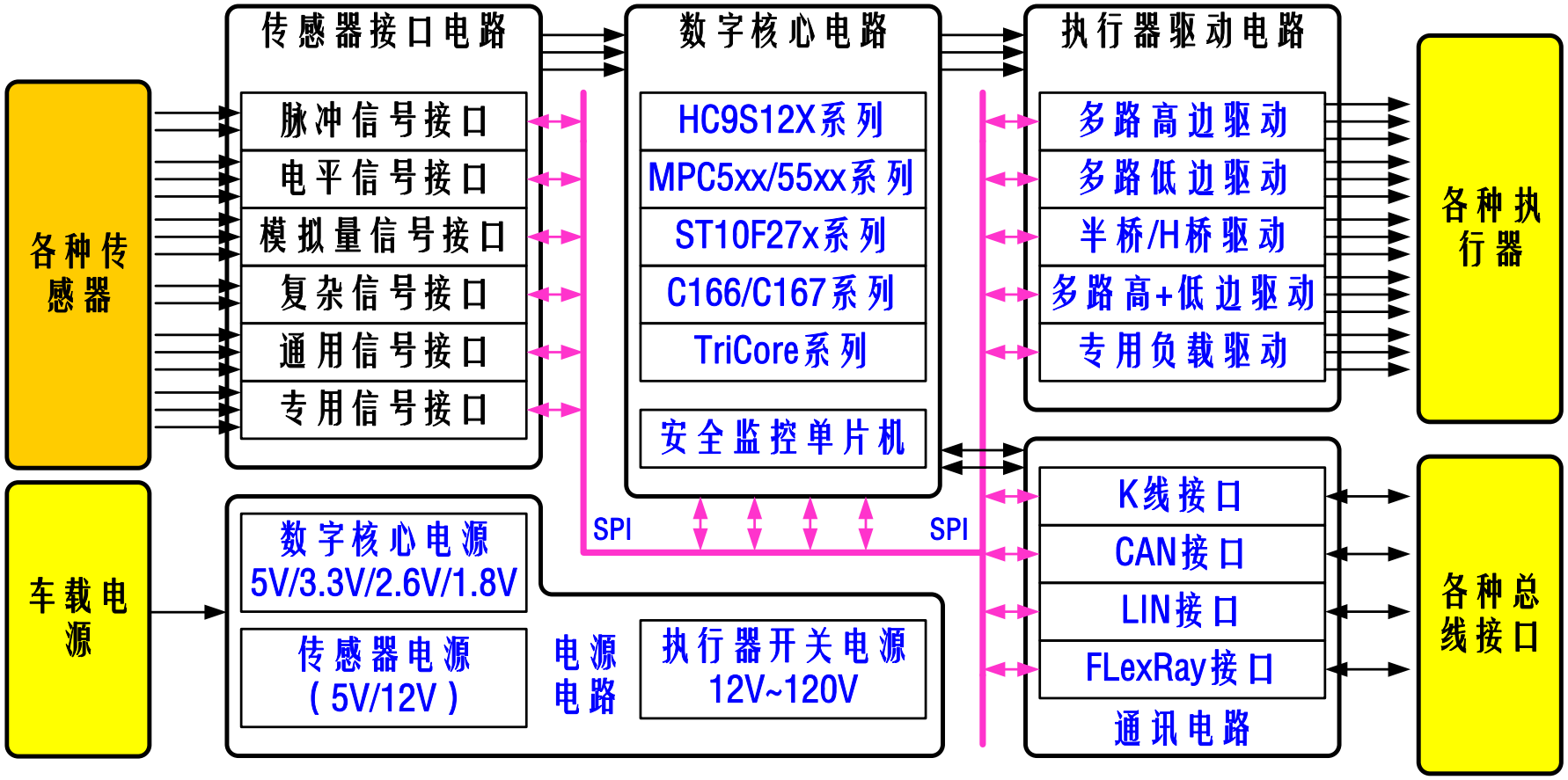
控制系统的硬件框架：Hardware framework of AECs





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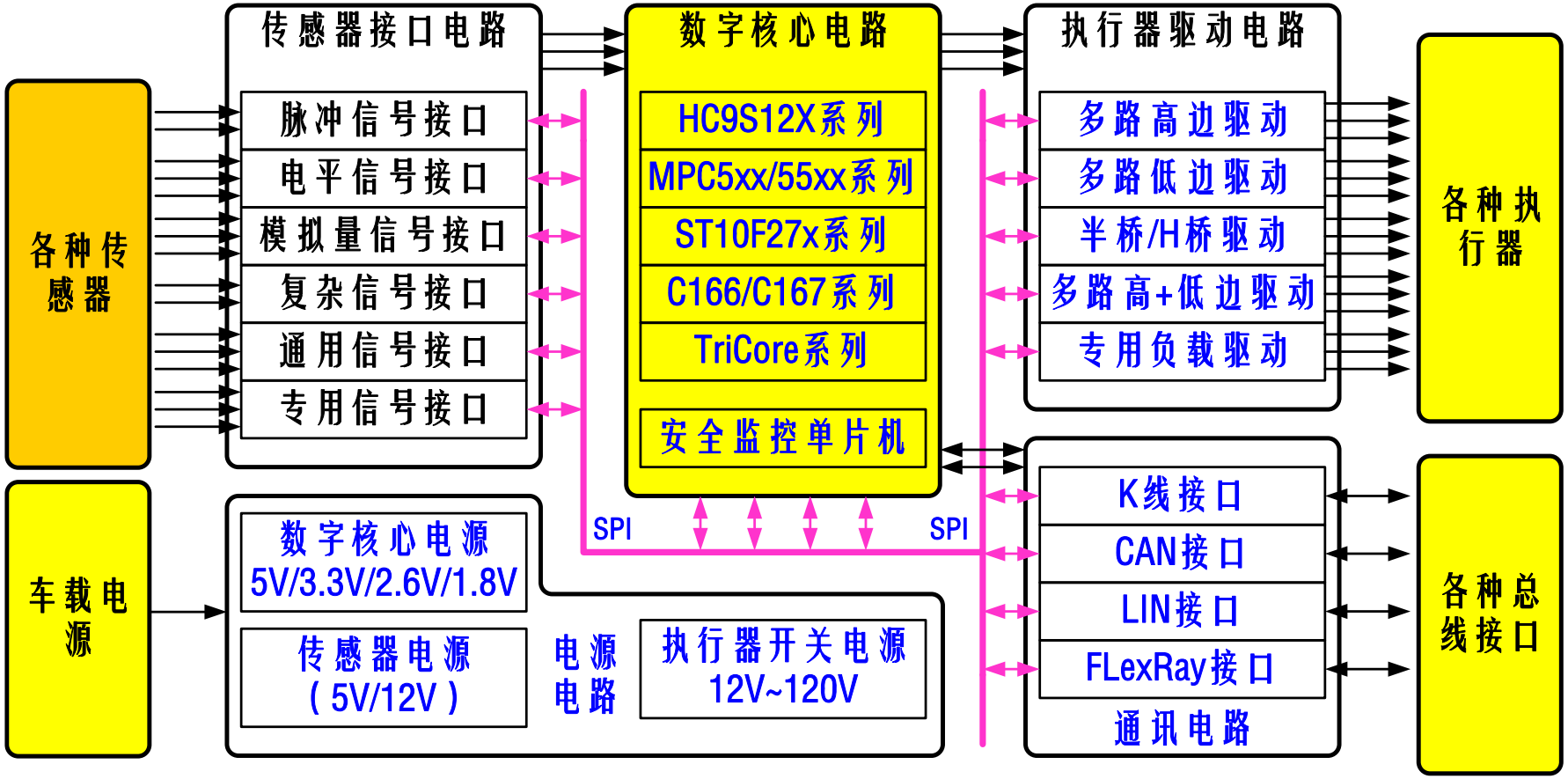
1) 确定传感器和执行器/Define Sensors and Actuators；





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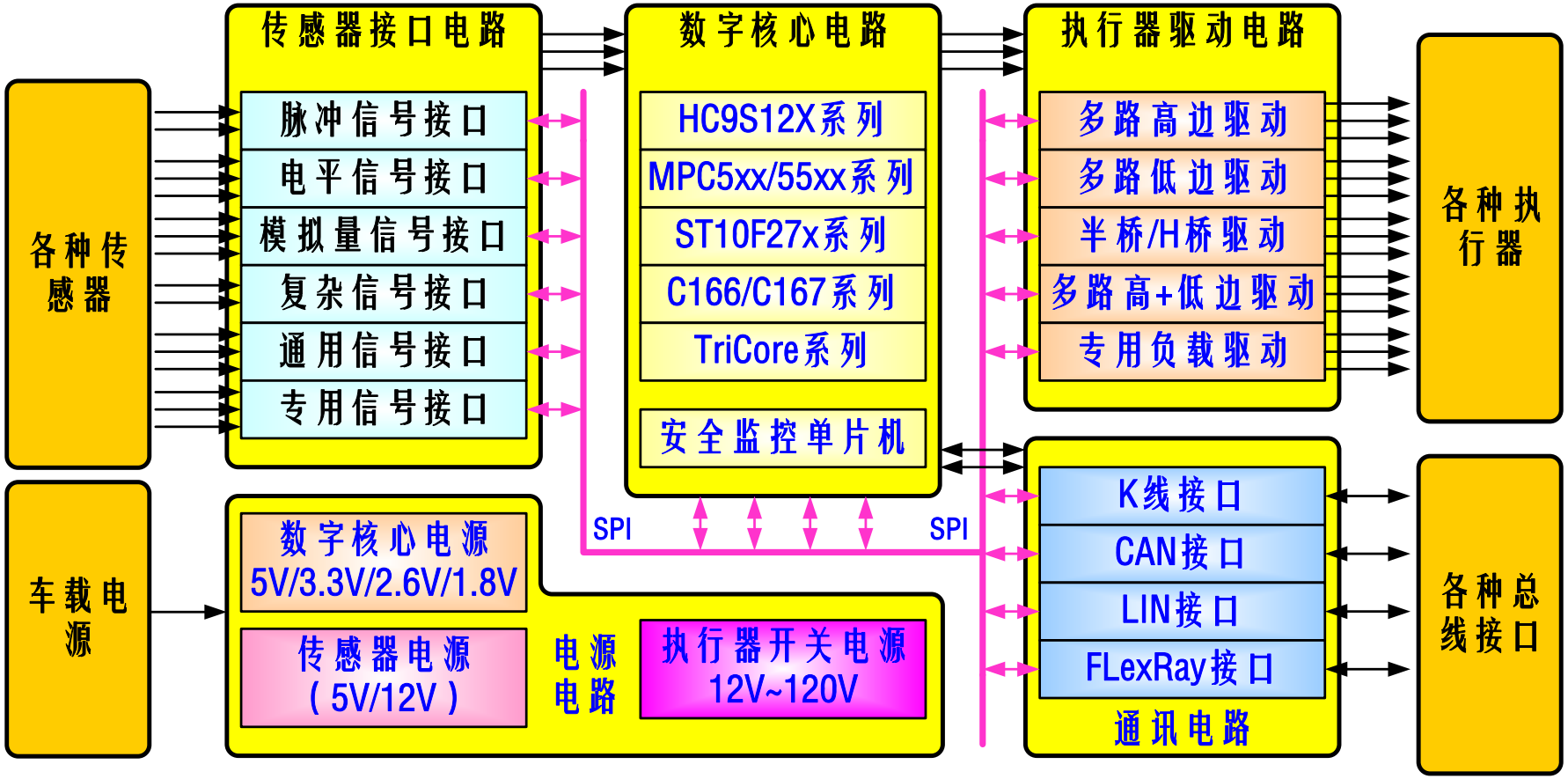
2) 根据控制算法和实时性的要求，确定单片机/ Select the MCU according to the control algorithm and real-time requirement.





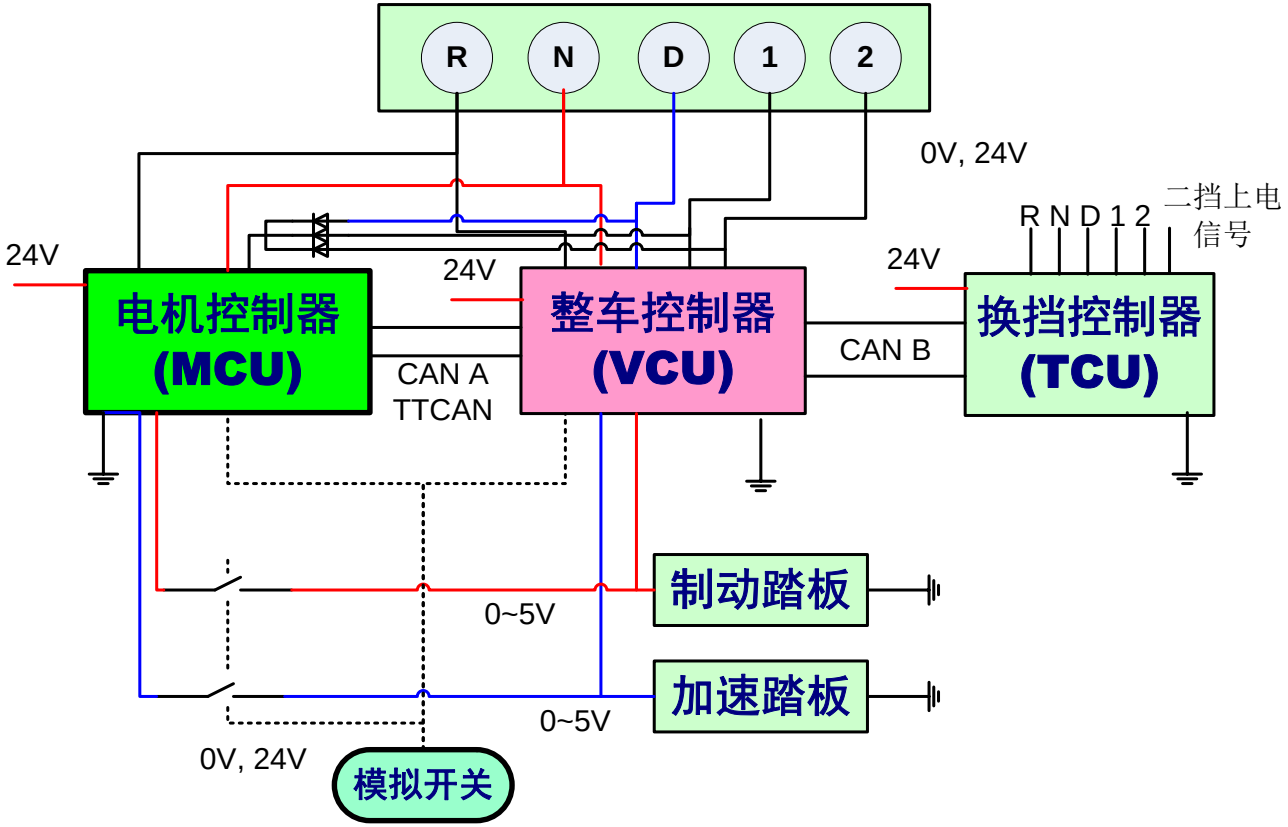
§4 控制系统的硬件设计 /Hardware Design of Control System

3) 将传感器、微处理器和执行器代入硬件设计系统的框架中 /Construct the framework with selected sensors, actuators, and MCUs





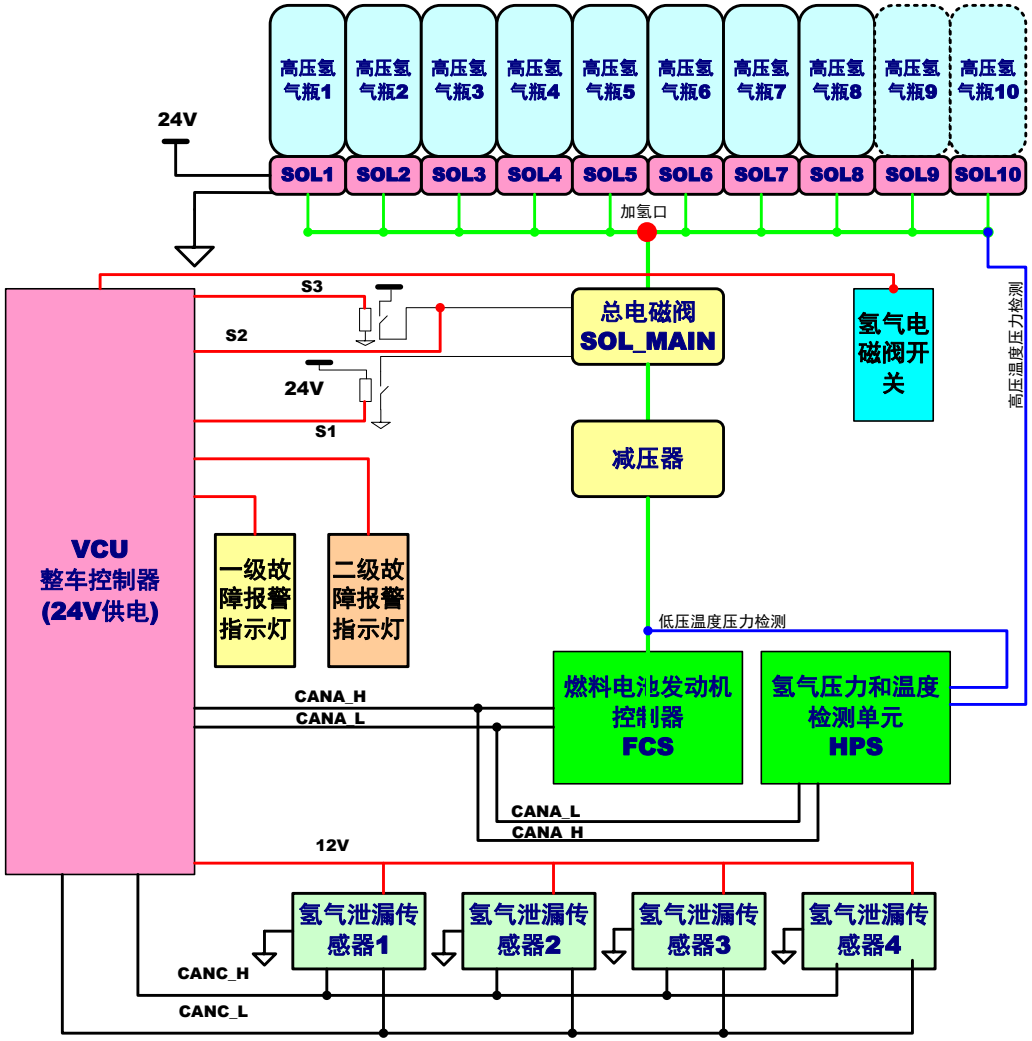
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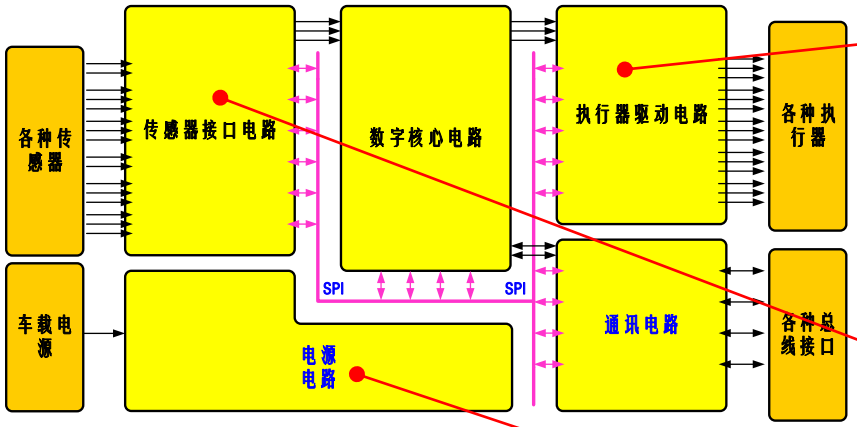
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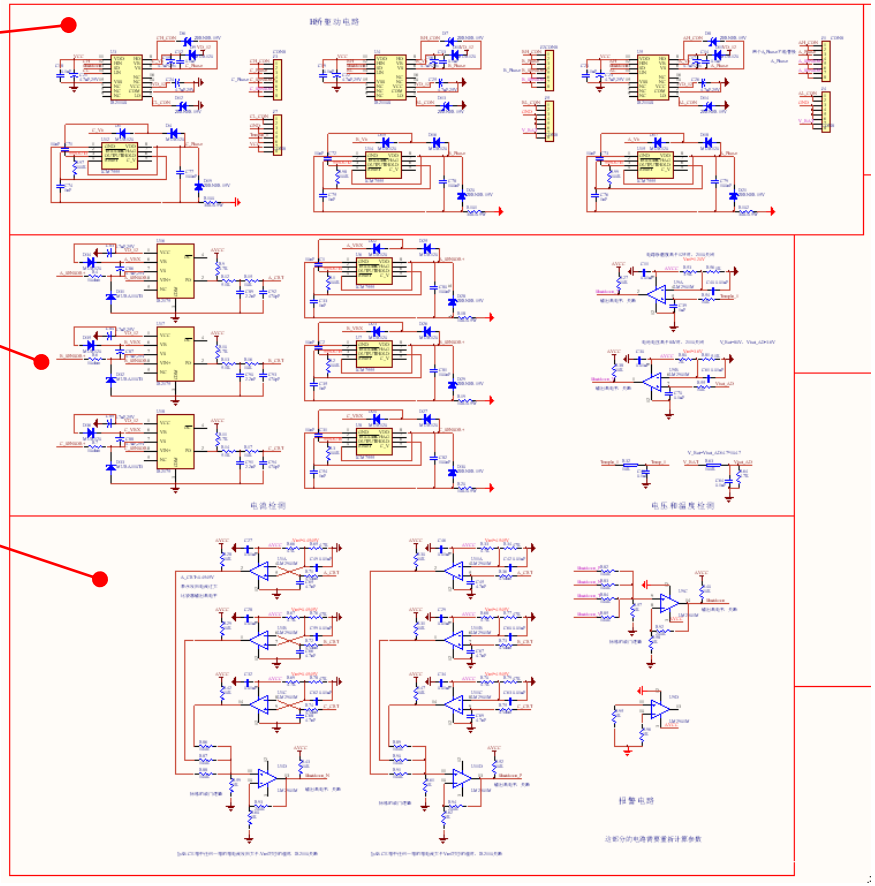




§4 控制系统的硬件设计 /Hardware Design of Control System

4) 原理图的细化：确定设计功能细节；数字和模拟信号处理；驱动电路的仿真分析/ Schematic circuit design: Function analysis, digital and signal process, simulation of drive circuit.



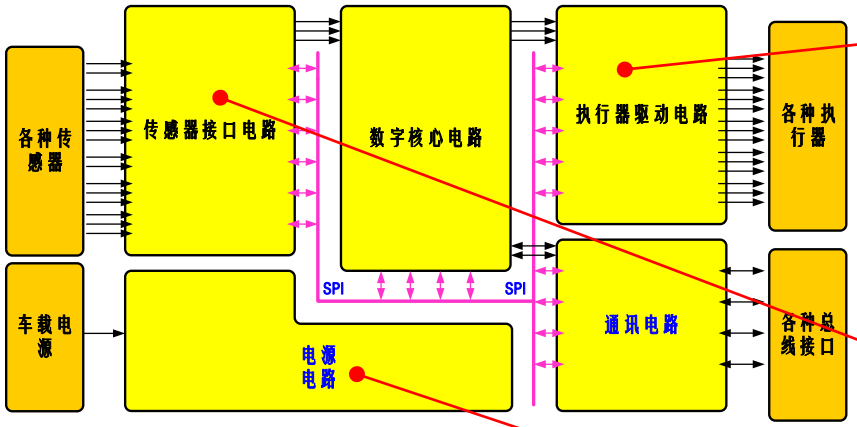


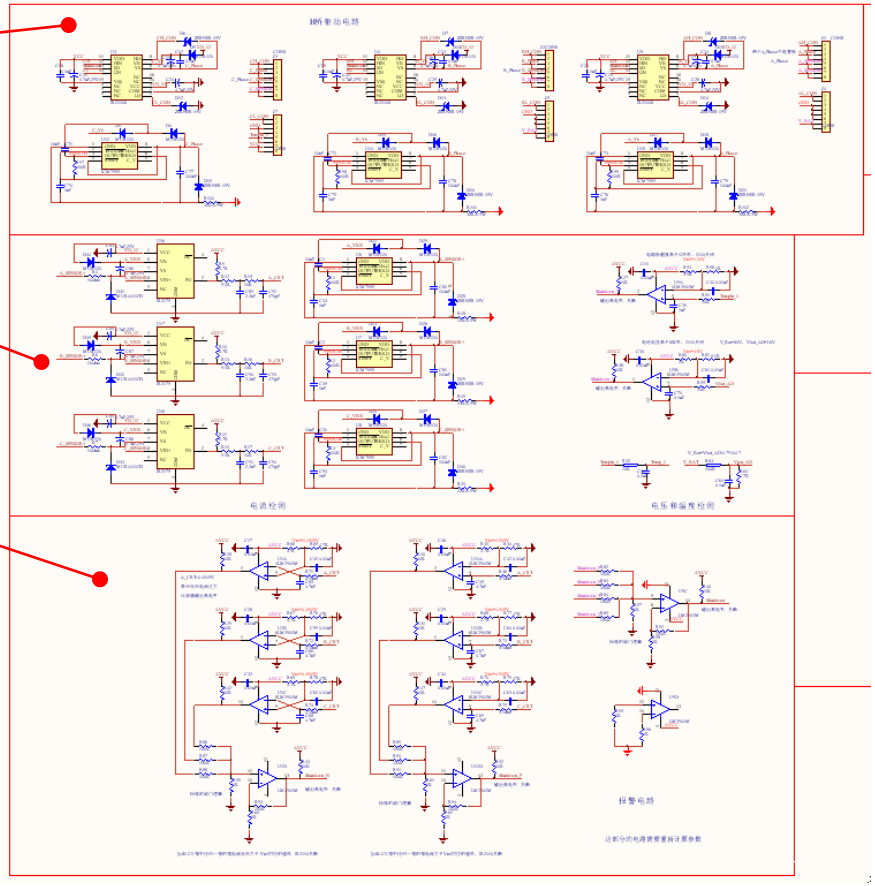
当电路比较简单时，所有电路可以画在一张图纸上/ The schematic circuit can be draw in one sheet if the ECU is simple.



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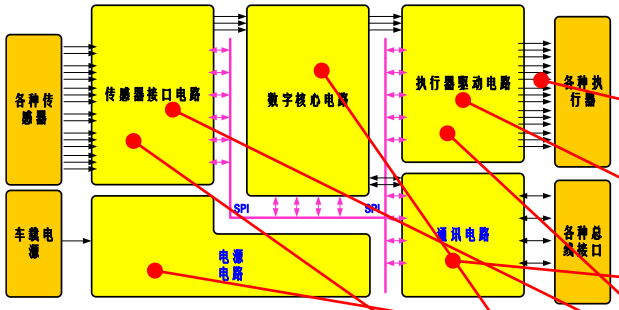


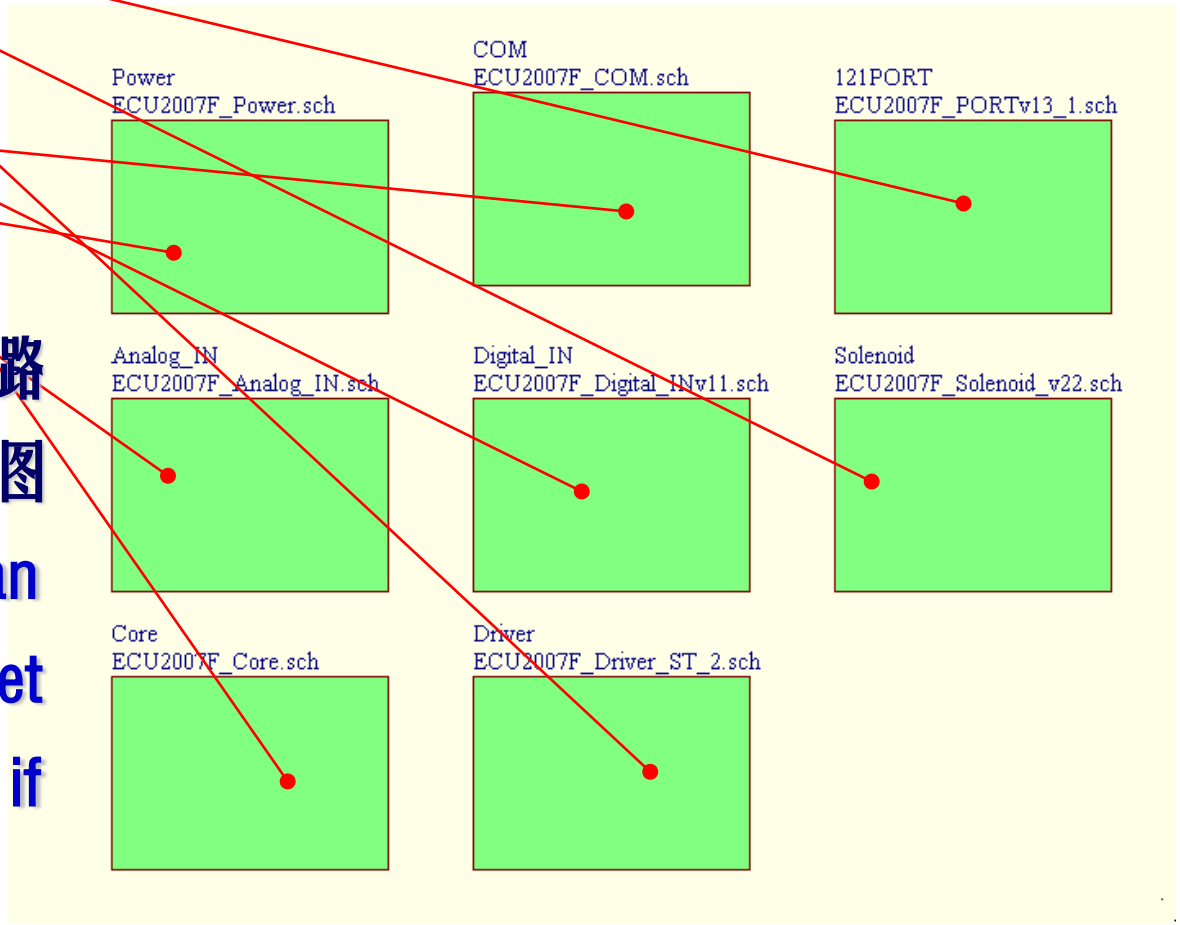


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4) 原理图的设计/ Schematic circuit design

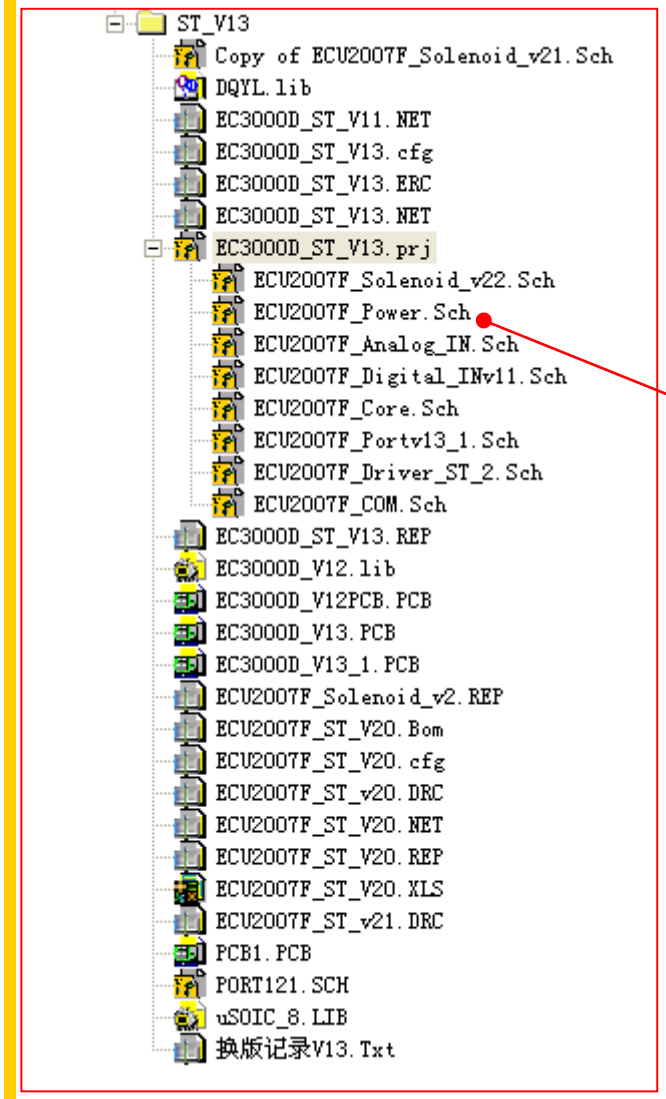


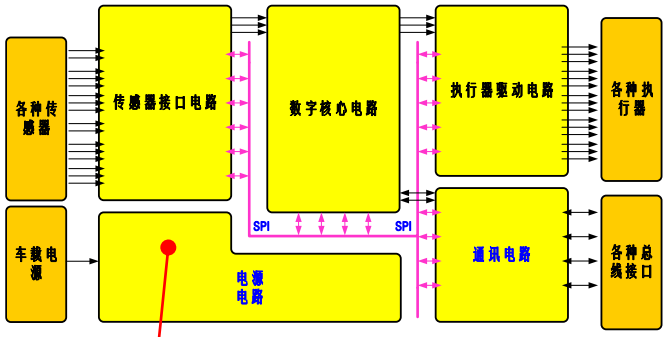


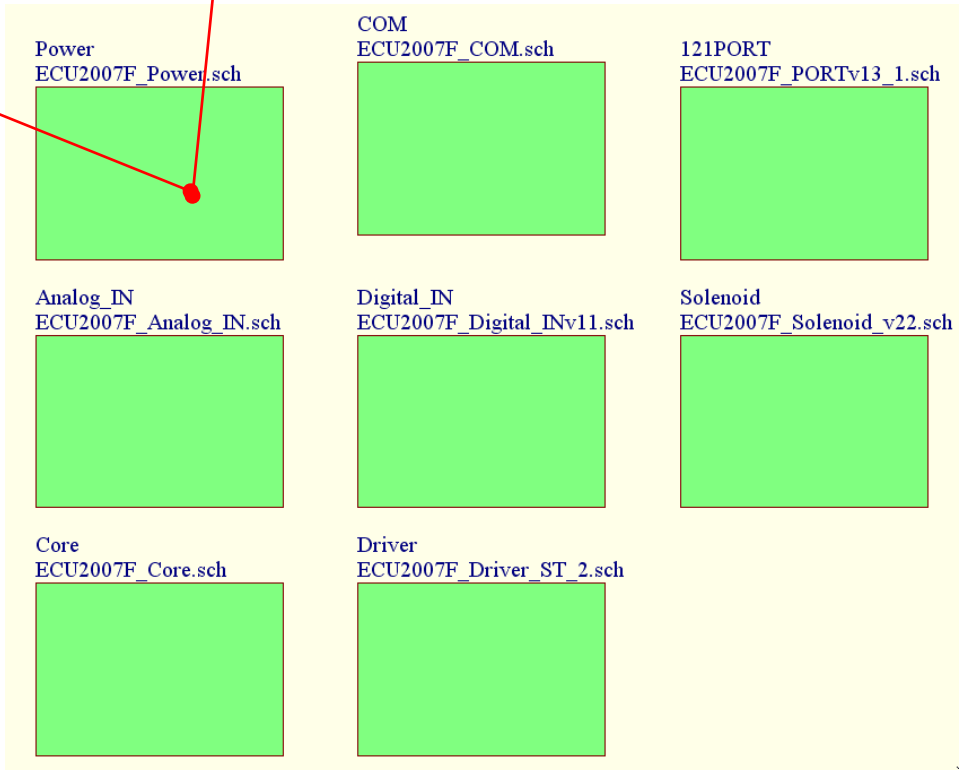
当电路比较复杂时，电路可以按照功能分好几张图/ The schematic circuit can be divided into multi-sheet according to the function if the ECU is complex.



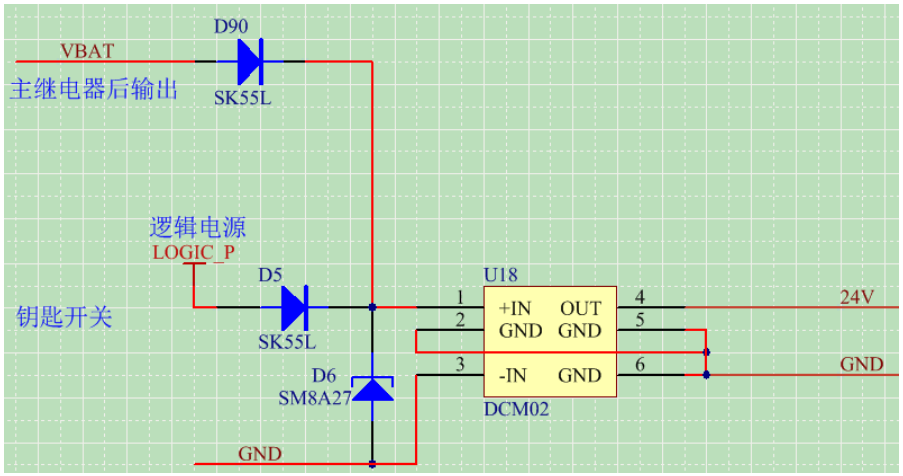
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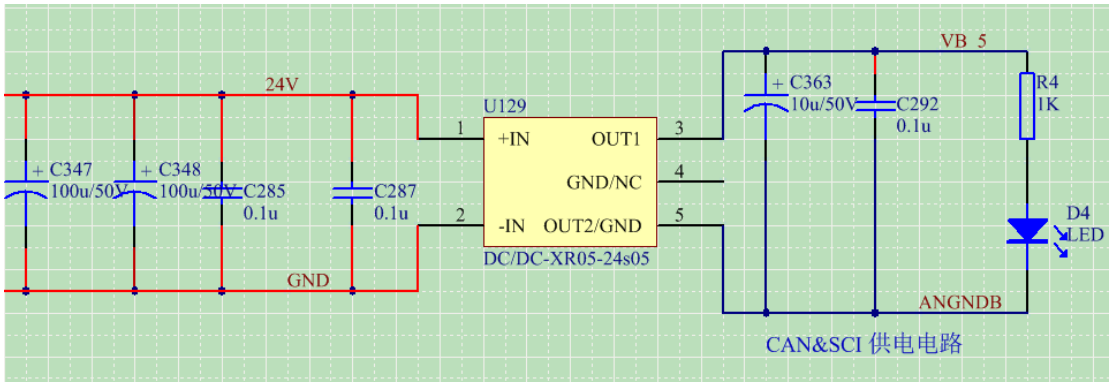






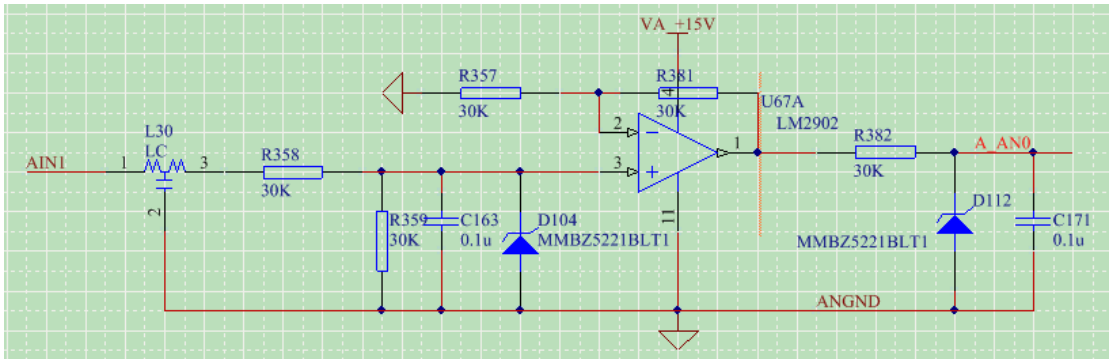
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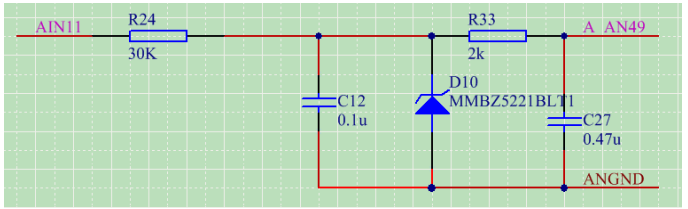


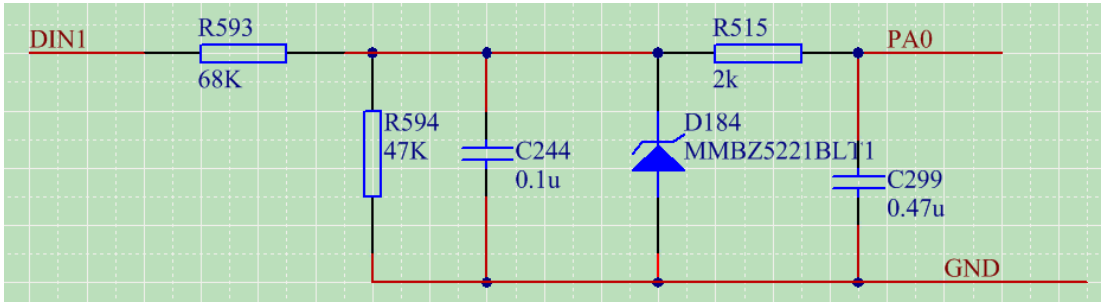




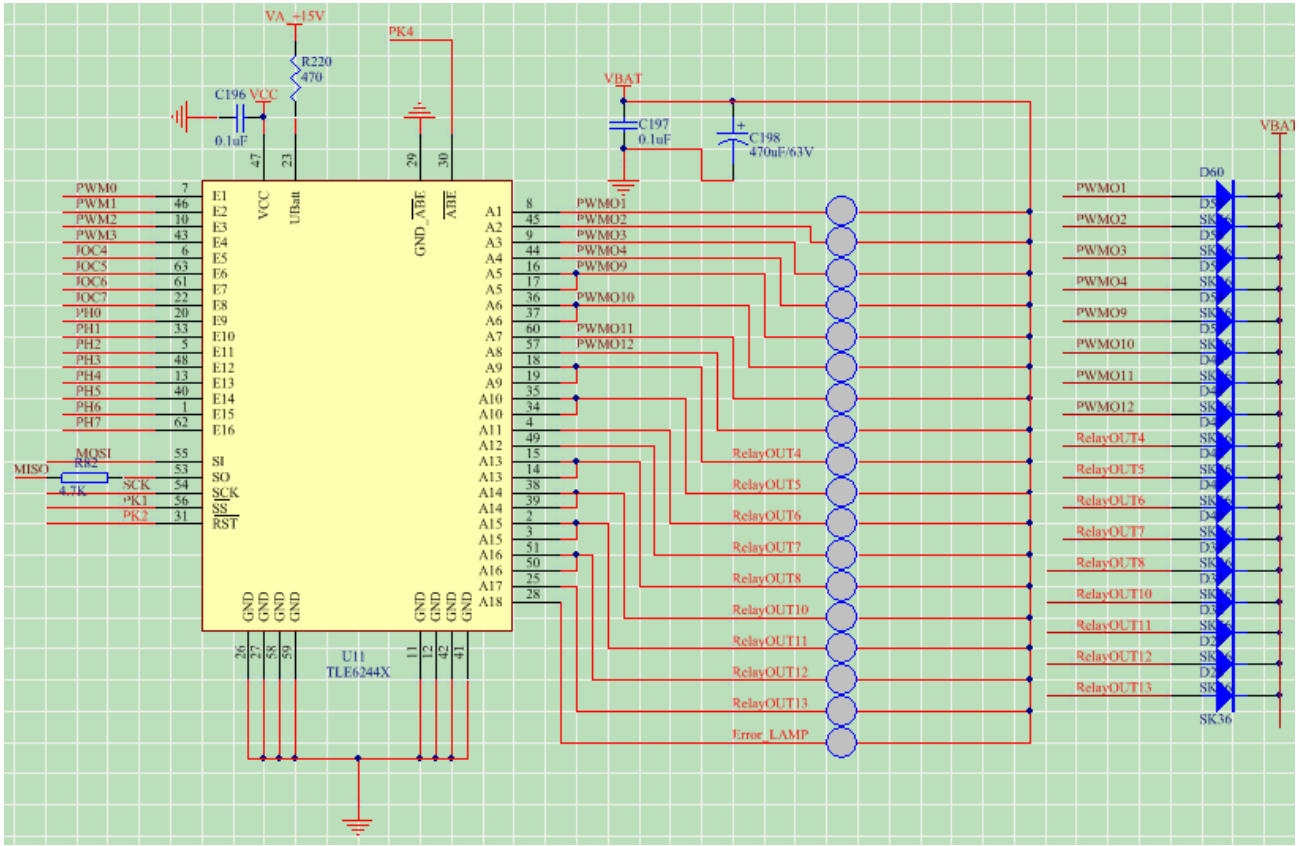
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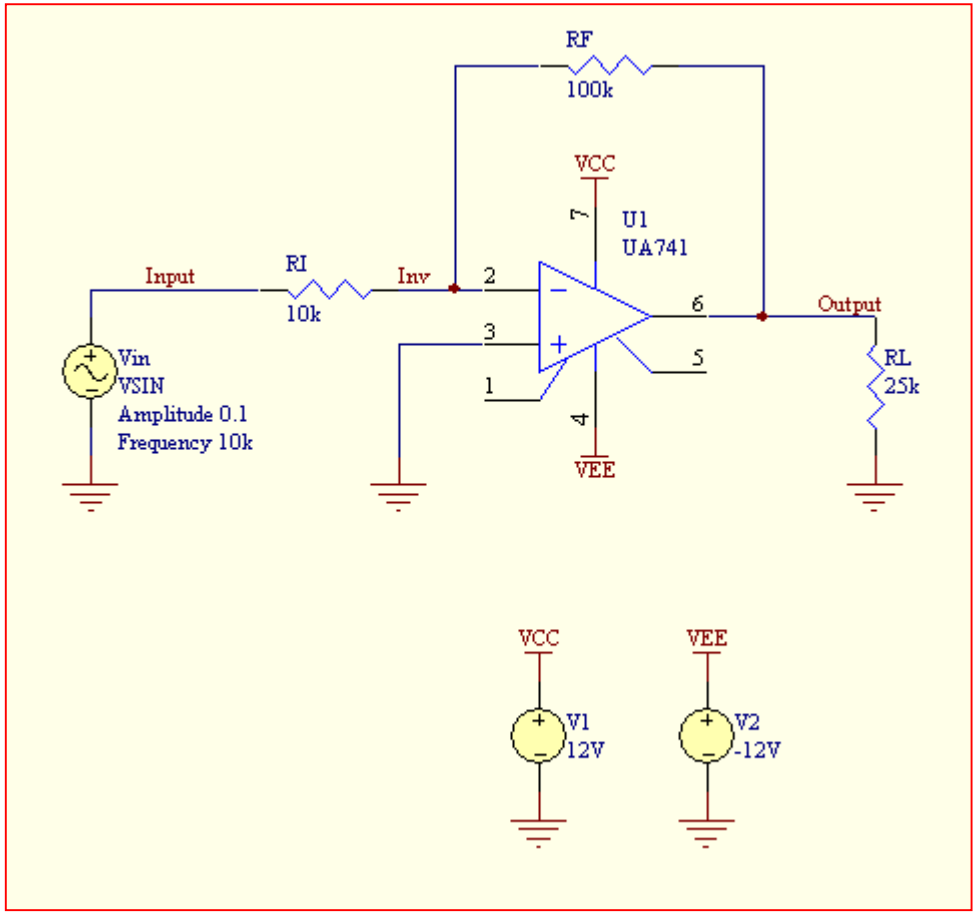


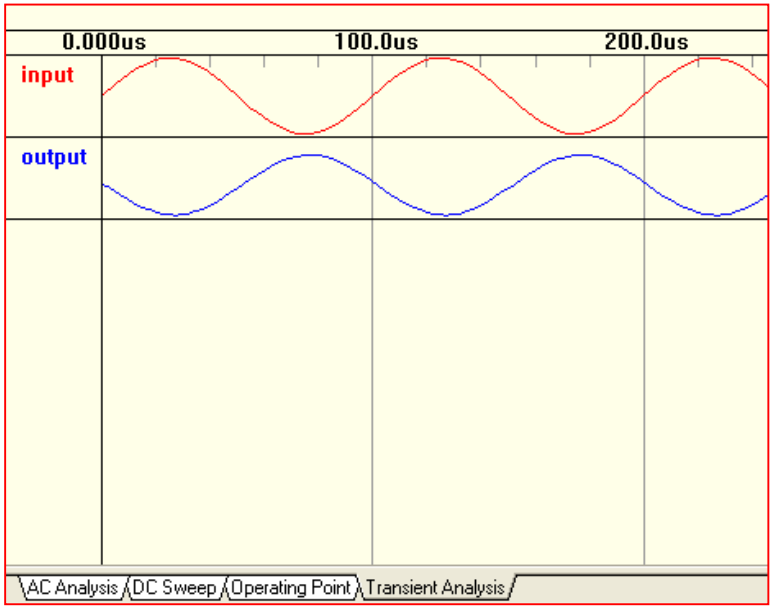


4) 原理图的设计/ Schematic circuit design



5) 电磁兼容性：审核并修改设计细节/ Electro-Magnetic Compatibility analysis and review







5) 电磁兼容性：审核并修改设计细节/ Electro-Magnetic Compatibility analysis and review: TVS design

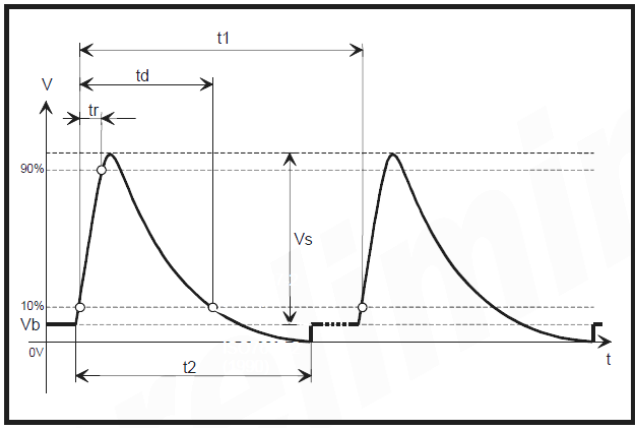
Vb = 27V +/–1.0V, Vs = +100V

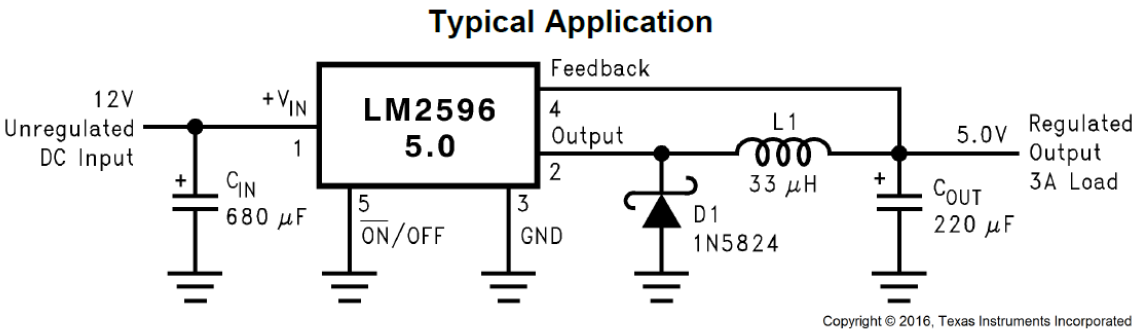
Ri = 10Ω

td = 0.05ms, tr <= 1us

t1 = 1s, t2 = 200ms

duration = 5000 impulses





(Fixed Output Voltage Versions)



5) 电磁兼容性：审核并修改设计细节/ Electro-Magnetic Compatibility analysis and review: TVS design

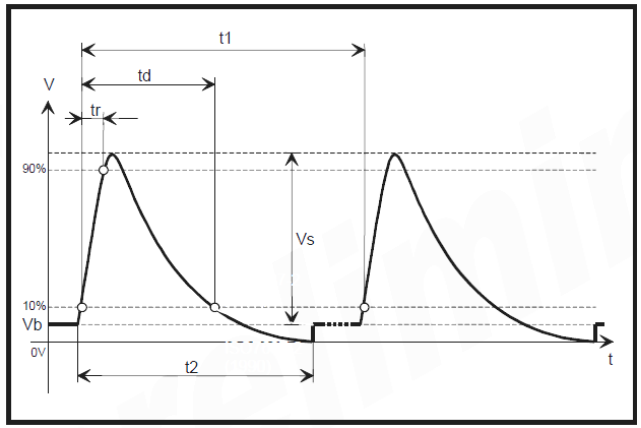
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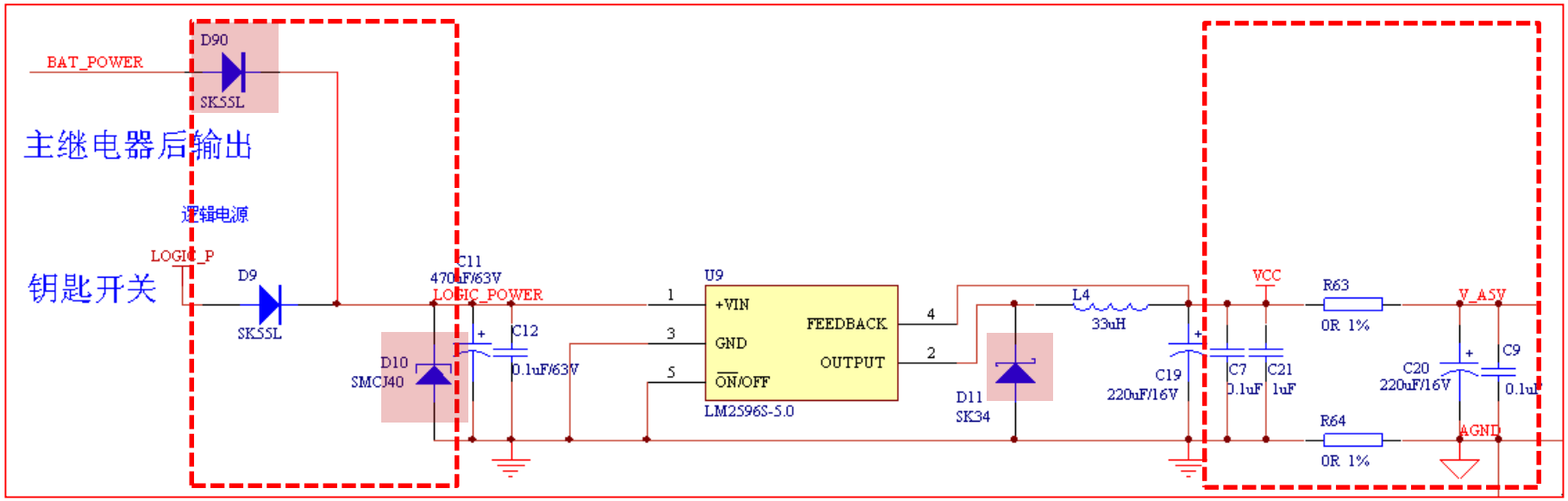
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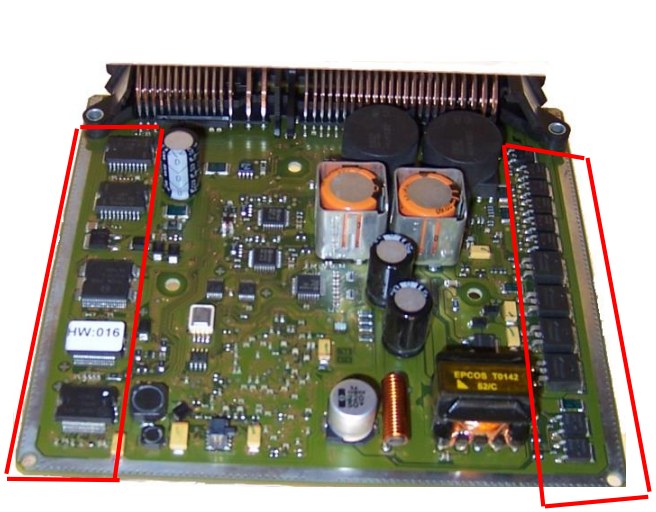


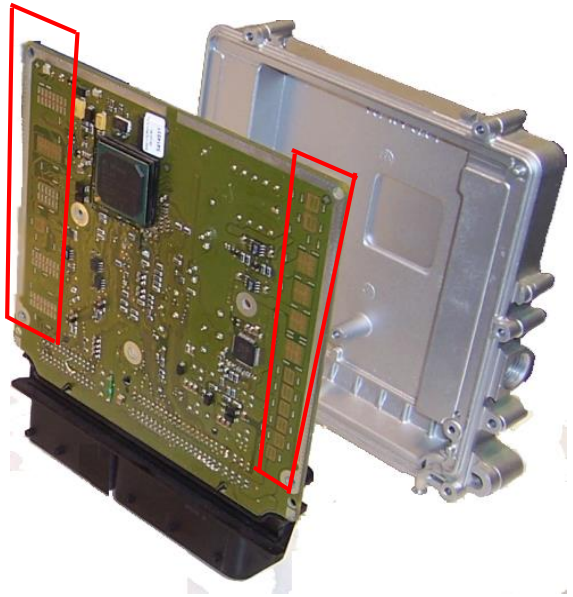




§4 控制系统的硬件设计 /Hardware Design of Control System

6) **设计布线图**：考虑振动、散热、可靠性、电磁兼容性、成本、安装、调试、版本升级和兼容……/ Publish Circuit Board design with considering vibration, power dissipation, reliability, EMC, cost, installation, debug, compatibility, update,……





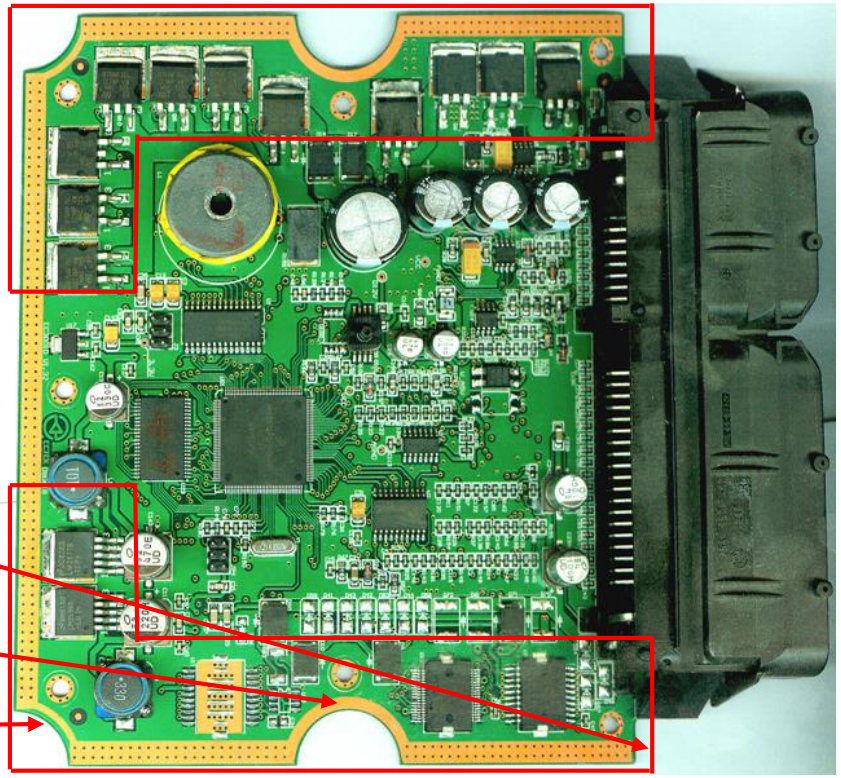
通常，功率电路设计在ECU的PCB的外围区域，一个方面是为了更好散热，另外一个也是为了获得更好的EMC性能/Generally, the drive circuit is assigned in the edge area of the PCB to obtain better power dissipation performance and better EMC performance.



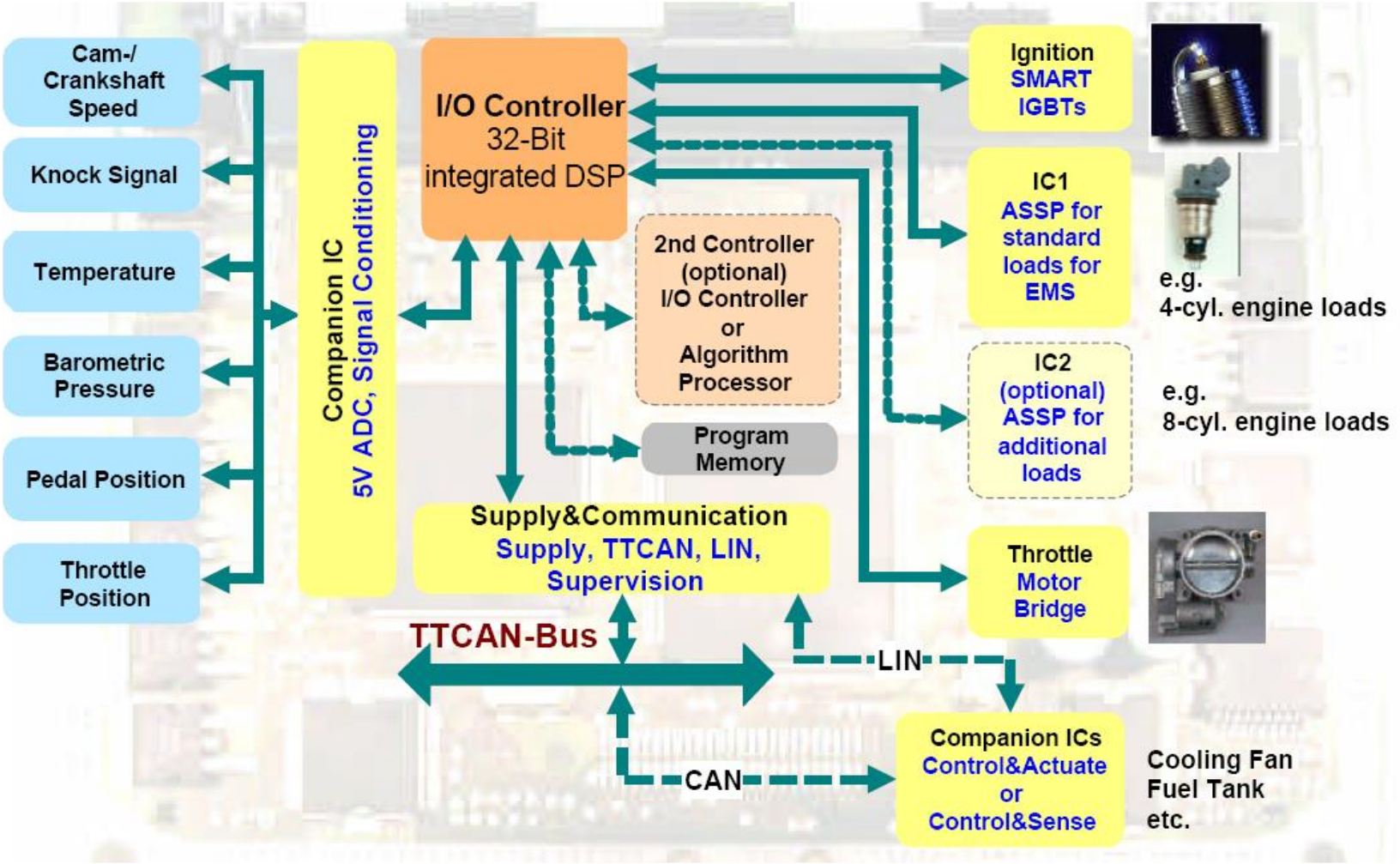
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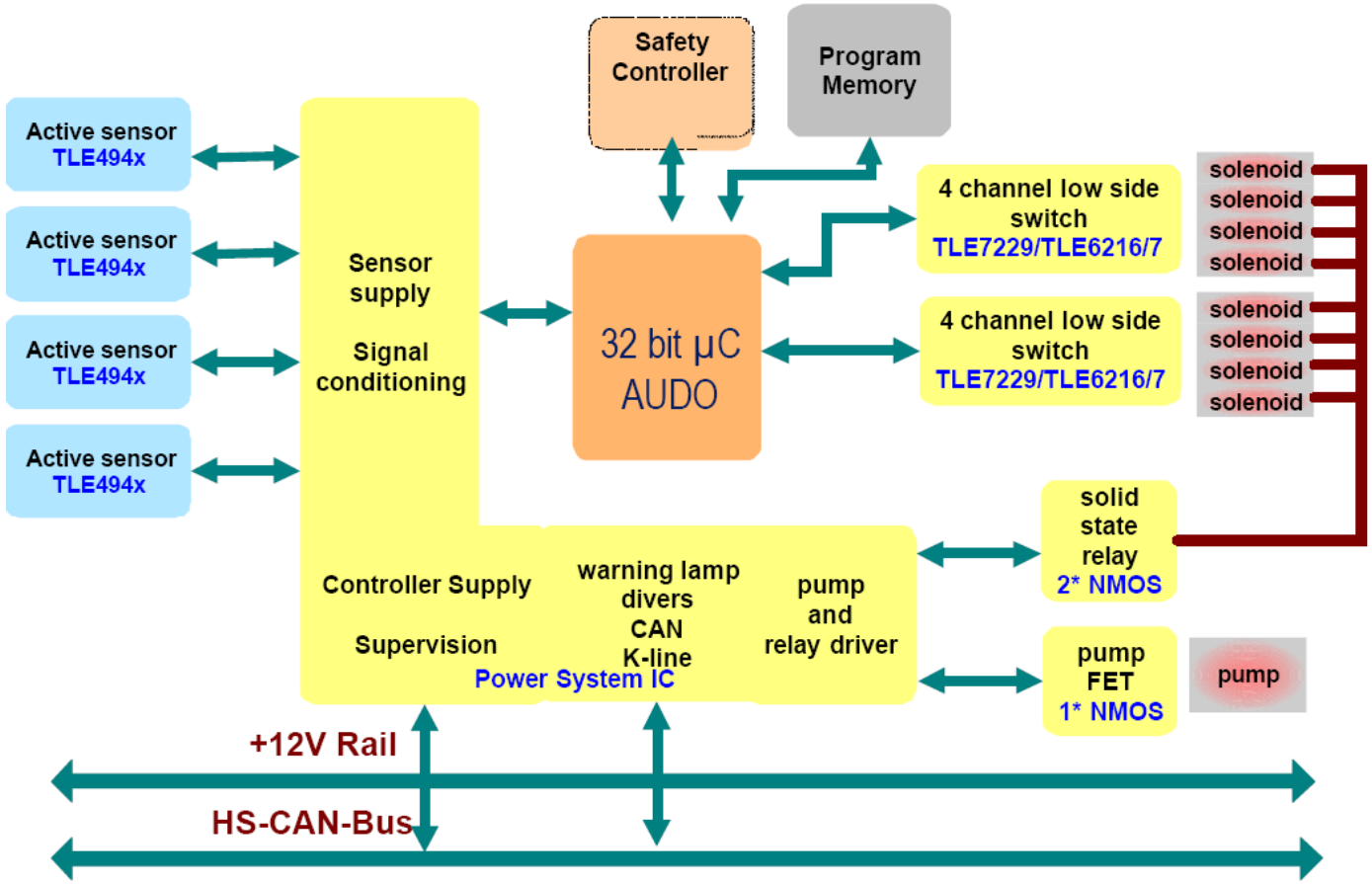


汽油机ECU / Gasoline Engine Management – ECU

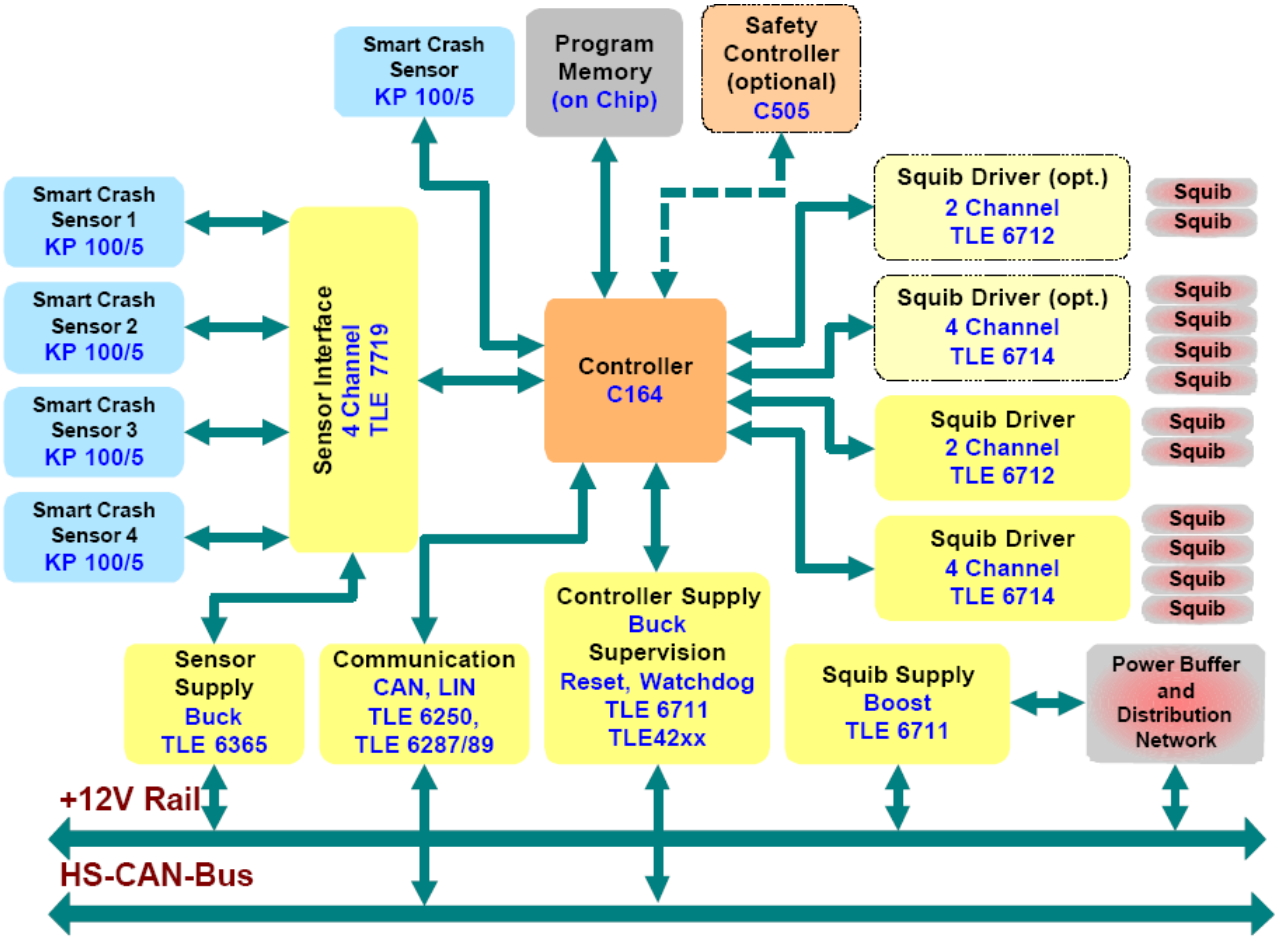


ASSP：Application Specific Standard Parts，专用标准产品

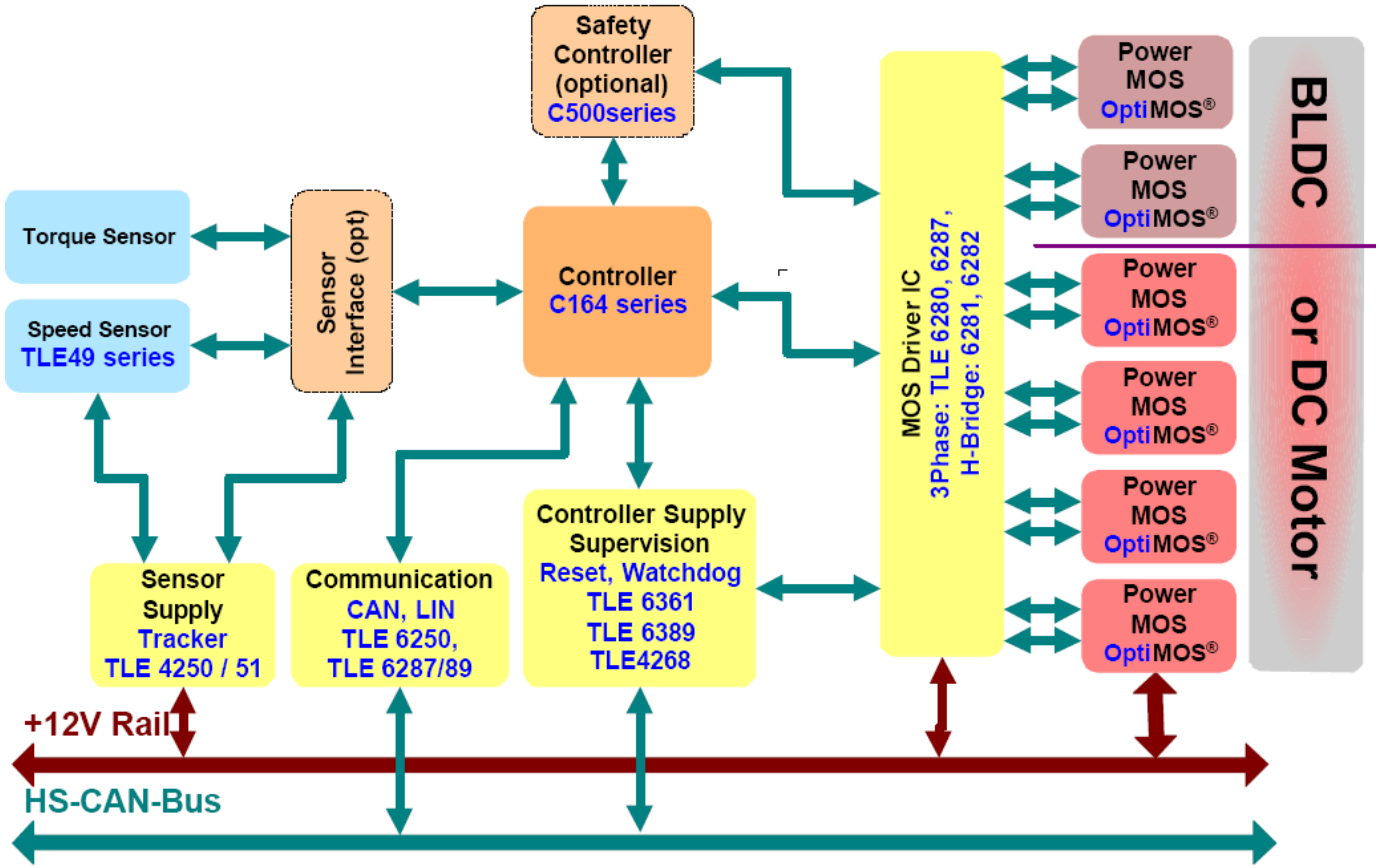
ABS系统框架 /System Partitioning for ABS Systems



成员约束系统/Restraint; System Block Diagram

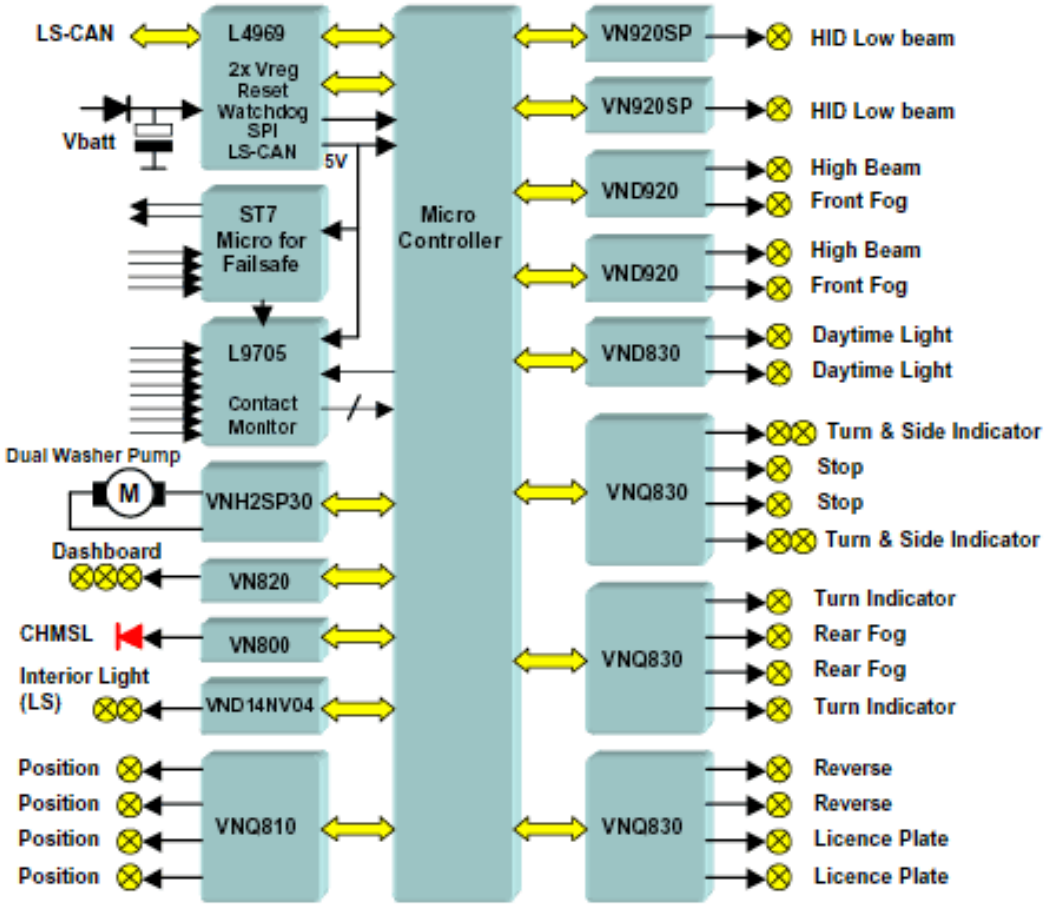


线控转向三相无刷直流电机驱动 /Steer-by-wire System
3-Phase BLDC Drive



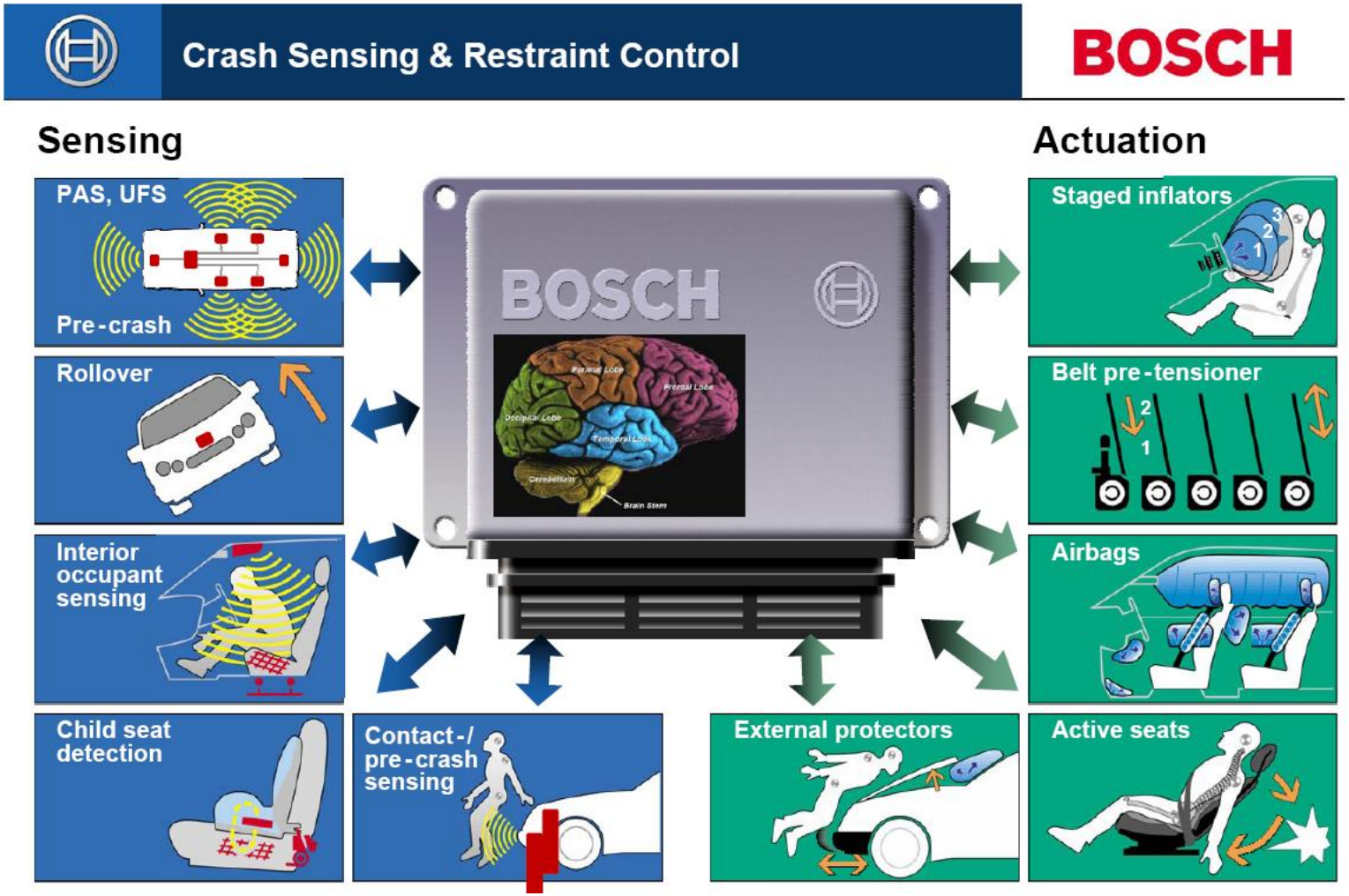


- ST的汽车电子光盘，运行microweb.exe





●博世被动安全控制系统ECU /Bosch passive control ECU

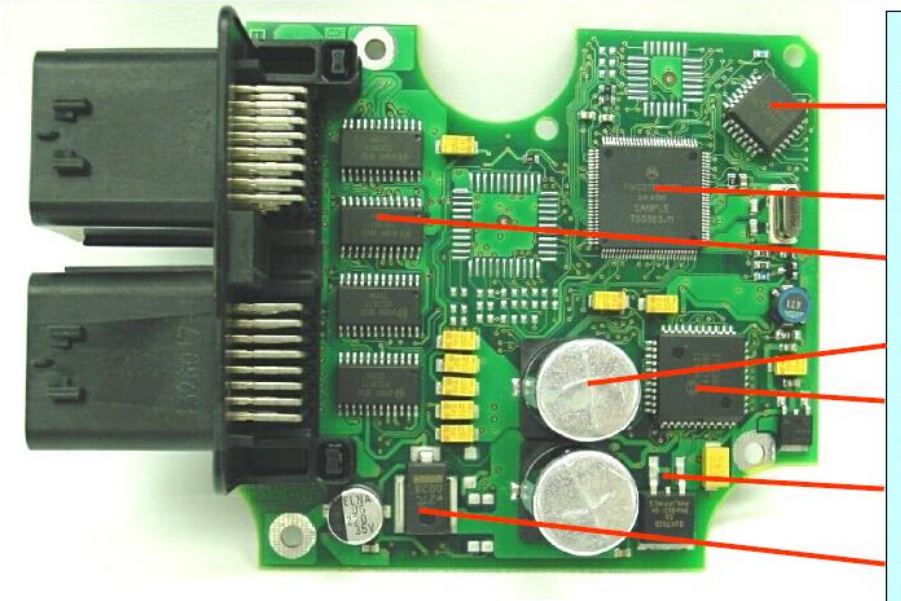




●博世被动安全控制系统ECU /Bosch passive control ECU

Central Restraint Control Unit
Key Components

BOSCH



Key Components:

2 accelerometers (saving sensor)
➔ 45° / 135° orientation

Microcontroller (32 bit)

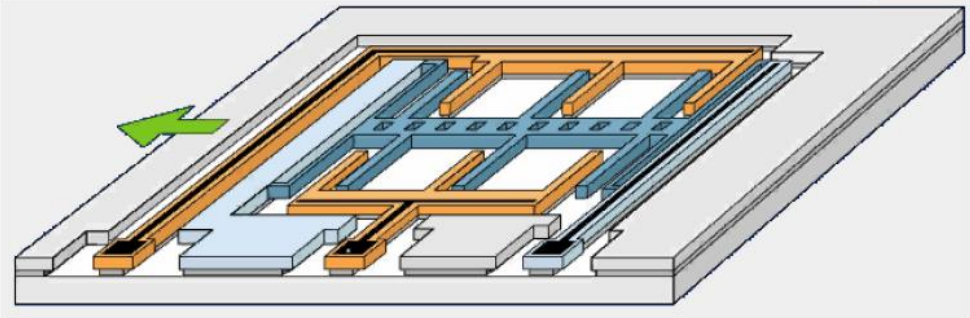
Firing loop IC' s (each for 4 loops)

Energy Reserve capacitors
➔ i.e. 200ms after battery loss

Peripheral IC' s w/ watchdog

Inductor for up/down convert
i.e. 1 s after ignition on

Diode for load dump protection
➔ against voltage peaks

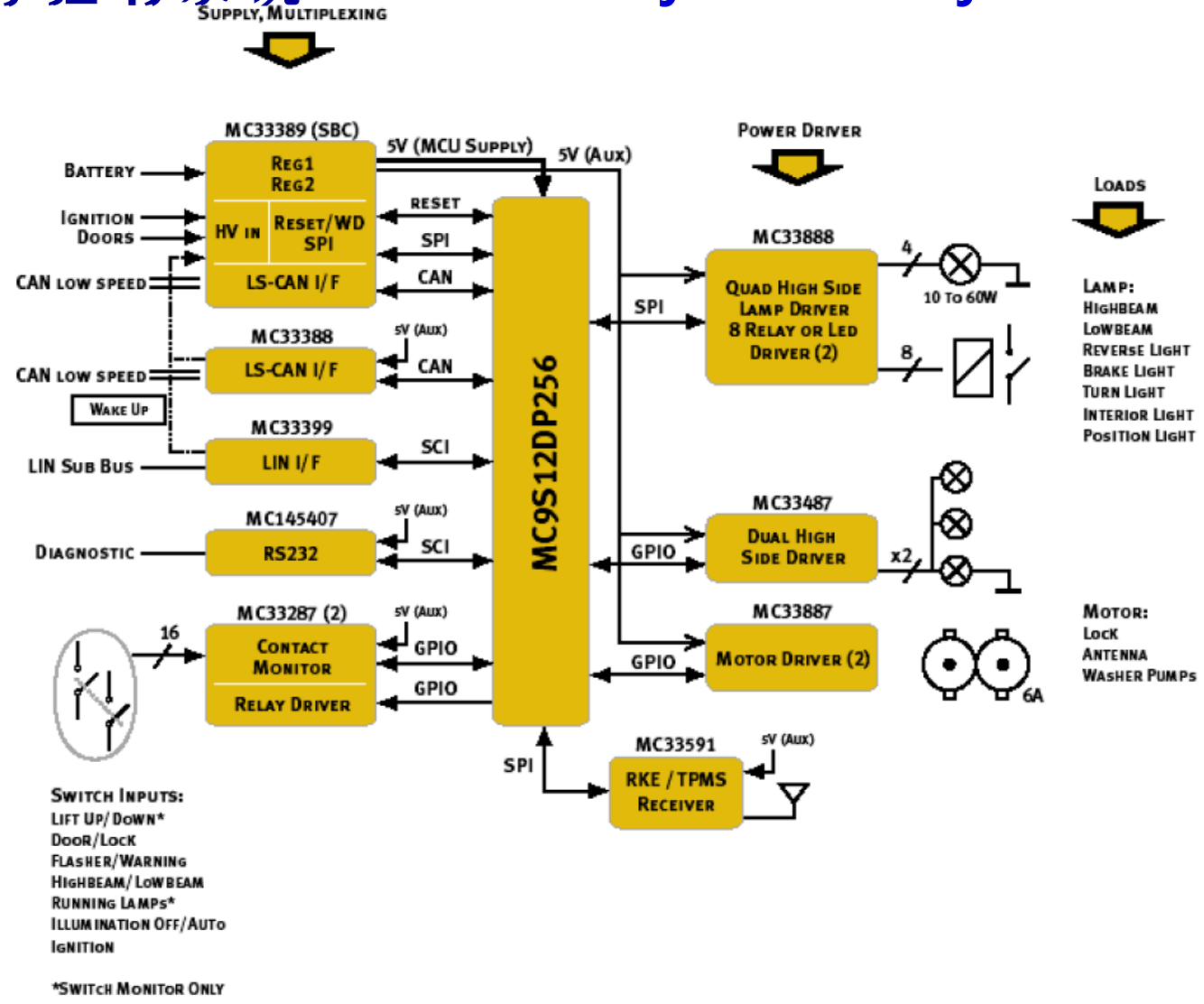


Micro-machined acceleration sensor: capacitive

8



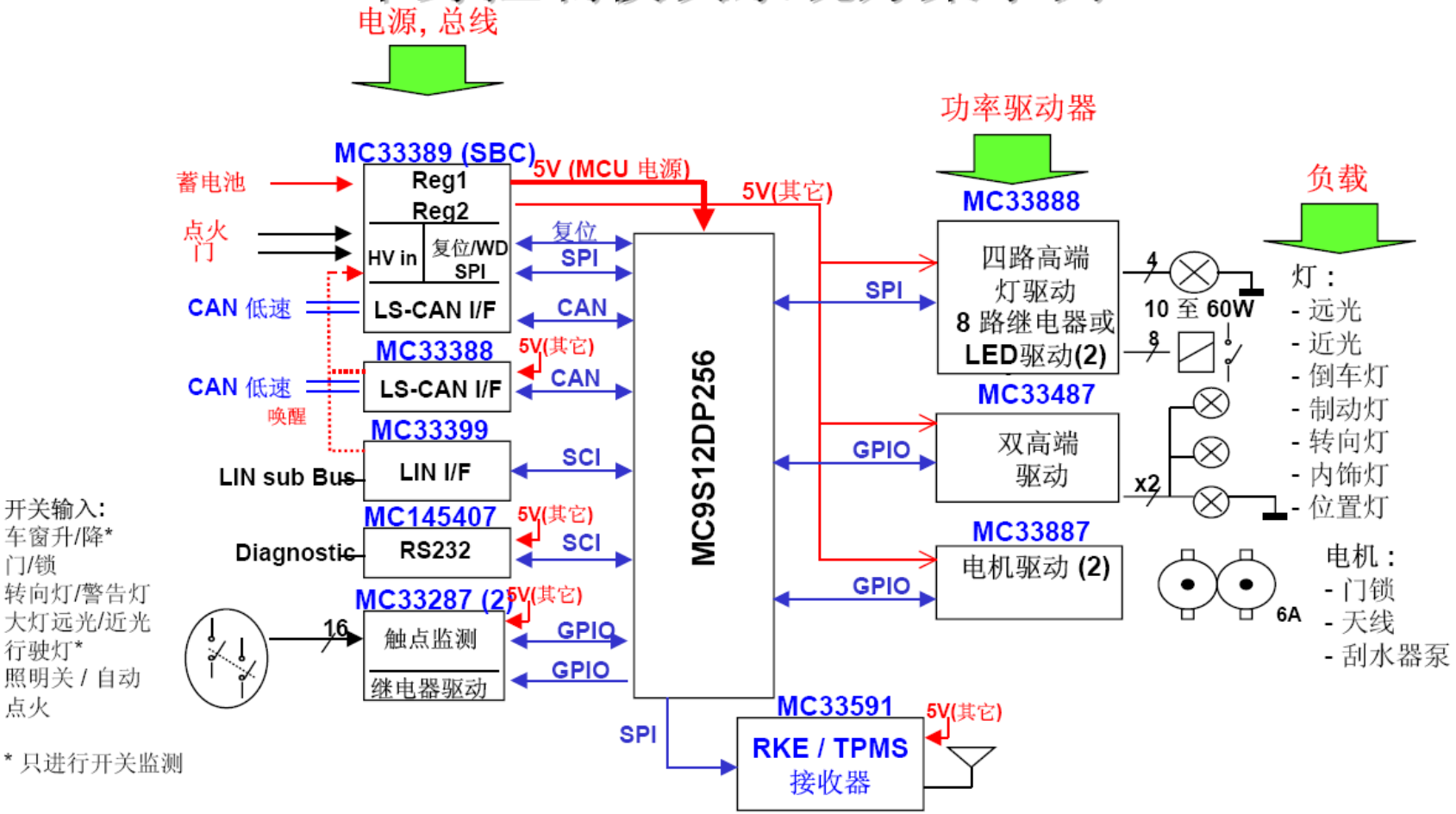
●车身控制系统 /Vehicle Body Control System



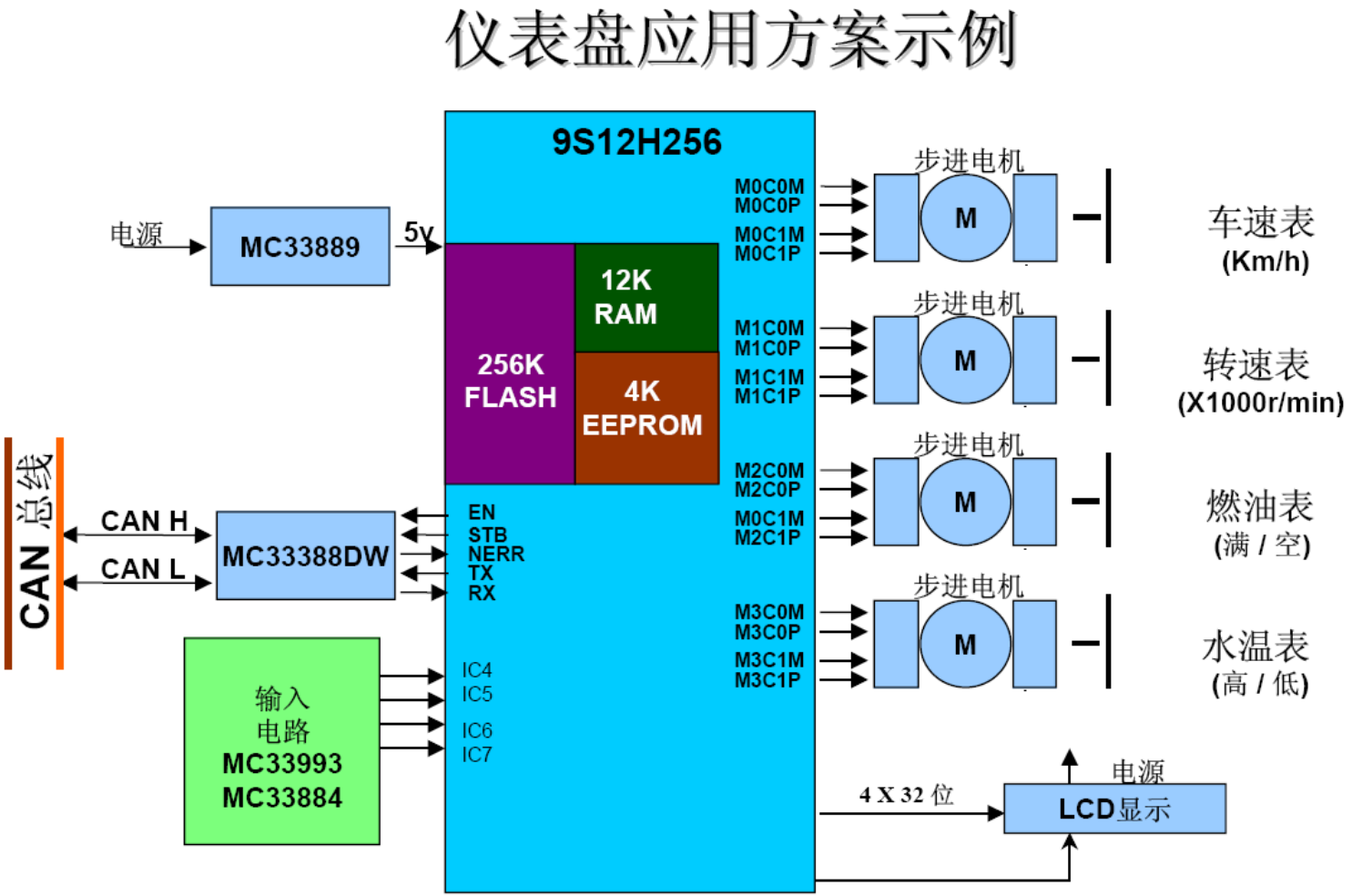


●车身控制系统 /Vehicle Body Control System

车身控制模块系统方案示例



●仪表显示 /Instrumentation





§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

学习参考文档/Protel study multimedia files:

（1）\Protel DXP 典型实例 老虎工作室教学文件

（2）\Protel DXP 入门与提高 老虎工作室教学文件

（3）\Protel DXP电路设计快速入门 张义和著 教学文件

（4）多媒体教学文件

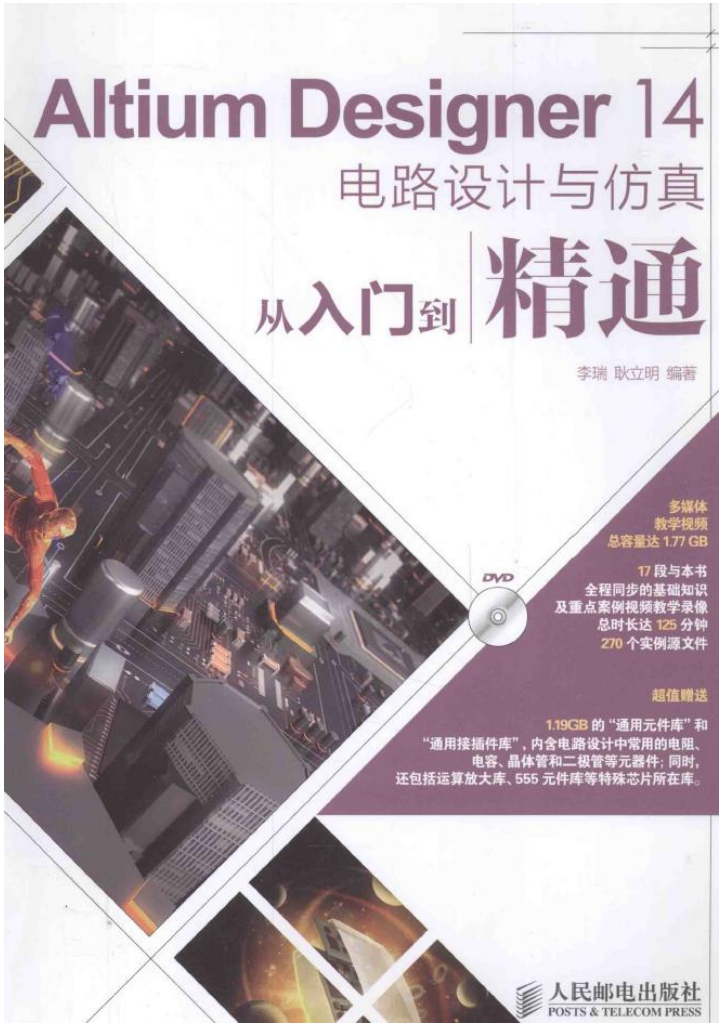






学习参考文档/Protel study multimedia files:

（5）Altium Designer 14 电路设计与仿真——从入门到精通





§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

推荐设计工具软件/ Recommended design tool:

- Altium Designer 2009，即DXP2009



原理图设计的基本步骤/ Basic Process of Schematic Design

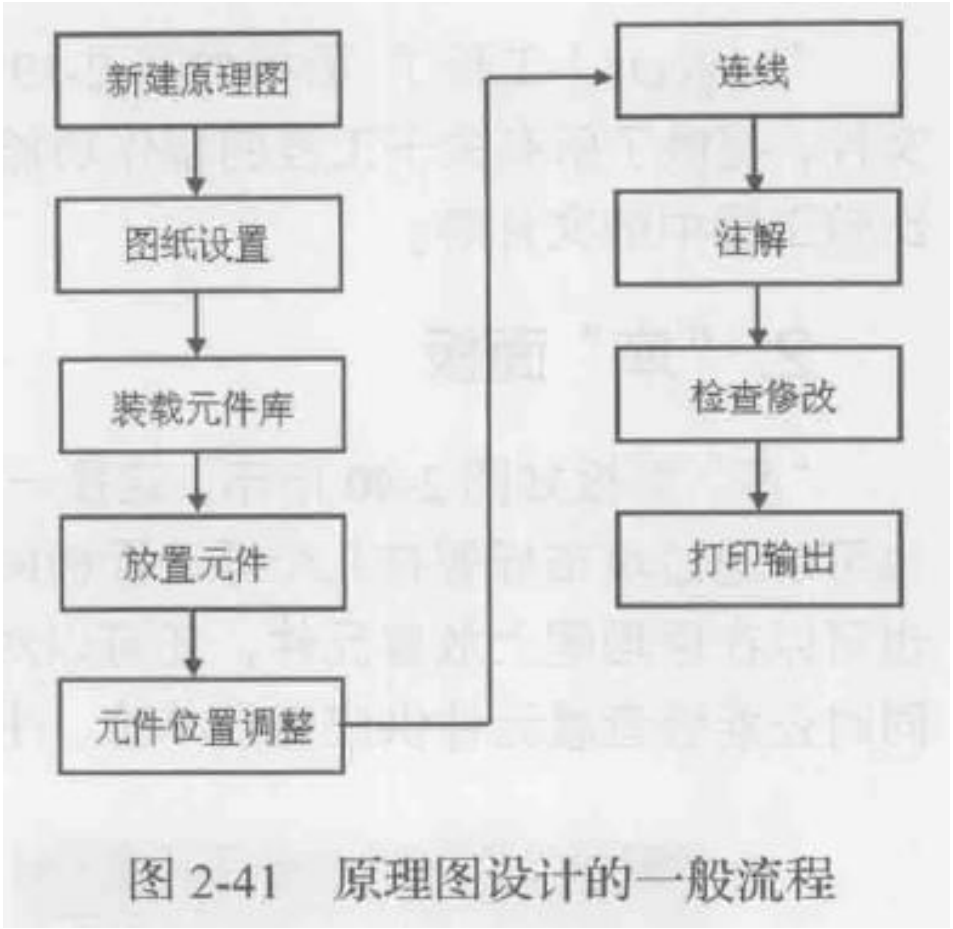


图 2-41　原理图设计的一般流程

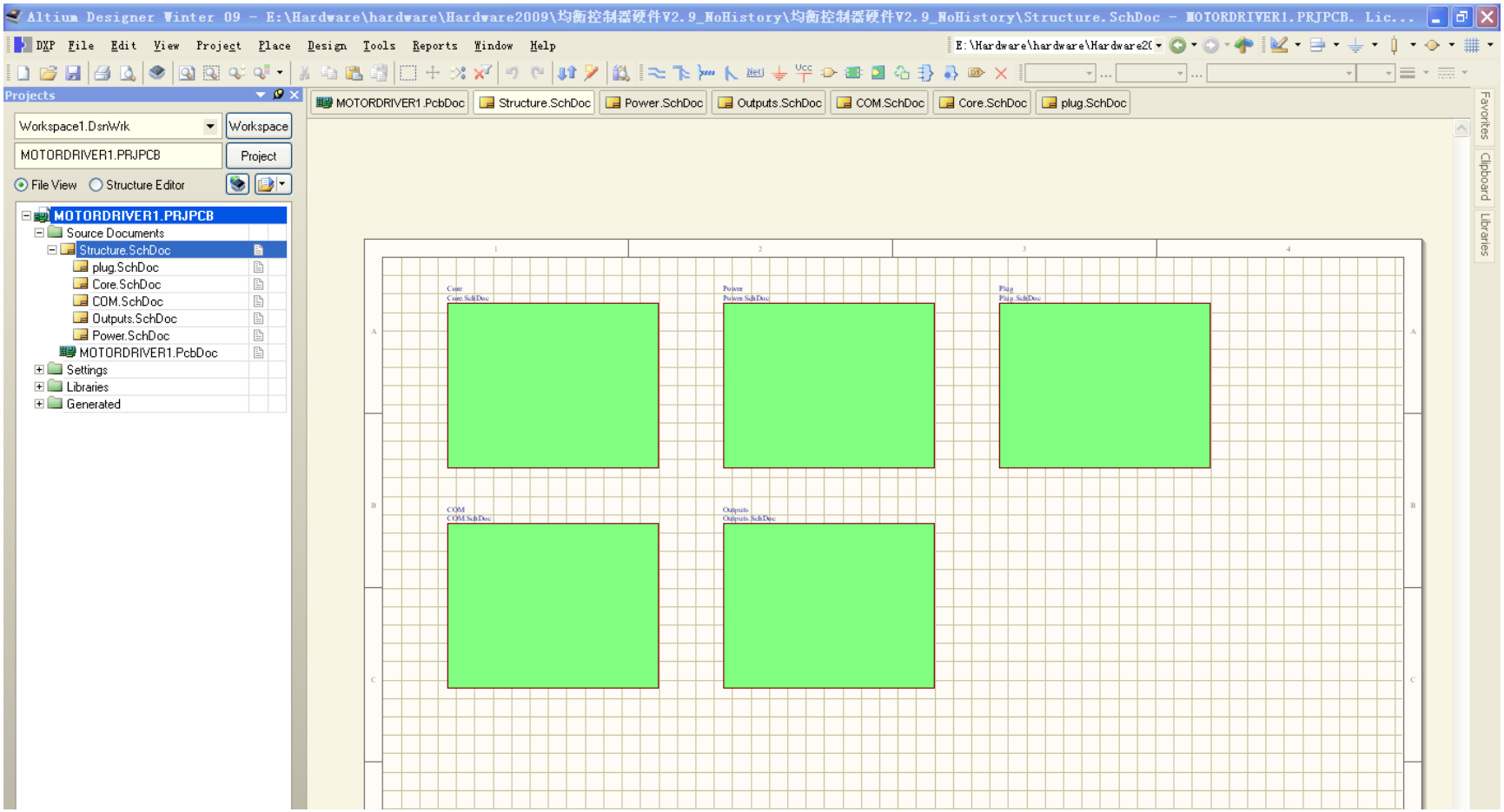


§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

原理图设计的基本步骤/ Basic Process of Schematic Design

1）生成设计文档的基本框架→根据ECU的基本框架相对应来生成；

/Create the basic design document according to the framework of ACEs.

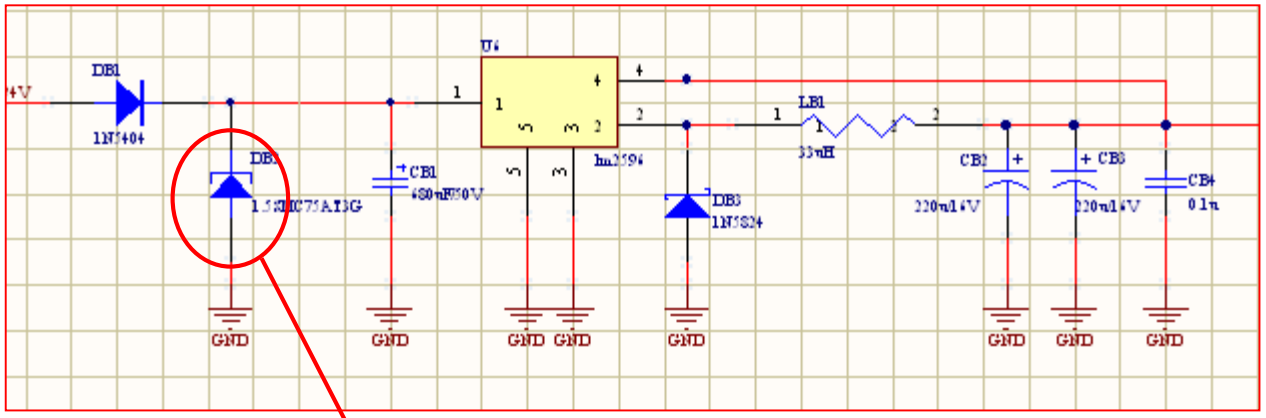




原理图设计的基本步骤/ Basic Process of Schematic Design

2）完成各个模块的原理图细化设计；确定每个器件的编号、封装和参数值；进行布线和编辑网络→功能设计；

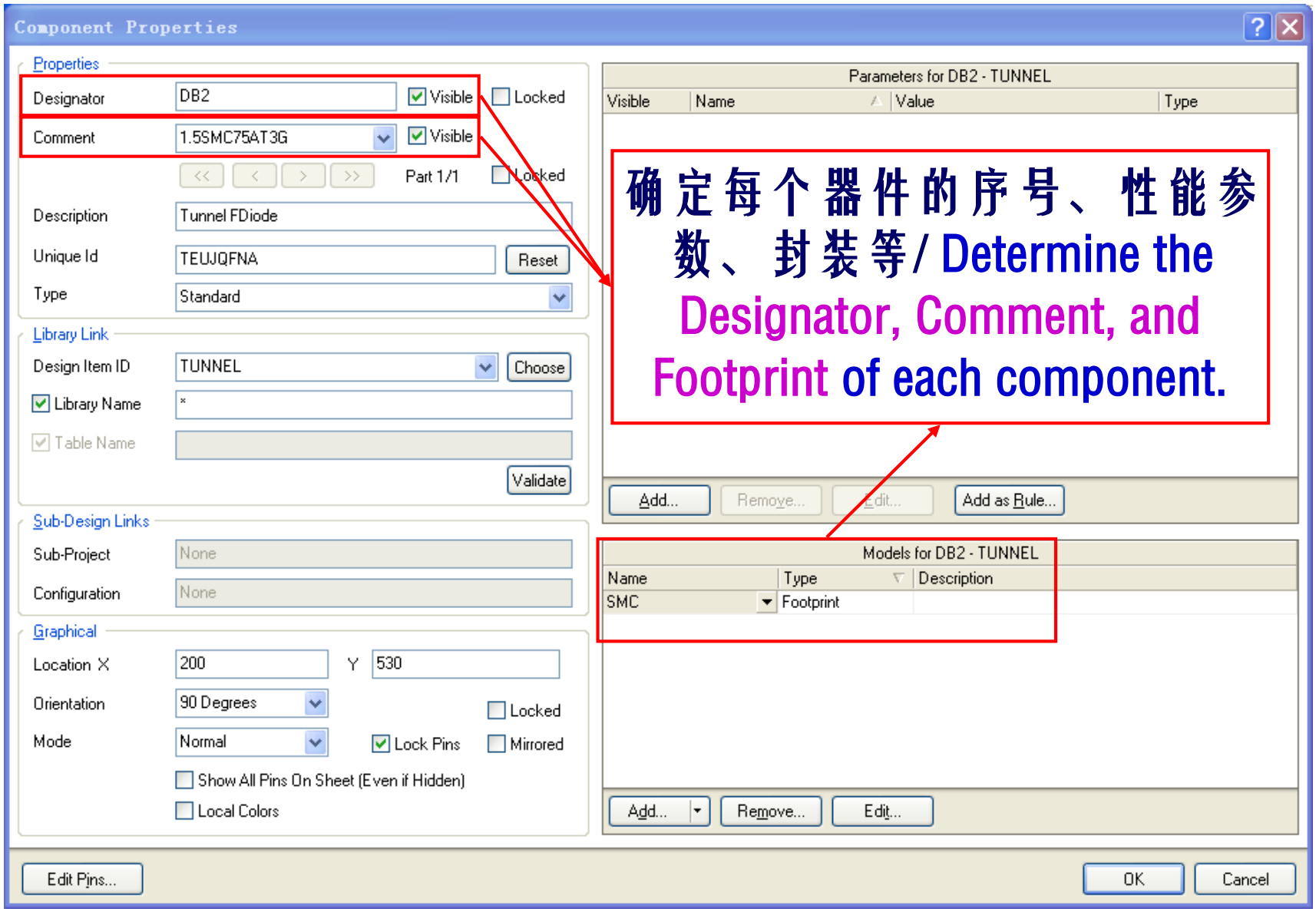
/Detail function design for schematic circuit, including the part number, footprint, parameter of each component and the whole netlist define.



确定每个器件的序号、性能参数、封装等/ Determine the Designator, Comment, and Footprint of each component



§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP



确定每个器件的序号、性能参数、封装等/ Determine the Designator, Comment, and Footprint of each component.



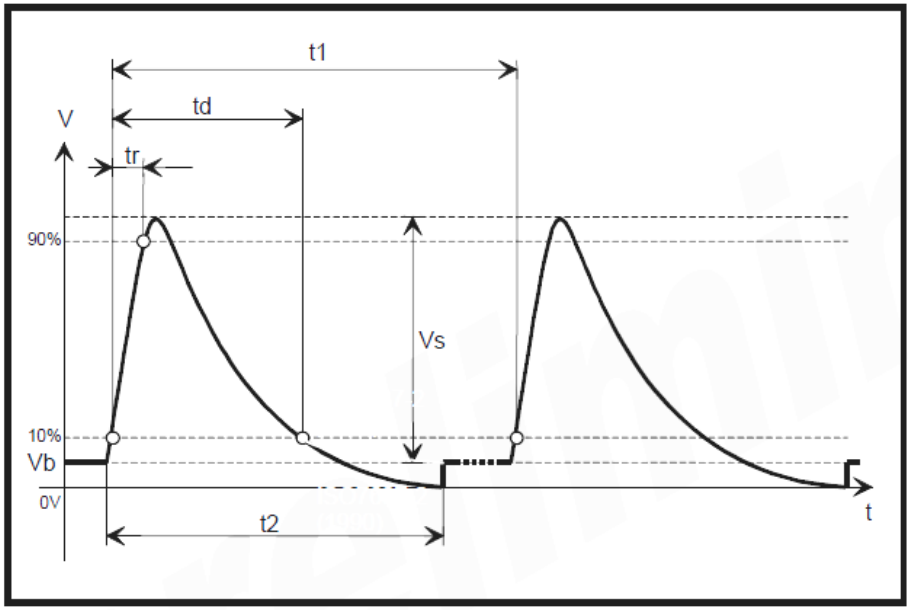
§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

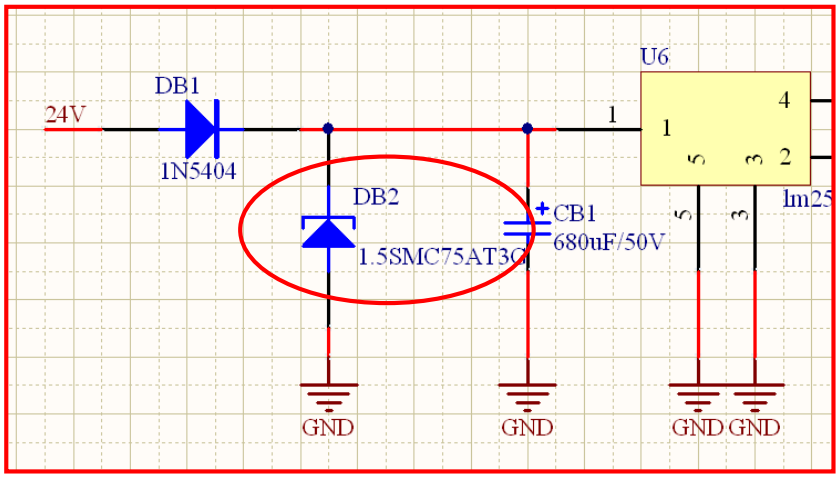
原理图设计的基本步骤/ Basic Process of Schematic Design

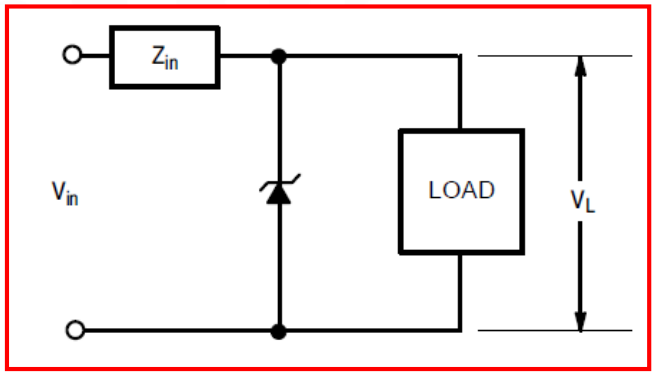
3）对原理图进行电磁兼容性能分析→EMC设计；

/EMC analysis and check on the schematic circuit. Example: TVS design

- Vb = 27V +/–1.0V, Vs = +100V
- Ri = 10Ω
- td = 0.05ms, tr<= 1us
- t1 = 1s, t2 = 200ms
- duration = 5000 impulses





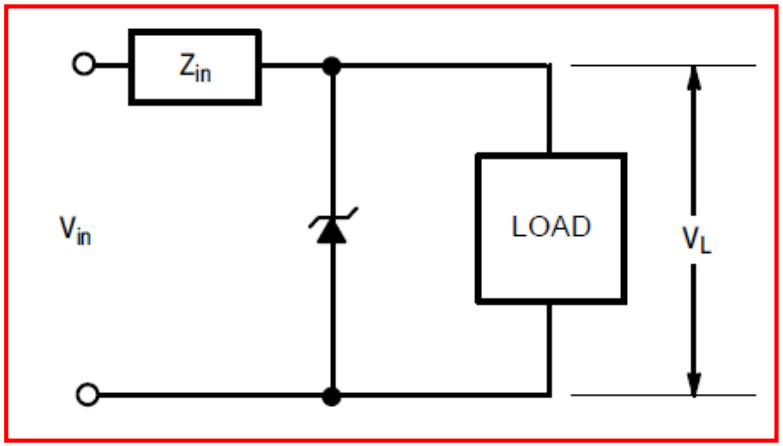


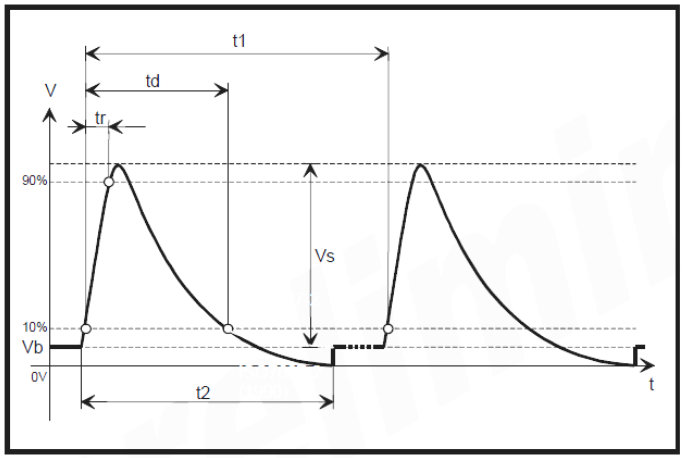
TVS: Transient Voltage Suppressor

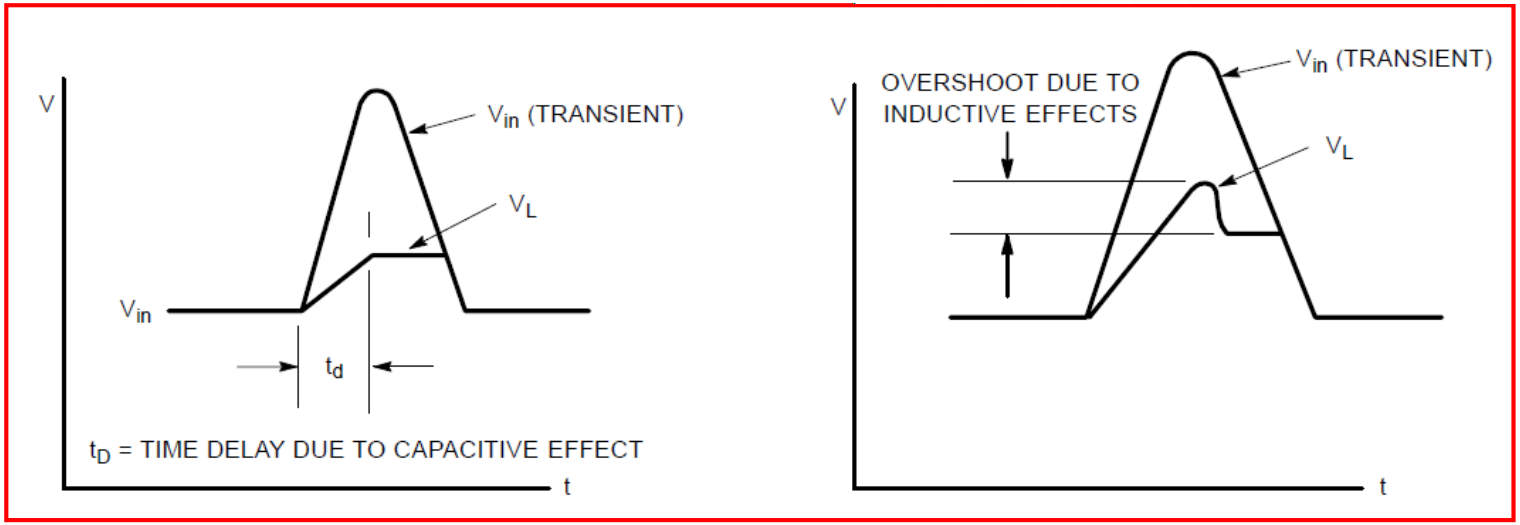
原理图设计的基本步骤/ Basic Process of Schematic Design

3）对原理图进行电磁兼容性能分析→EMC设计；

/EMC analysis and check on the schematic circuit. Example: TVS design







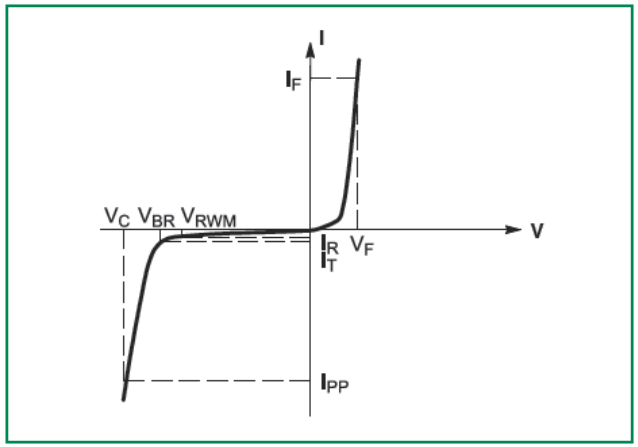


§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

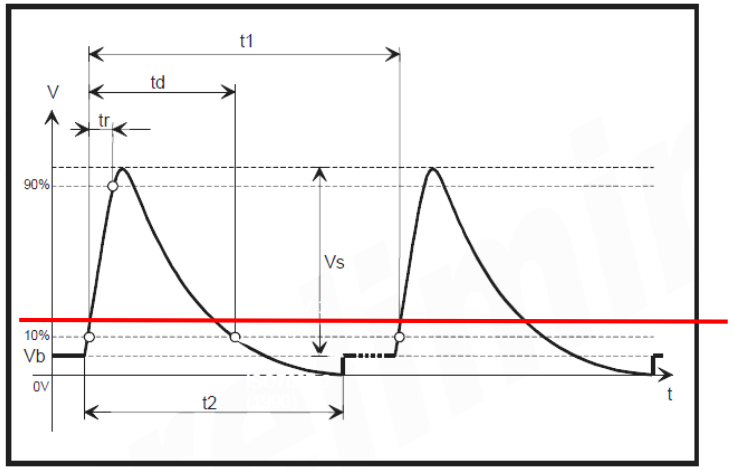
原理图设计的基本步骤/ Basic Process of Schematic Design

3）对原理图进行电磁兼容性能分析→EMC设计；

/EMC analysis and check on the schematic circuit. Example: TVS design



Symbol	Parameter
I_{PP}	Maximum Reverse Peak Pulse Current
V_C	Clamping Voltage @ I_{PP}
V_{RWM}	Working Peak Reverse Voltage
I_R	Maximum Reverse Leakage Current @ V_{RWM}
V_{BR}	Breakdown Voltage @ I_T
I_T	Test Current
I_F	Forward Current
V_F	Forward Voltage @ I_F





§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

原理图设计的基本步骤/ Basic Process of Schematic Design

3）对原理图进行电磁兼容性能分析→EMC设计；

/EMC analysis and check on the schematic circuit. Example: TVS design

Device*	Device Marking	V_{RWM} (Note 6)	I_R @ V_{RWM}	Breakdown Voltage				V_C @ I_{PP} (Note 8)		ΘV_{BR}
				V_{BR} V (Note 7)			@ I_T	V_C	I_{PP}	
		V	μA	Min	Nom	Max	mA	V	A	%/°C
1.5SMC6.8AT3, G	6V8A	5.8	1000	6.45	6.8	7.14	10	10.5	143	0.057
1.5SMC7.5AT3, G	7V5A	6.4	500	7.13	7.5	7.88	10	11.3	132	0.061
1.5SMC8.2AT3, G	8V2A	7.02	200	7.79	8.2	8.61	10	12.1	124	0.065
1.5SMC9.1AT3	9V1A	7.78	50	8.65	9.1	9.55	1	13.4	112	0.068
1.5SMC10AT3	10A	8.55	10	9.5	10	10.5	1	14.5	103	0.073
1.5SMC11AT3	11A	9.4	5	10.5	11	11.6	1	15.6	96	0.075
1.5SMC12AT3, G	12A	10.2	5	11.4	12	12.6	1	16.7	90	0.078
1.5SMC13AT3, G	13A	11.1	5	12.4	13	13.7	1	18.2	82	0.081
1.5SMC15AT3, G	***15A***	***12.8***	***5***	***14.3***	***15***	***15.8***	***1***	***21.2***	***71***	***0.084***
1.5SMC16AT3, G	16A	13.6	5	15.2	16	16.8	1	22.5	67	0.086
1.5SMC18AT3, G	18A	15.3	5	17.1	18	18.9	1	25.2	59.5	0.088
1.5SMC20AT3, G	20A	17.1	5	19	20	21	1	27.7	54	0.09
1.5SMC22AT3, G	22A	18.8	5	20.9	22	23.1	1	30.6	49	0.092
1.5SMC24AT3, G	***24A***	***20.5***	***5***	***22.8***	***24***	***25.2***	***1***	***33.2***	***45***	***0.094***
1.5SMC27AT3, G	27A	23.1	5	25.7	27	28.4	1	37.5	40	0.096
1.5SMC30AT3, G	30A	25.6	5	28.5	30	31.5	1	41.4	36	0.097
1.5SMC33AT3, G	***33A***	***28.2***	***5***	***31.4***	***33***	***34.7***	***1***	***45.7***	***33***	***0.098***
1.5SMC36AT3, G	***36A***	***30.8***	***5***	***34.2***	***36***	***37.8***	***1***	***49.9***	***30***	***0.099***
1.5SMC39AT3, G	***39A***	***33.3***	***5***	***37.1***	***39***	***41***	***1***	***53.9***	***28***	***0.1***
1.5SMC43AT3, G	***43A***	***36.8***	***5***	***40.9***	***43***	***45.2***	***1***	***59.3***	***25.3***	***0.101***
1.5SMC47AT3, G	***47A***	***40.2***	***5***	***44.7***	***47***	***49.4***	***1***	***64.8***	***23.2***	***0.101***
1.5SMC51AT3, G	51A	43.6	5	48.5	51	53.6	1	70.1	21.4	0.102
1.5SMC56AT3, G	56A	47.8	5	53.2	56	58.8	1	77	19.5	0.103
1.5SMC62AT3, G	62A	53	5	58.9	62	65.1	1	85	17.7	0.104
1.5SMC68AT3, G	68A	58.1	5	64.6	68	71.4	1	92	16.3	0.104
1.5SMC75AT3, G	***75A***	***64.1***	***5***	***71.3***	***75***	***78.8***	***1***	***103***	***14.6***	***0.105***
1.5SMC82AT3, G	82A	70.1	5	77.9	82	86.1	1	113	13.3	0.105



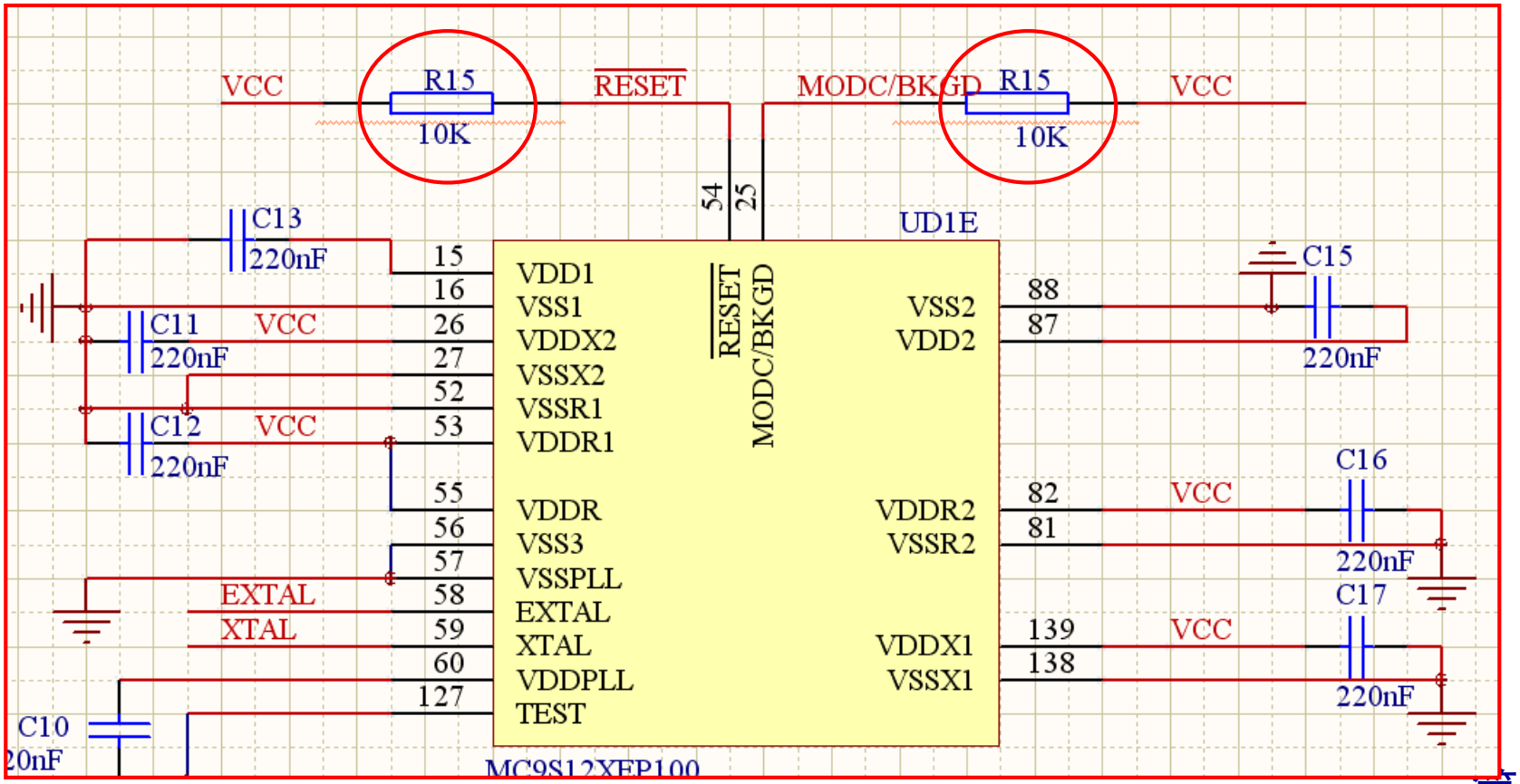
§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

原理图设计的基本步骤/ Basic Process of Schematic Design

4）对整个系统的原理图文档进行ERC校验；

/Implement the Electric Rule Check on the schematic circuit.

R15 same designator! Error is indicated with small red wave.





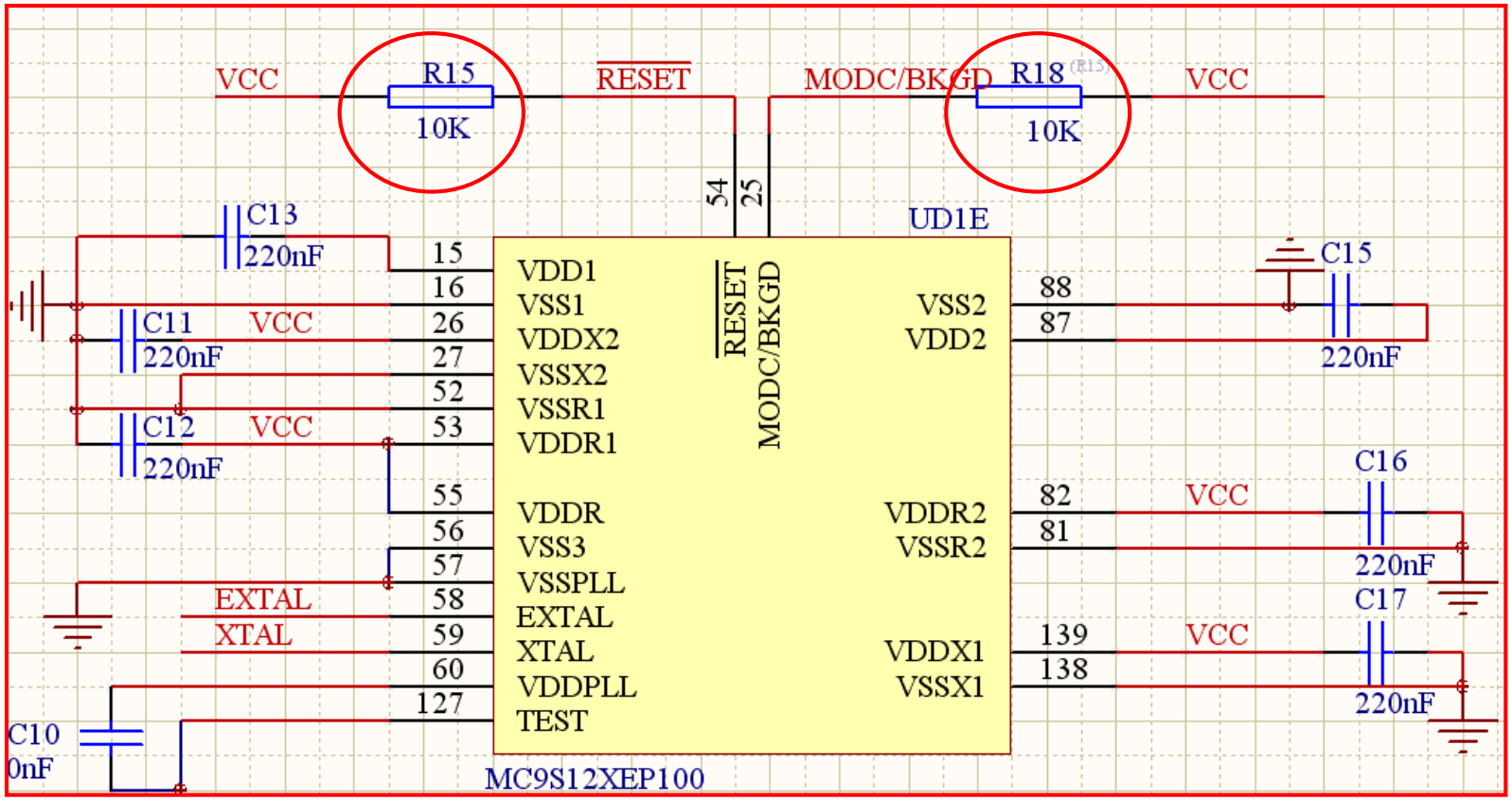
§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

原理图设计的基本步骤/ Basic Process of Schematic Design

4）对整个系统的原理图文档进行ERC校验；

/Implement the Electric Rule Check on the schematic circuit.

R15 same designator! With change one designator, error is removed.



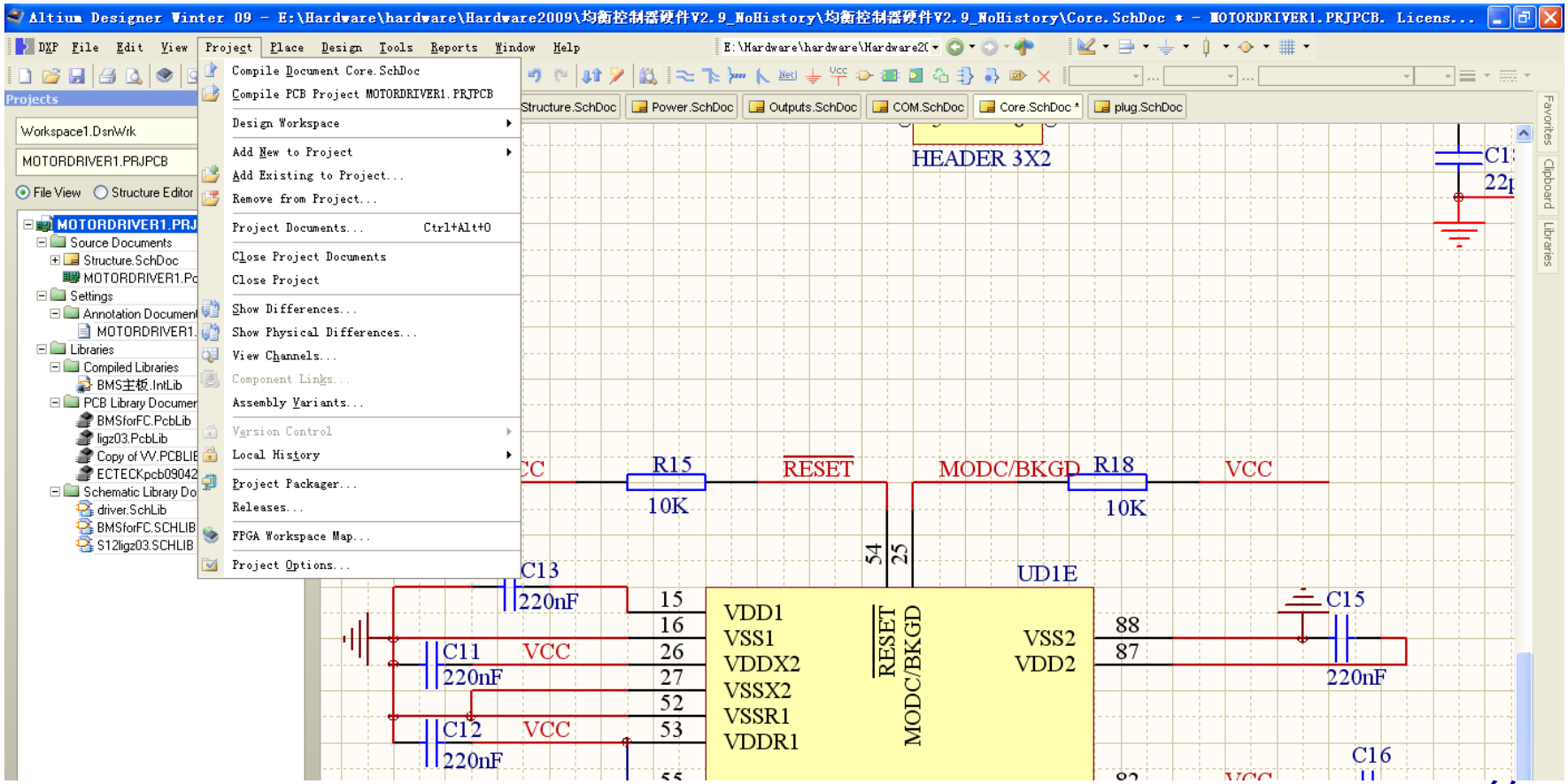


§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

原理图设计的基本步骤/ Basic Process of Schematic Design

5）生成网络表和器件清单→为PCB设计和生产准备；

/ Generate the Netlist and Boom List , prepare for PCB design and purchase components. Compile Document

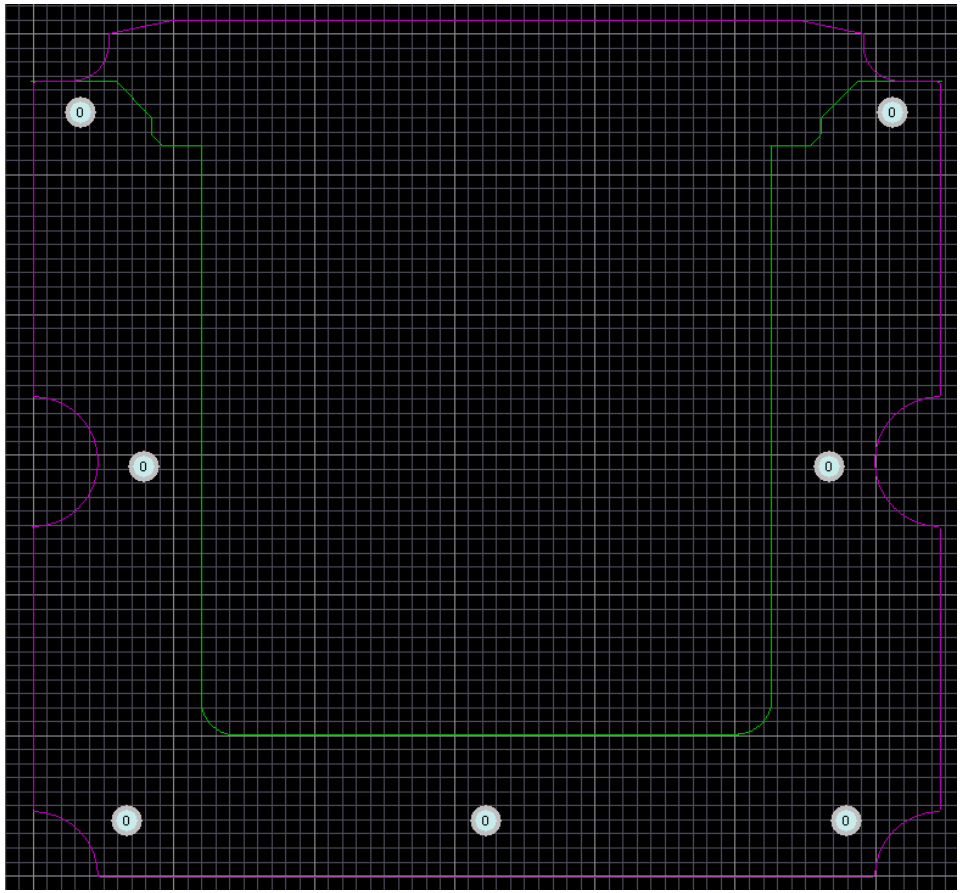




§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

●布线图设计的基本步骤

1）根据实际的应用要求确定PCB板子的尺寸和基本工艺：复杂程度；散热；成本；工艺；接插件 /Define PCB dimensions according to installation, cost, welding process, plug type and so on.



通过Keepout layer定义电路板的边界和尺寸，通过Mechanical layer确定特殊区域(下表面需要散热）。首先把电路板的安装孔位置定好。

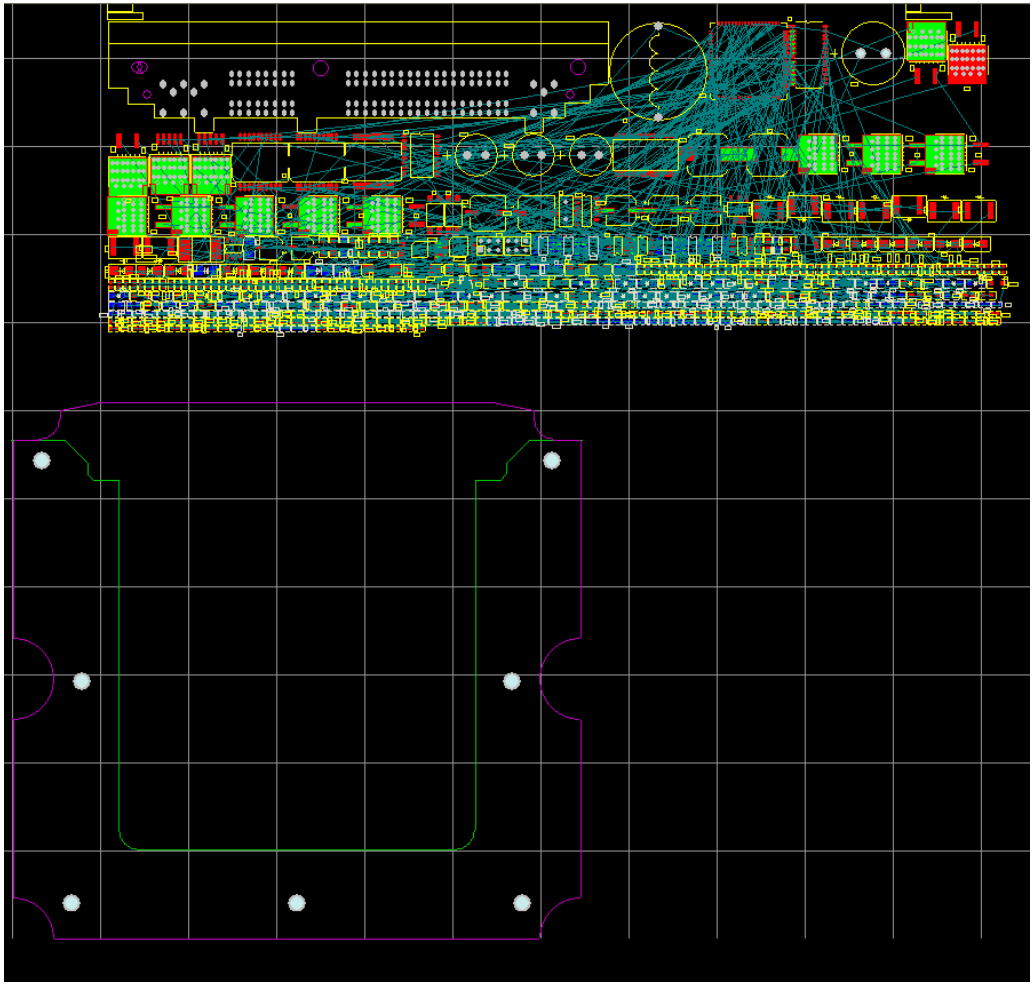
/First define the size and dimension of the PCB in keepout layer, the heat dissipation area needs define in mechanical layer, then specify the installation hole.



§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

●布线图设计的基本步骤

2）根据原理图的网络表，调入器件，检查封装/ Download netlist which generated by schematic design, check the footprint of each component.



调入器件之后，器件自动位于电路板的外侧，便于布置器件操作。同时可以显示出网络连接。

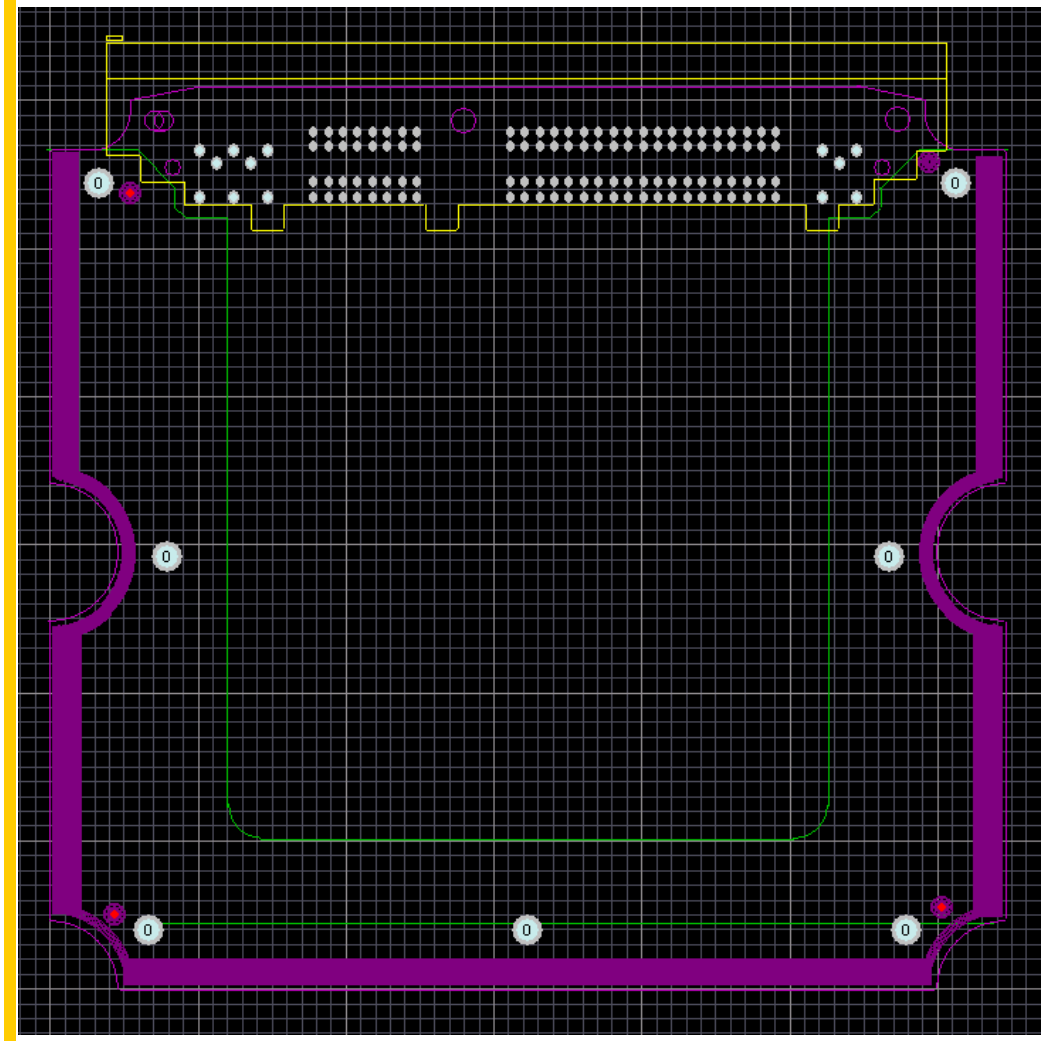
/After import the netlist, the valid component will display in the outside area of keepout layer. The net connect will display the relationship of the chips.



§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

●布线图设计的基本步骤

3）布局：决定器件如何摆放/ Component placement–Outline architecture



首先确定PCB对外接口，包括接插件、和机壳接触区域的范围，生产是电路版的定位标志等。

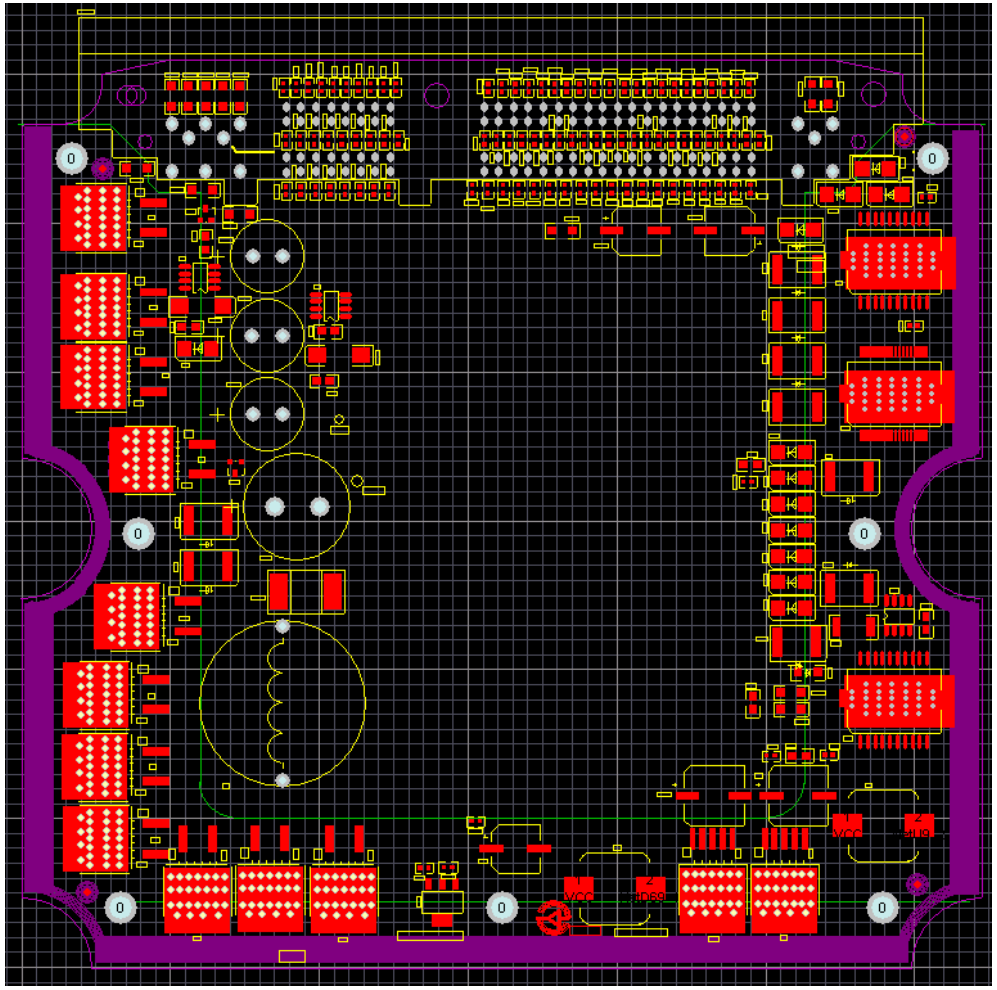
/The Installation interface of PCB needs definition firstly. This includes the plug position, contact area with shell, production alignment mark and so on.



§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

●布线图设计的基本步骤

3）布局：决定器件如何摆放/ Component placement–Outline architecture



然后根据散热和电磁兼容等要求，首先布置好功率驱动部分，与PCB散热区域基本对应，接插件下的退藕电容和初步布置。

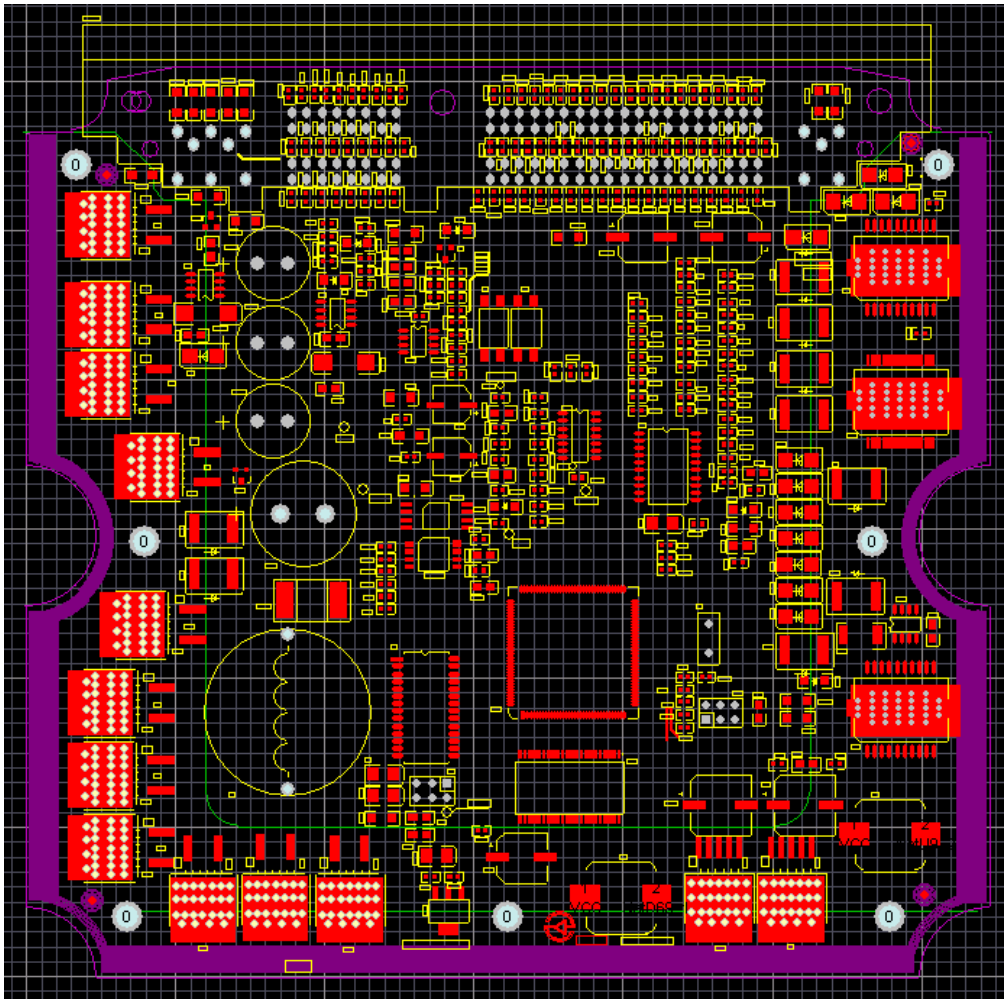
/The components of drive circuit are necessary to be placed at first. They are placed in the area of good dissipation capability. The de–noise capacitors need also place at first.



§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

●布线图设计的基本步骤

3）布局：决定器件如何摆放/ Component placement–Outline architecture



最后布置弱电（逻辑和模拟信号）部分，其中模拟信号优先于数字信号的布局。

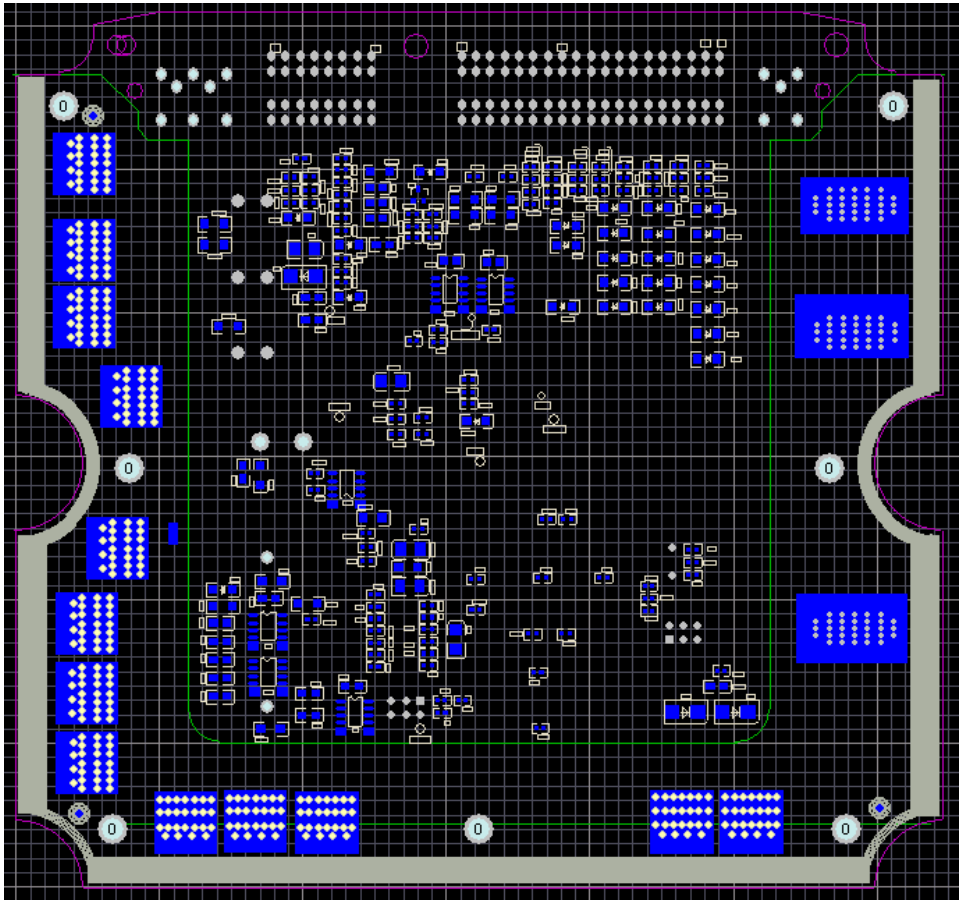
/Last place the component related with logic and analogue circuit. The analogue signal priors then the digital signal.



§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

●布线图设计的基本步骤

3）布局：决定器件如何摆放/ Component placement–Outline architecture



背面的布局，器件放置均在散热贴合面之外的区域。检查散热器件的下表面与机壳接触面积是否足够。

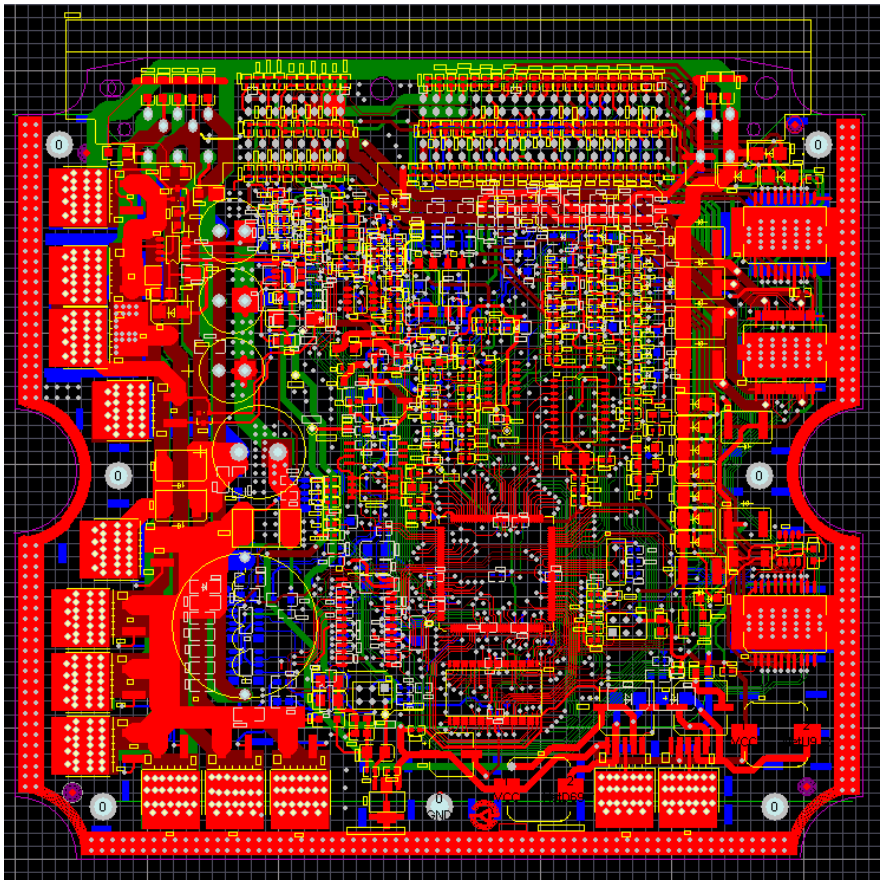
/On the bottom layer， all components are placed outside the heat dissipation area. Check whether the bottom area of the power components is sufficient or not to contact with shell.



§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

●布线图设计的基本步骤

4）布线：制定规则（DRC）；例如最小的线宽、器件的间距、过孔的大小以及电磁兼容性处理等/ Define the Rules for PCB Routing Check: including clearance, via size, and EMC problems.



布线时先走功率大电流线，保证线宽和间距，然后再布小电流细线，功率线和逻辑线尽量不要近距离平行走线。

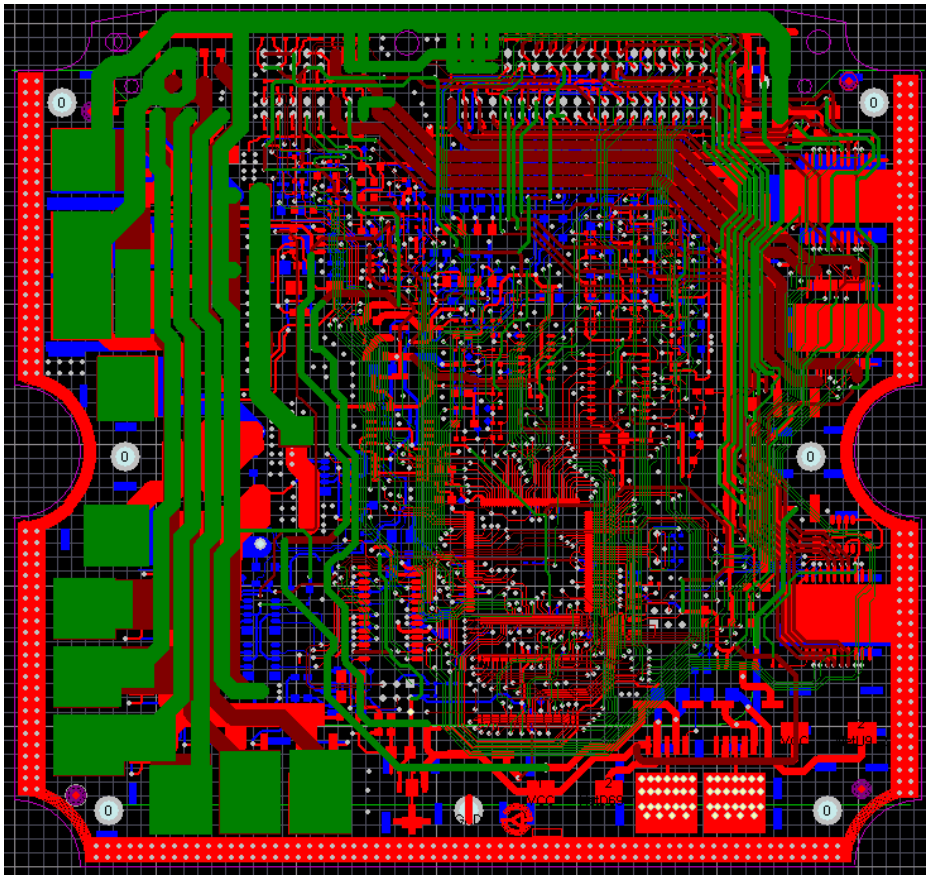
/First route the large current track, the width and clearance of this track needs ensure before route the weak signal's track. It is recommend that do not parallel the logic and power tracks in long distance.

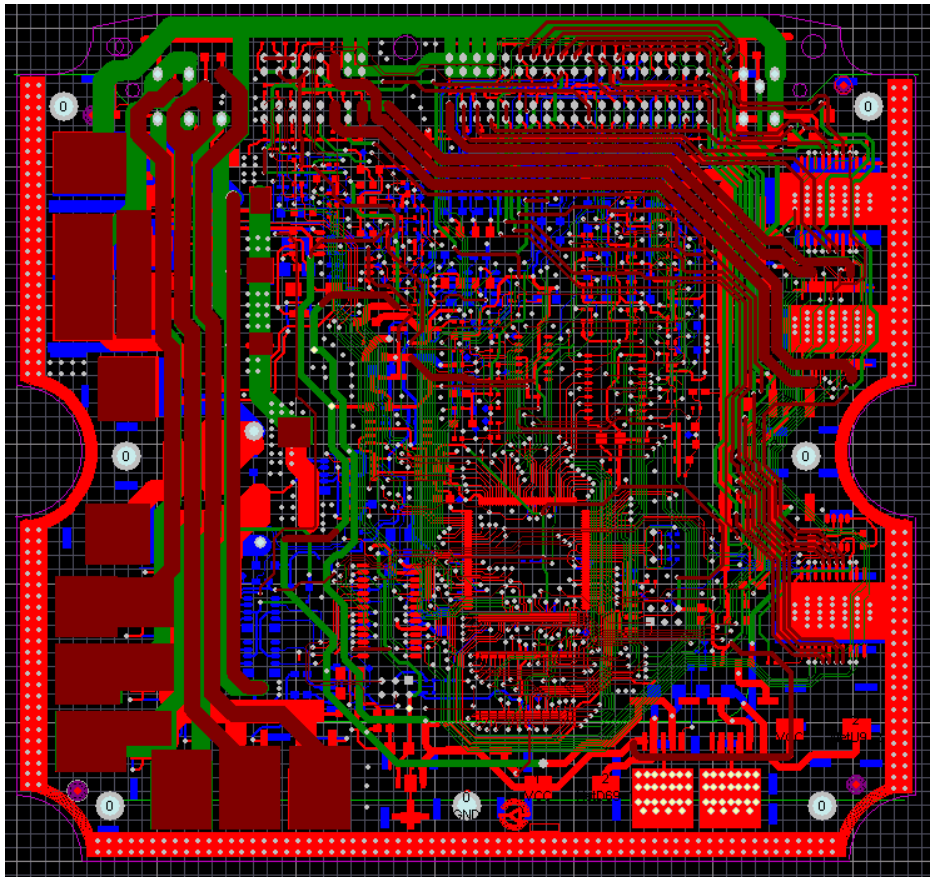


§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

- 布线图设计的基本步骤

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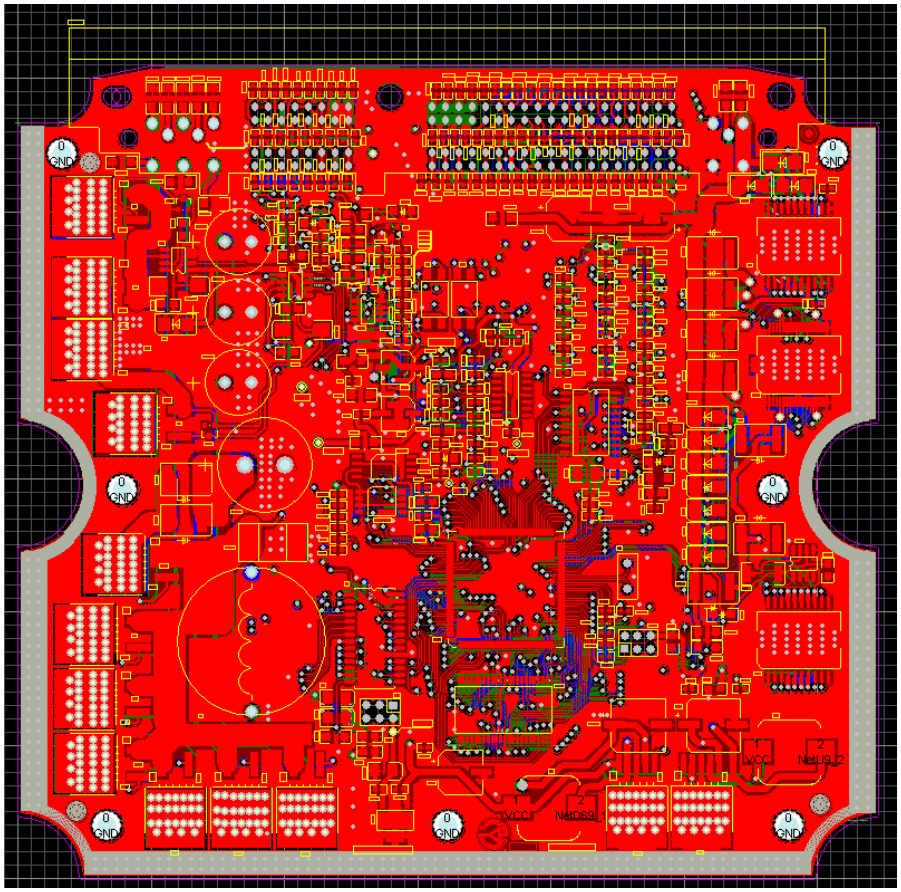


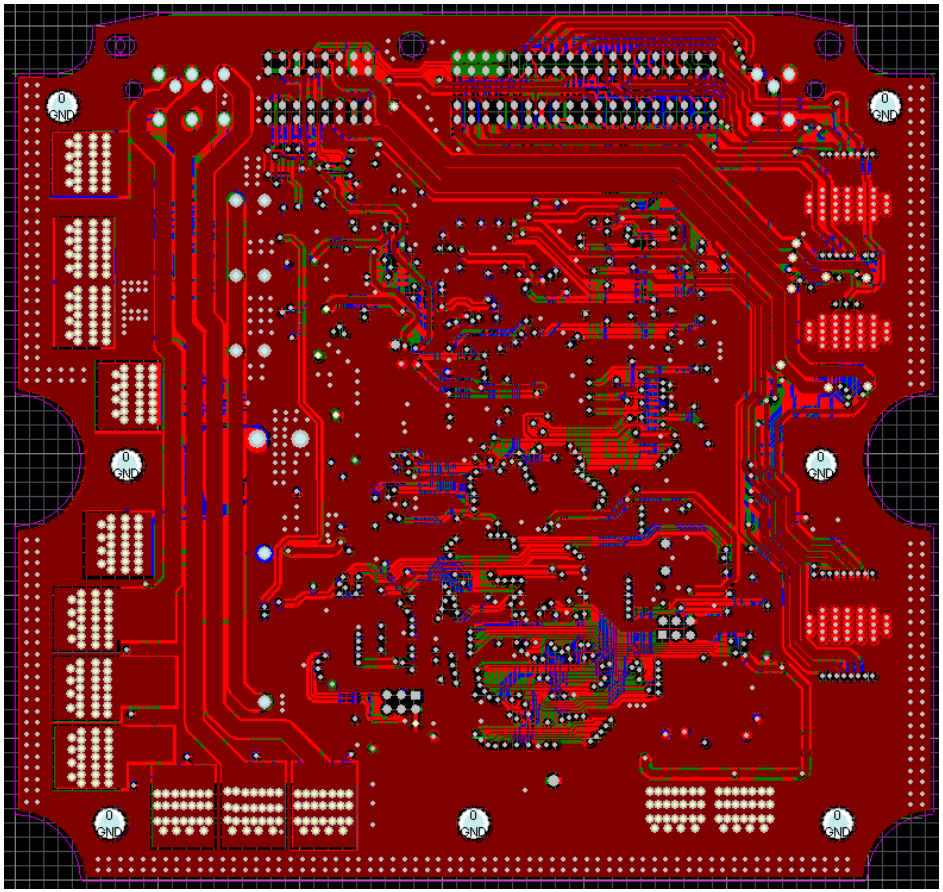


§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

- **布线图设计的基本步骤**

  4）布线：制定规则（DRC）；例如最小的线宽、器件的间距、过孔的大小以及电磁兼容性处理等/ Define the Rules for PCB Routing Check: including clearance, via size, and EMC problems.



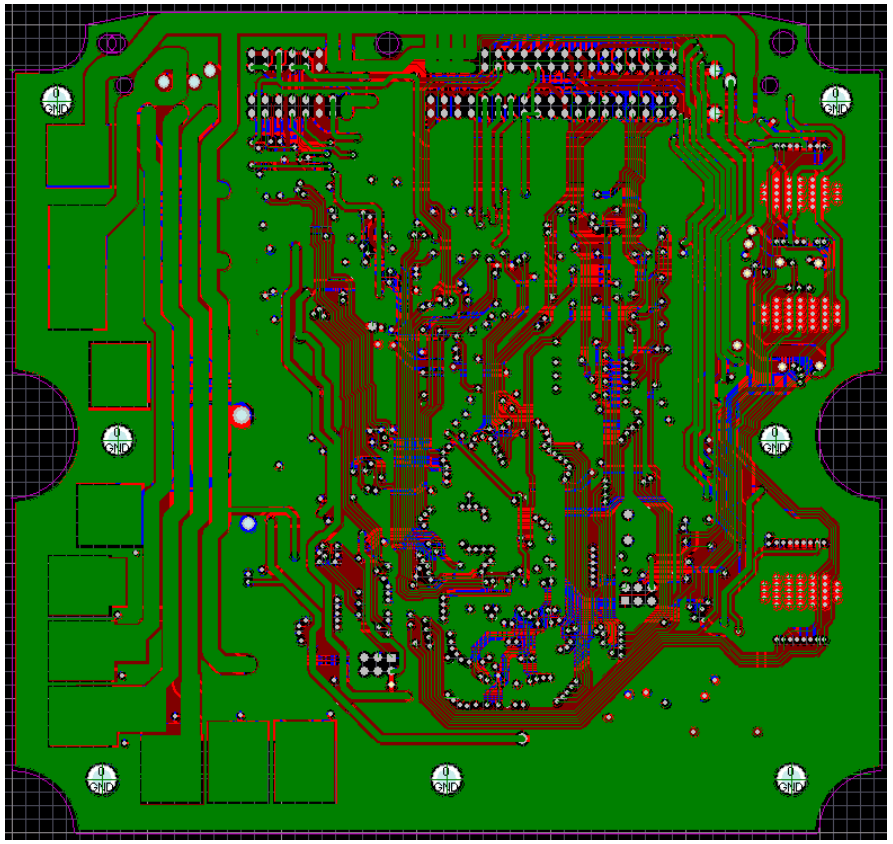


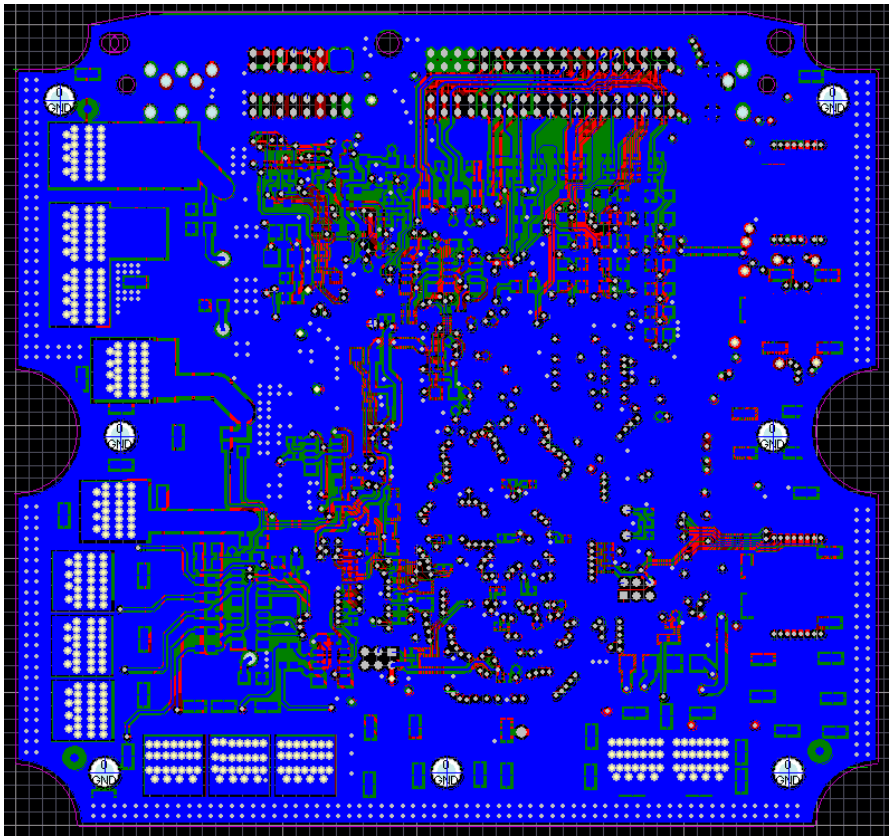


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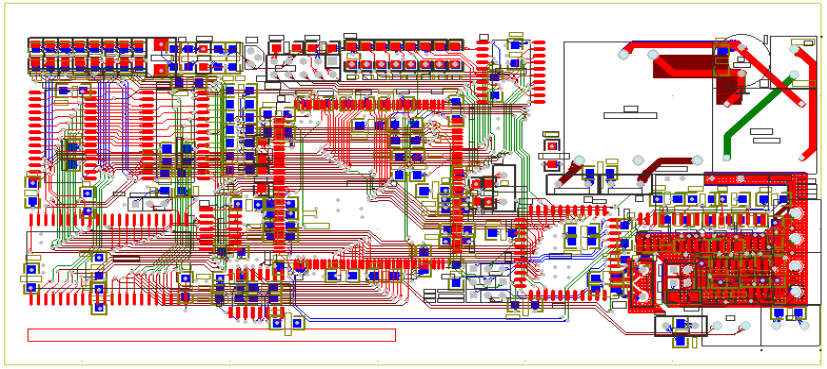


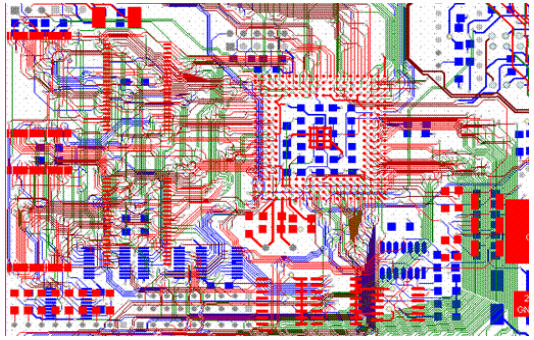
§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP

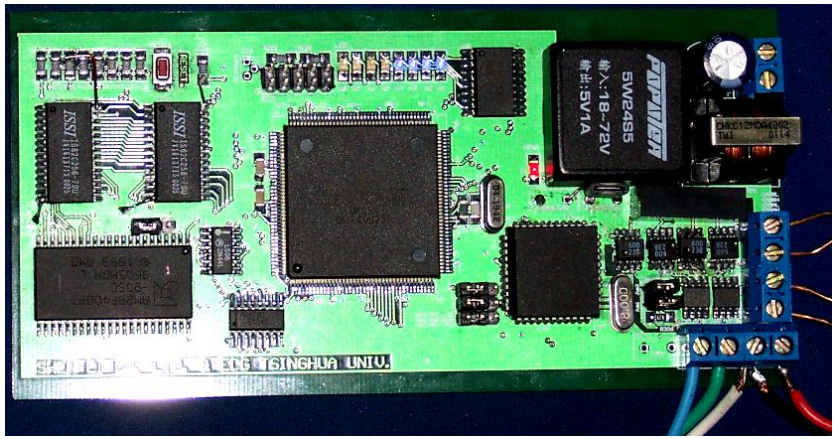
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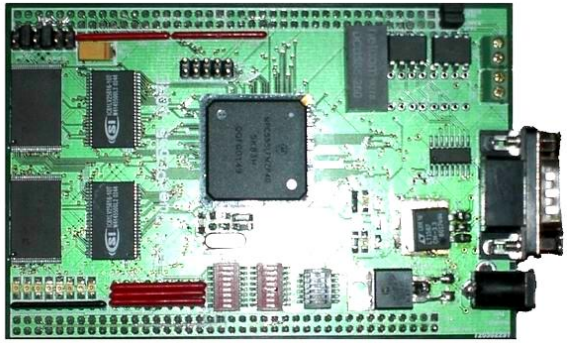
5）进行最后网络校对和DRC校验/Compare the netlist and DRC verify.

6）光绘制版/ Photo draw and PCB manufacture.









DXP Altium 2009演示

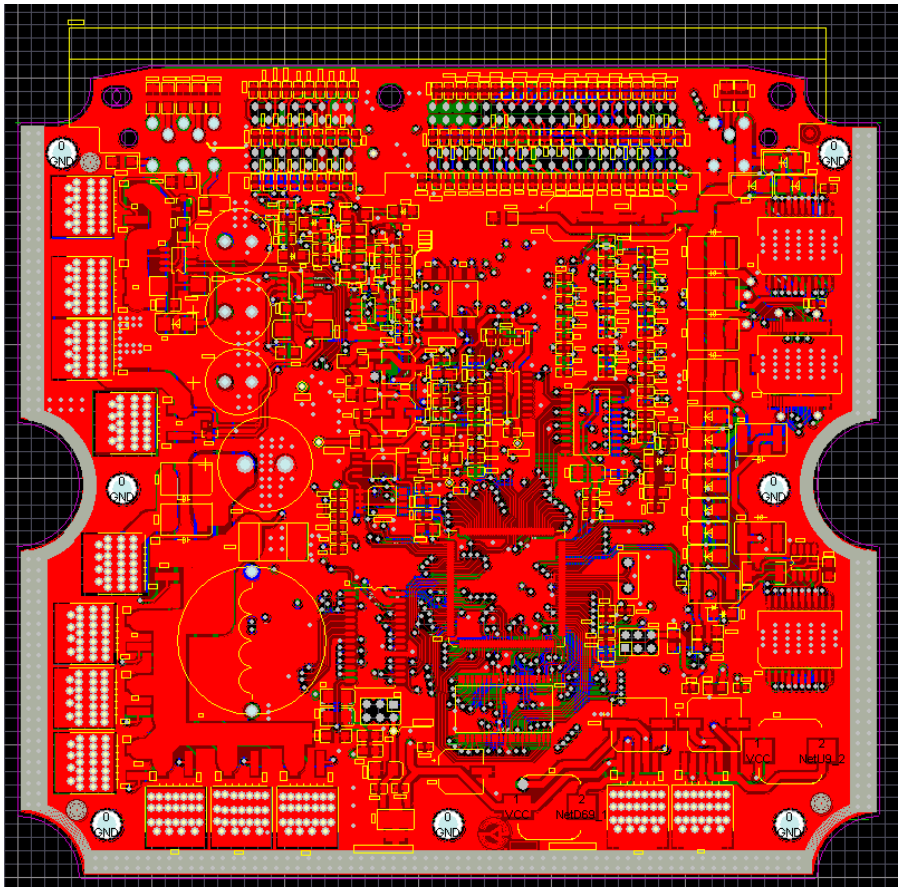


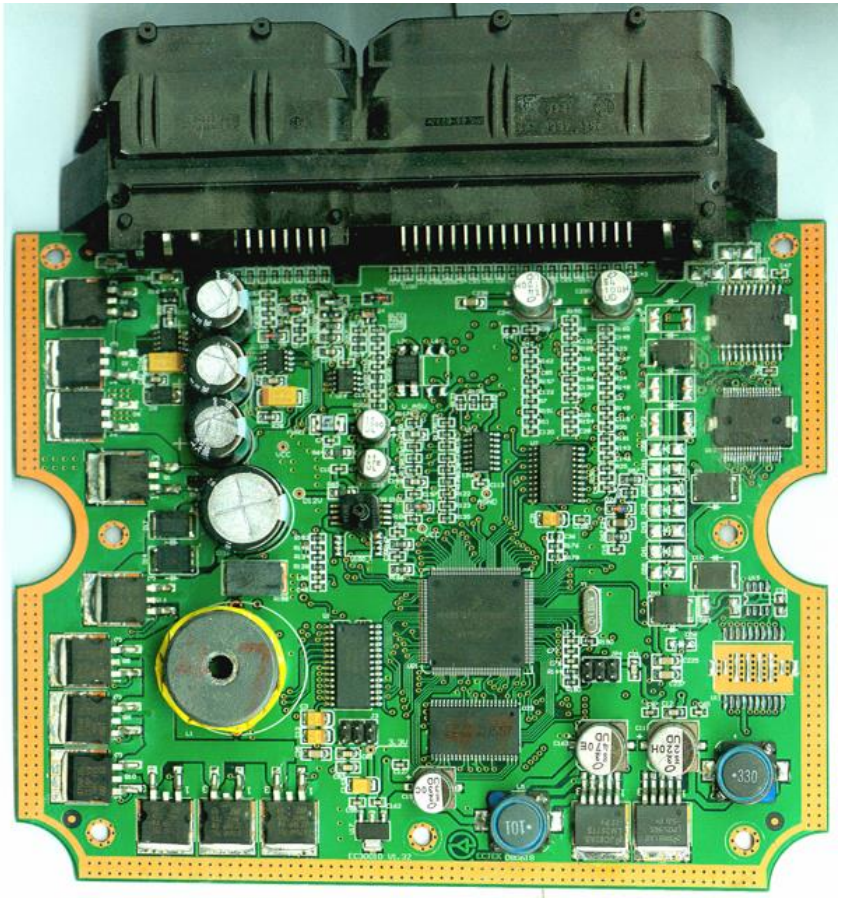
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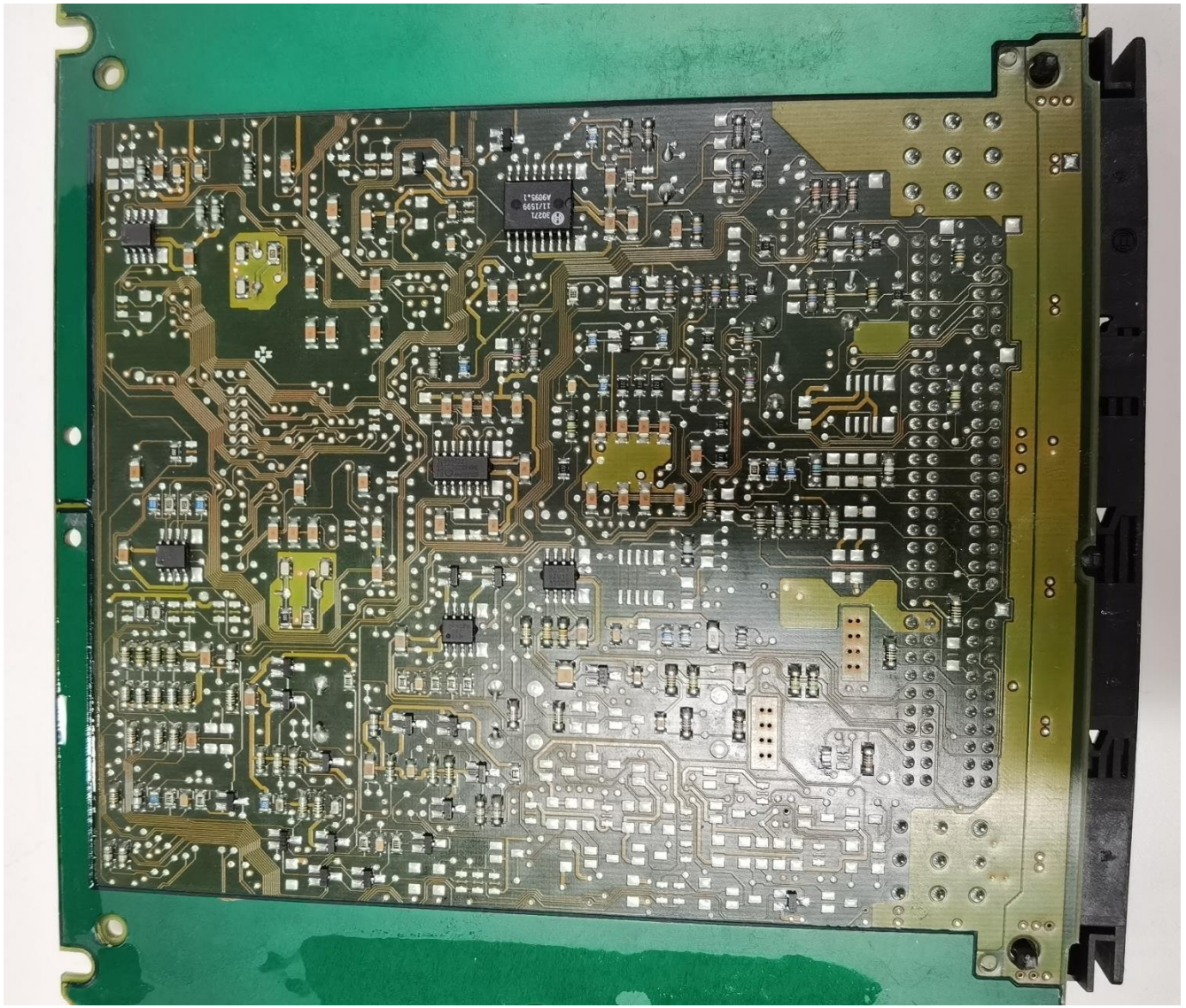
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§4.2 硬件设计工具PROTEL DXP的使用 / Hardware design tools: PROTEL DXP





柴油机共轨喷射系统ECU/
Diesel Common Rail Fuel Injection System

BMW轿车

柴油机

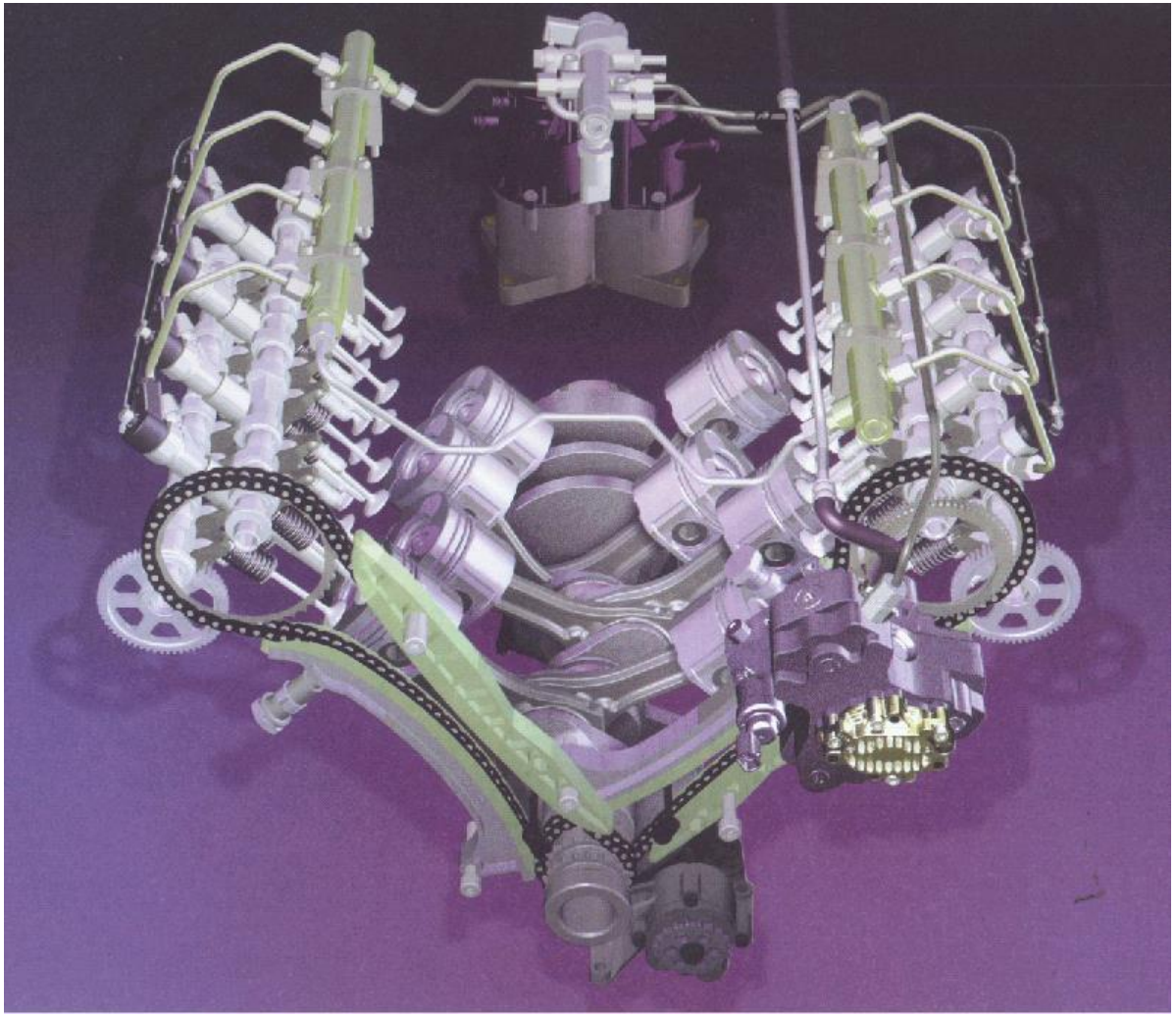




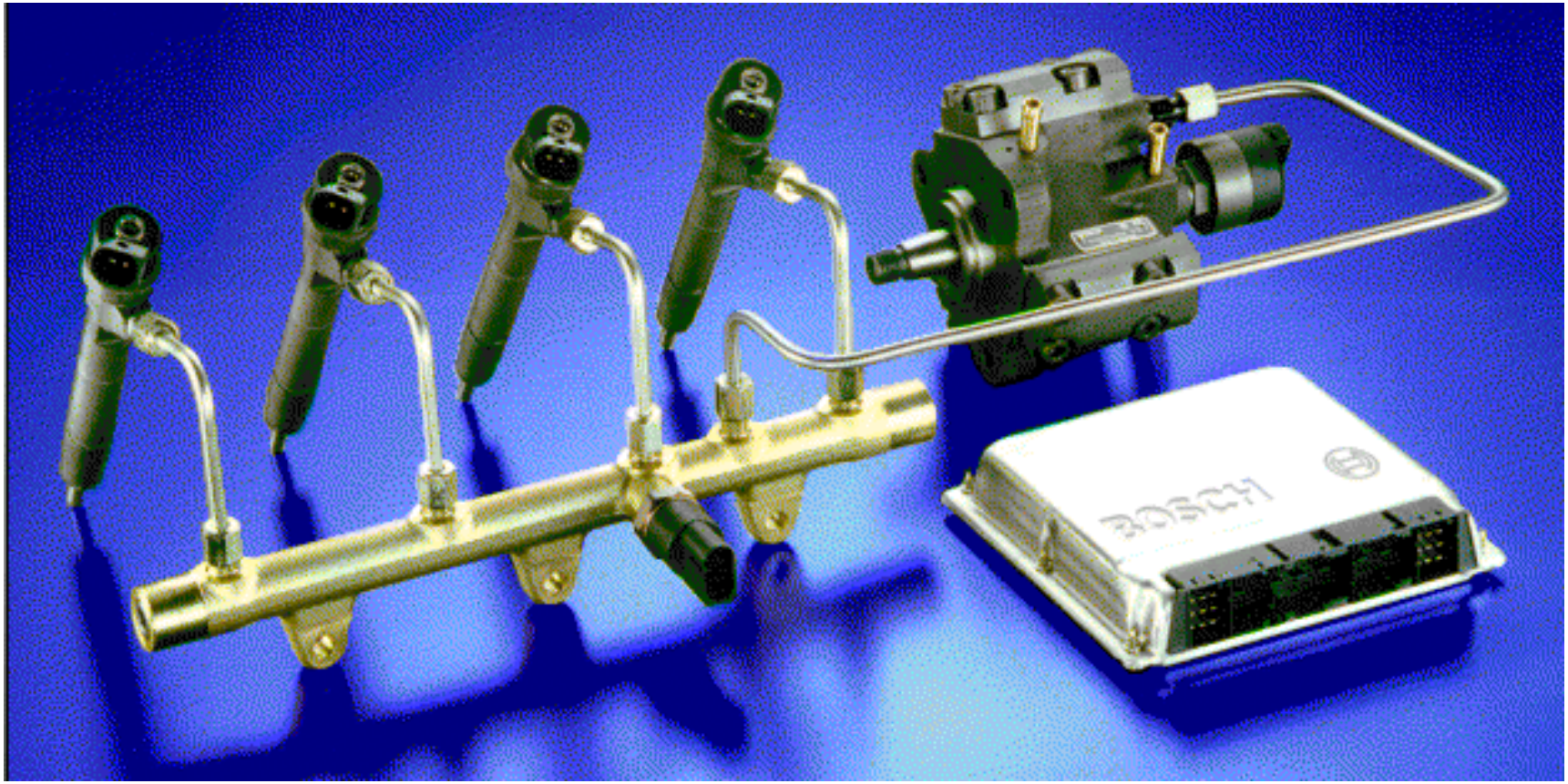
柴油机共轨喷射系统ECU/
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BMW轿车

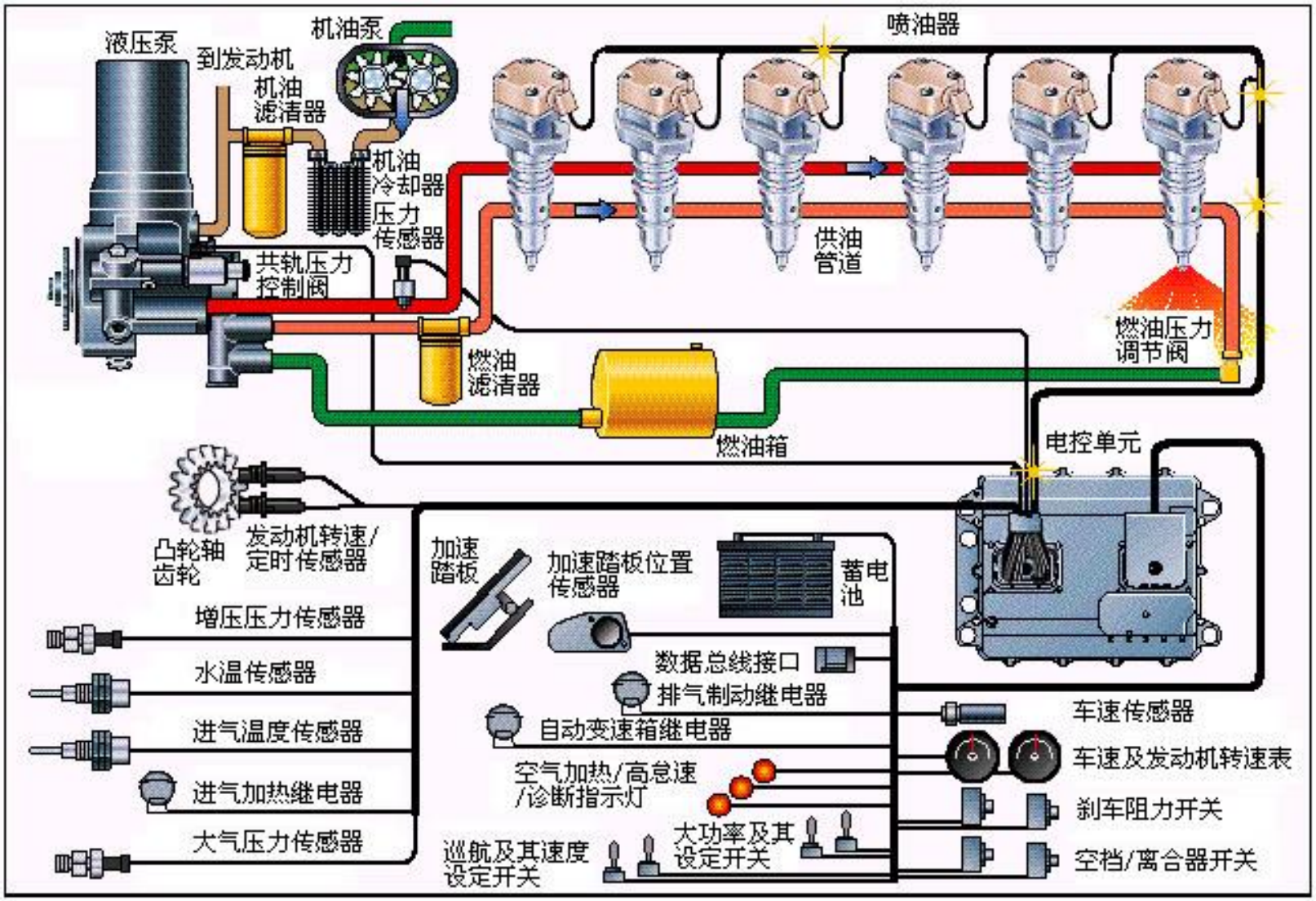
柴油机



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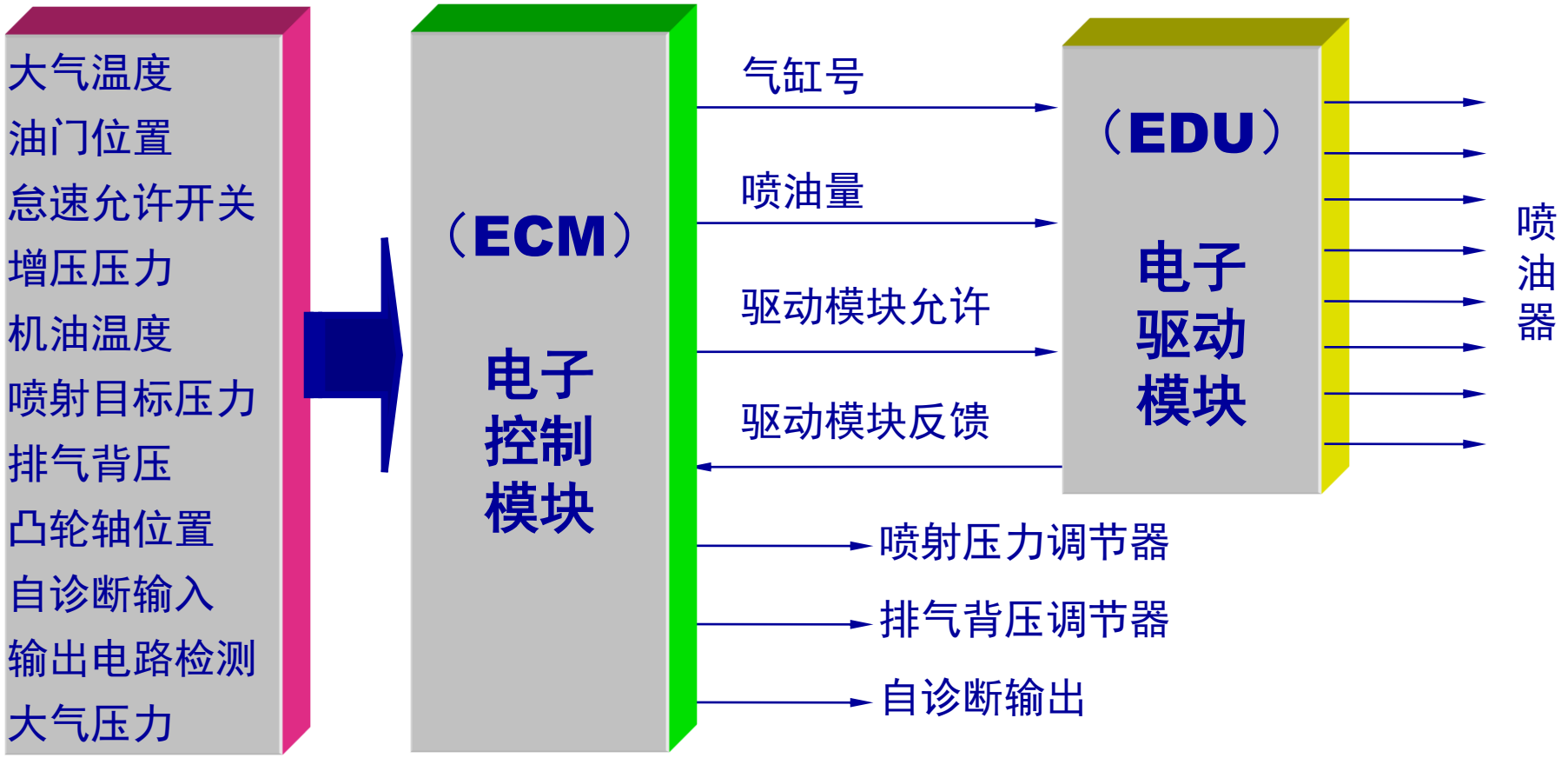
柴油机共轨喷射系统ECU/
Diesel Common Rail Fuel Injection System



液力活塞增压式共轨系统(Hydraulic Electronic Unit Injector, HEUI)

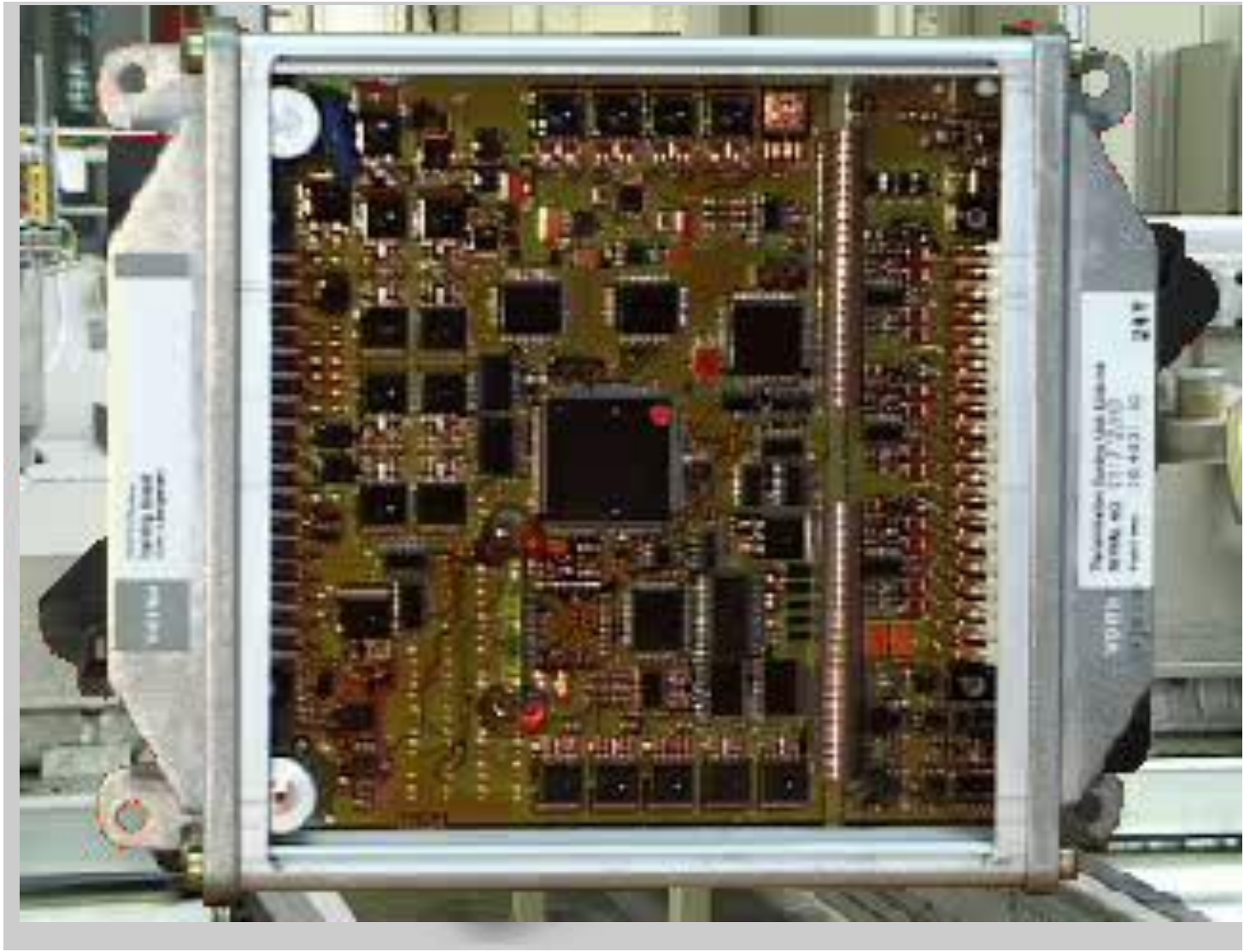


柴油机共轨喷射系统ECU/
Diesel Common Rail Fuel Injection System



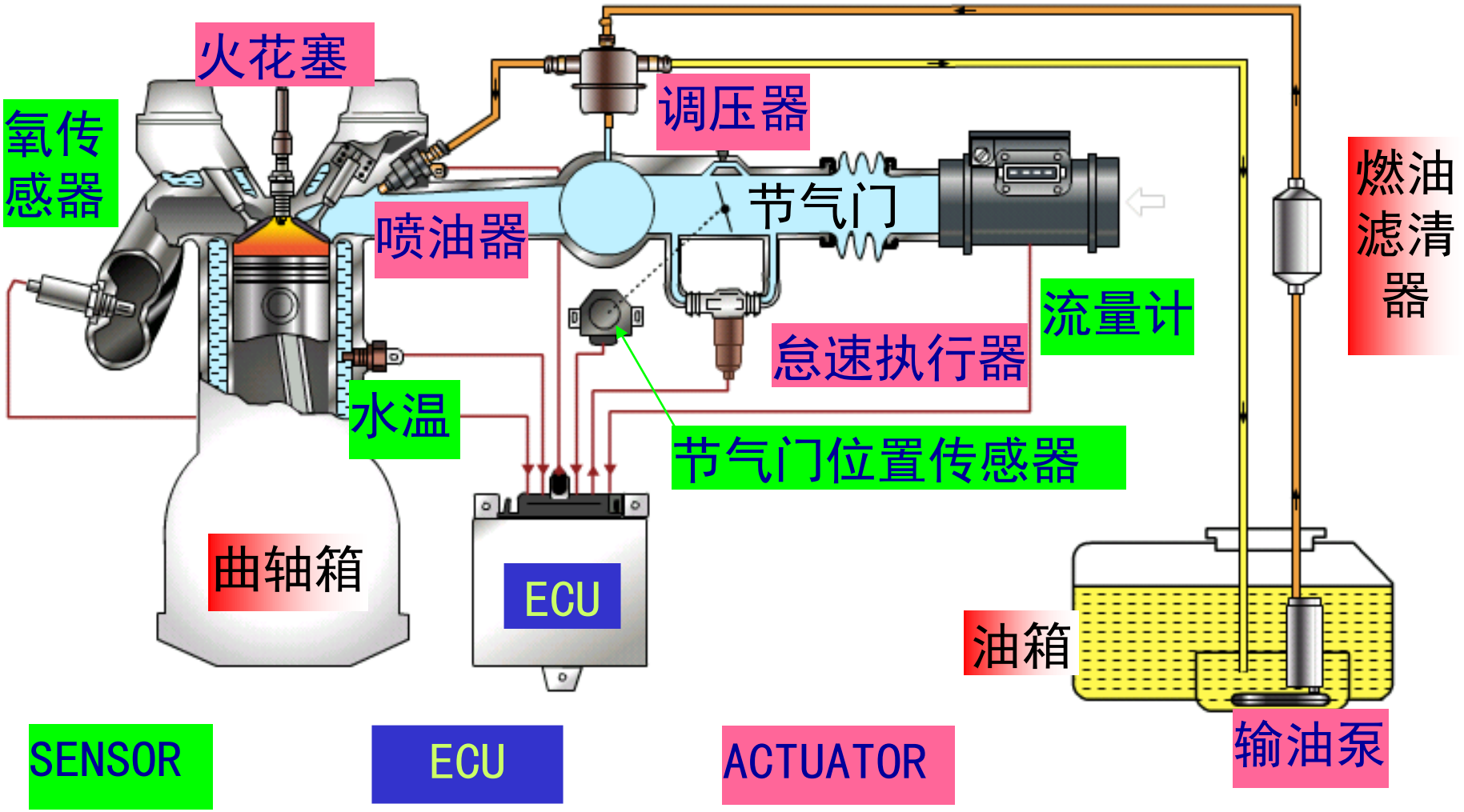


自动变速箱的ECU
Gearbox Control Unit



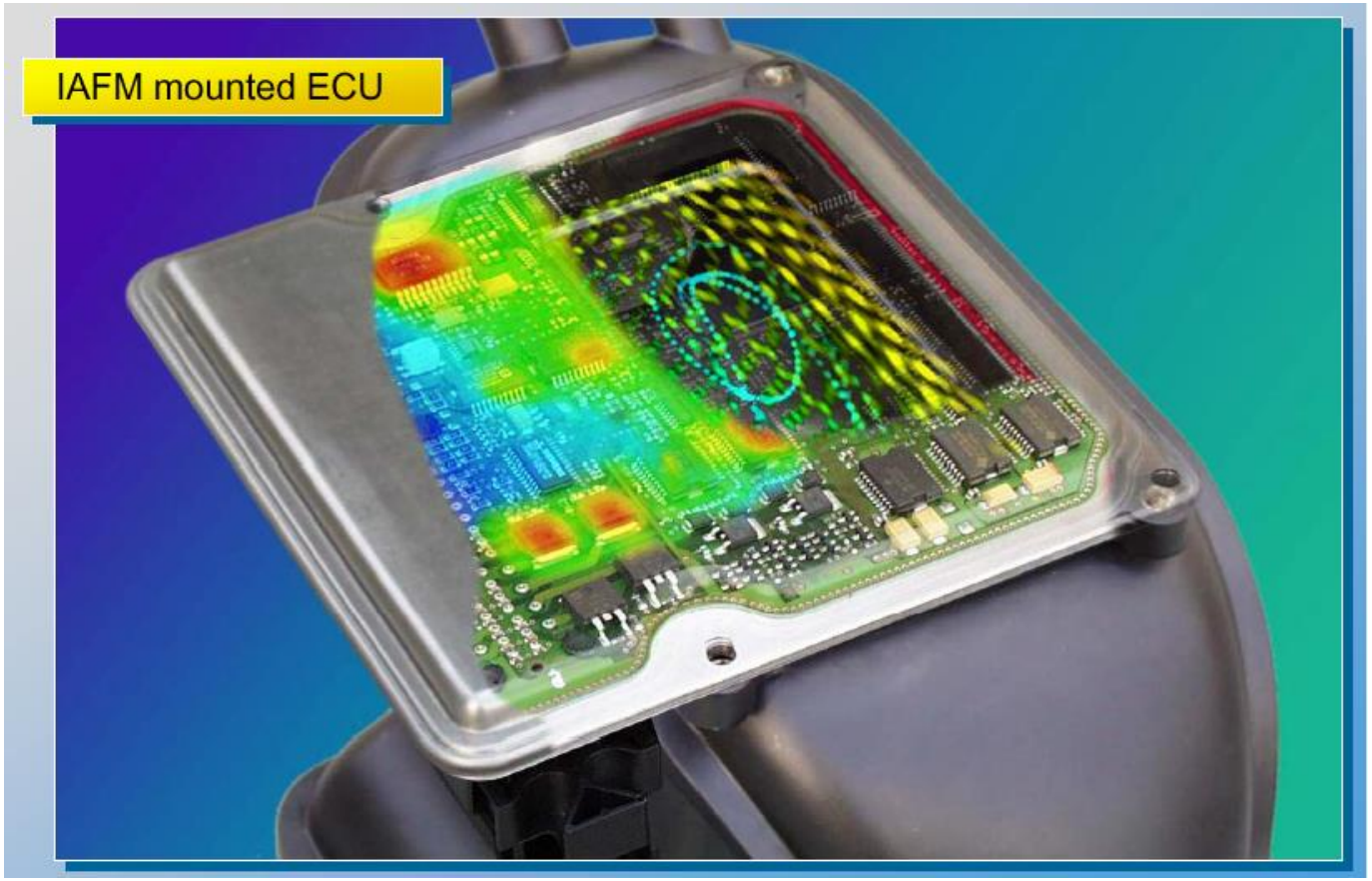


汽油机ECU
/ECU of Gasoline Engine





汽油机ECU
/ECU of Gasoline Engine



IAFM: Intake Air-Fuel Module / Integrated Air-Fuel Module